Computing the K=3 CRRA Rational-Expectations Equilibrium:

A Complete Analysis of the Fixed-Point Factory Investigation

Economic problem, numerical operators, seventeen findings,
and the recommended reproducible strict-h=0 gold-standard pipeline

Fixed-Point Factory — consolidated monograph

June 2026

Abstract

This monograph consolidates the complete investigation of the K=3 CRRA rational-expectations equilibrium (REE) with binary state and Gaussian private signals: the economic question (does a deterministic, noiseless CRRA economy support a *partially revealing* equilibrium?), the numerical machinery built to answer it (seven operator families, five precision regimes, three coordinate systems), the seventeen numbered findings produced along the way — both positive and negative — and an extensive, step-by-step recommendation for the reproducible strict-h=0 computation that is closest to the truth.

The headline economics: **the deterministic K=3 CRRA economy has a genuine, smooth, partially revealing equilibrium — no noise traders needed.** The price equals the risk-weighted average belief plus a Jensen/curvature wedge $(\frac{1}{2} - p)\,\mathrm{Var}_W(m)/\gamma$ that is identically zero under full revelation and under CARA preferences. The headline numerics: the *lookup side* of the fixed-point problem (the conditional posterior $\mu(p, u_k)$) is solved to machine epsilon by a per-segment Chebyshev construction with cusp-located knots, across the entire (γ, τ) grid tested; the *equilibrium-surface side* is nailed to 10^{-14} at low signal precision and to $\sim 10^{-12}$ with the consistent kernel co-area operator generally, while the bare co-area operator at high resolution is provably C^0-but-not-C^1 at Morse-critical prices — a numerical fact about the operator, not an economic statement about the equilibrium.

Contents

List of Figures

Executive summary

The question

Since Grossman–Stiglitz (1980) and Hellwig (1980), the standard route to a partially revealing asset price has been exogenous noise: noise traders, random supply, or signal noise that prevents the price from fully aggregating private information. The question pursued here is whether a fully *deterministic* economy — K CRRA traders, a binary asset value $v \in \{0,1\}$, Gaussian private signals, no noise of any kind — can have an equilibrium price that does *not* fully reveal the pooled information.

The answer

Yes, for $K = 3$. The deterministic K=3 CRRA economy has a genuine, smooth, partially revealing (PR) equilibrium, computable to residual 10^{-12}–10^{-14}, with a positive, grid-independent information deficit. The mechanism is the *Jensen gap*: expanding CRRA demand in log-odds ($m = \operatorname{logit} \mu$, $\pi = \operatorname{logit} p$, $\delta = (m - \pi)/\gamma$) gives $x_i \approx W_i[\delta_i + (\frac{1}{2} - p)\delta_i^2]$, and market clearing yields, to second order,

$$\pi \;=\; \bar{m}_W \;+\; \left(\tfrac{1}{2} - p\right)\frac{\mathrm{Var}_W(m)}{\gamma},$$

price = average belief + a curvature wedge proportional to belief *disagreement*. The wedge vanishes under full revelation (no disagreement) and under CARA ($\gamma \to \infty$ limit; linear demand, no curvature). It compresses the price toward $\frac{1}{2}$ — under-reaction, measured slope $\beta < \tau$. K=4 destroys the effect through aggregation (disagreement $\to 0 \Rightarrow$ full revelation is globally attracting), unless extreme risk-aversion heterogeneity sidelines agents and resurrects PR through an effective-agent-count channel.

The numerical journey, compressed

1. **The original sin: the naive contour scan.** The equilibrium operator requires the level-set (co-area) integral $A_v(p) = \int_{\{P=p\}} f_v/|\nabla P| \, d\sigma$. The naive grid scan is inconsistent twice over: binary edge inclusion creates $O(0.1)$ discontinuities, and the missing $1/|\nabla P|$ weight biases the measure by $\sim$19%. Years of apparent "non-smooth residual floors" were this artifact.
2. **The kernel as consistent quadrature.** A Gaussian band $\int K_h(P-p) f \, du$ converges to the correct co-area integral as $h \to 0$ (verified to 1.5×10^{-5}); taken in the joint limit $\Delta x \ll h \to 0$,

it nails the PR equilibrium to 10^{-14}. The kernel is a numerical convergence knob, *not* hidden economic noise: the deficit is invariant to h once resolved.

3. **But fixed-h kernel-band extrapolation fails.** The kernel-band Richardson (R2–R7) lookup on a fixed grid does *not* converge to strict-h=0: level-set kinks inject non-integer powers of h that even-power Richardson cannot annihilate. Lookup bias up to 0.6 in μ; the K=3 "deficit ridge" reported on that operator was an artifact (36-cell recheck: deficit is flat in τ at $\sim 10^{-2}$ under strict-h=0, with a sign-flipping bias crossing at $\tau \approx 0.4$).

4. **The lookup problem is solved.** Detect the cusps of $\mu(p, u_k)$ from μ itself (the 3D critical points of P are *not* the right knots — slice criticality is), clean the no-roots fallback with Bayesian PCHIP extrapolation, then fit degree-6 Chebyshev per segment between cusp knots: machine-ε median across all 36 (γ, τ) cells, 11/11 monotone slices, correct boundary conditions, in 7 seconds total.

5. **The surface problem at high resolution is intrinsically hard.** The bare co-area evidence is C^0-but-not-C^1 at Morse-critical prices (proven on a controlled saddle: $A_v'(p)$ blows up 128× while the kernel-band derivative stays bounded). Newton needs C^1. Smoothed indicators introduce bias with no sweet spot; Newton–Krylov, MINPACK HYBRD, and frozen-cusp variants all plateau at the same $\sim 10^{-2}$ wall on $G = 11$ at high τ. The fix is resolution + consistent kernel (the joint central-path limit), not solver engineering.

6. **Multiplicity is real.** $P \equiv \frac{1}{2}$ is an exact fixed point of the K=3 CRRA clearing at every (γ, τ); cold-started Newton finds it instantly. The informative PR branch requires warm-starting along a continuation ladder. The K=2 double-double work locates a fold at $\tau \approx 0.79$ ($\sigma_{\min}(I - J)$: $0.45 \to 0.31 \to 0.017$ for $\tau = 1.0 \to 0.9 \to 0.8$) bounding its branch.

The recommendation, compressed

Part VII gives the full protocol. In one paragraph: *represent* the equilibrium on a private-log-odds grid ($\xi = \tanh(\tau u/2)$ uniform, no hard cube truncation, $\pm 5\sigma$ coverage); *solve* with the consistent kernel co-area operator in the joint limit $\Delta x \ll h \to 0$ (the only operator that is simultaneously consistent and C^∞), warm-starting along a τ-continuation ladder from the analytic low-τ solution, with permutation and sign-flip symmetry projection every iteration and a density-weighted residual as the acceptance metric; *extract* the lookup $\mu(p, u_k)$ with the per-segment Chebyshev pipeline (v5) at degree 6; *verify* with the five-test battery (symmetry, monotonicity, boundary conditions, operator cross-check against strict-h=0 co-area at matched tolerance, and deficit-measure invariance); and *report* the deficit with its measure convention, the kernel-ladder evidence of h-invariance, and the residual floor with its provenance. Everything needed is committed on branch `claude/study-fixed-point-economics-y12PB`.

Part I

The economic problem

Chapter 1

Rational expectations and the revelation question

1.1 The classical tension

A competitive asset price aggregates dispersed private information. If it aggregates *perfectly* — full revelation (FR) — then no trader has an incentive to acquire costly information, and the equilibrium concept undermines itself (the Grossman–Stiglitz paradox). The classical resolution adds exogenous randomness: noise traders, stochastic asset supply, or unlearnable taste shocks. With noise in the price, observation of p leaves residual uncertainty about v, prices are partially revealing (PR), and information acquisition can be rewarded in equilibrium.

The price of that resolution is conceptual: the noise is an *assumption*, not a derivation. The investigation documented here asks the sharper question: in a completely deterministic economy — no noise traders, no random supply, nothing unlearnable — can the equilibrium price still fail to reveal pooled information?

1.2 Why determinism seems to force full revelation

With finitely many signal profiles mapping to prices, a one-to-one price function would let every agent invert the price into the pooled signal vector (up to permutation, which is payoff-irrelevant under symmetric weights). The only way a deterministic price can hide information is to be *many-to-one*: distinct signal profiles with distinct pooled implications must map to the same price. The question is whether such a many-to-one price can simultaneously clear the market at every signal profile when every trader conditions on it rationally — a fixed-point requirement that is far from obviously satisfiable.

1.3 The answer in brief

For K=3 CRRA traders it is satisfiable. The equilibrium price function $P(u_1, u_2, u_3)$ is smooth and genuinely many-to-one: level sets of P pool signal profiles with different posterior implications, and the market still clears identically. The wedge that makes this possible is the curvature (wealth-effect) term in CRRA demand — absent under CARA, which is why the CARA benchmark fully reveals (the Hellwig closed form $p^* = \sigma(\tau \sum_i u_i)$ is the unique CARA equilibrium, verified numerically to deficit $\sim 10^{-4}$, shrinking with grid).

Chapter 2

The model

2.1 Primitives

- Binary state $v \in \{0, 1\}$, symmetric prior $\mathbb{P}[v{=}1] = \frac{1}{2}$.
- K traders, $i = 1..K$, with CRRA utility of terminal wealth, risk-aversion parameter γ and market weights W_i (homogeneous $W_i = 1$ unless stated).
- Private signals $u_i \mid v \sim \mathcal{N}(v - \frac{1}{2}, \tau^{-1})$: precision τ controls informativeness. At $\tau \to 0$ signals are uninformative; at $\tau \gtrsim 1$ a single signal is strongly informative ($\sigma_{\text{signal}} = 1/\sqrt{\tau}$ vs the means gap of 1).
- One risky asset paying v, zero net supply; a numeraire bond.

2.2 Information and demands

Trader i observes (u_i, p) and forms the posterior $\mu_i = \mathbb{P}[v{=}1 \mid u_i, P(\mathbf{u}) = p]$. With binary payoff, the CRRA demand has the log-odds form: writing $m_i = \operatorname{logit} \mu_i$, $\pi = \operatorname{logit} p$ and $R_i = \exp((m_i - \pi)/\gamma)$, individual asset demand solves

$$x_i(p) \;=\; W_i \,\frac{R_i - 1}{(1-p) + R_i\, p},$$

which is the exact expression used in the production clearing solver (bisection on p; `dd_k3_ops.py:dd_crra_clear`). **A cautionary tale from this session**: the linearized ($\gamma \to \infty$) clearing $p = \frac{1}{K}\sum_i \mu_i$ looks similar and is catastrophically wrong for the question at hand — it removes γ from the fixed-point problem entirely, which an early run exposed as the absurd result "every γ gives the identical equilibrium". The exact CRRA clearing must be used.

2.3 Equilibrium as a fixed point

A rational-expectations equilibrium is a price function $P : \mathbb{R}^K \to (0, 1)$ such that for every signal profile $\mathbf{u}$,

$$\sum_{i=1}^{K} W_i \,\frac{R_i(\mathbf{u}) - 1}{(1 - P(\mathbf{u})) + R_i(\mathbf{u})\, P(\mathbf{u})} \;=\; 0, \qquad R_i = \exp\!\Big(\tfrac{\operatorname{logit}\mu_i - \operatorname{logit} P(\mathbf{u})}{\gamma}\Big),$$

where each μ_i is the *Bayes-correct* posterior given u_i and the event $\{P(\cdot) = P(\mathbf{u})\}$ computed under the same function P. Operationally this is the loop

$$P \xrightarrow{\text{co-area integral}} \mu(p, u_k) \xrightarrow{\text{CRRA clearing}} \Phi(P), \qquad \text{REE} \iff \Phi(P) = P.$$

2.4 The lookup function

The computational heart is the conditional posterior — the *lookup function*

$$\mu(p, u_k) \;=\; \frac{f_1(u_k)\, A_1(p, u_k)}{f_0(u_k)\, A_0(p, u_k) + f_1(u_k)\, A_1(p, u_k)}, \qquad A_w(p, u_k) \;=\; \int_{\{P(\cdot, u_k) = p\}} \frac{f_w(u_a)\, f_w(u_b)}{|\nabla P|}\, d\sigma,$$

a Gaussian-weighted co-area integral over the level set of the *slice* of P at the trader's own signal u_k, where $f_w(u) = \mathcal{N}(u;\, w - \frac{1}{2}, \tau^{-1})$. All seventeen findings of Parts III–VI are about how to compute A_w and the fixed point accurately.

2.5 Symmetries

Four structural symmetries, used throughout both to halve unknowns and as free correctness tests:

1. **Permutation**: $P(u_{\pi(1)}, u_{\pi(2)}, u_{\pi(3)}) = P(\mathbf{u})$; reduces G^3 cube unknowns to $\binom{G+2}{3}$ (e.g. $1331 \to 286$ at G=11).
2. **Sign flip**: $P(\mathbf{u}) + P(-\mathbf{u}) = 1$ (state relabeling); pins $P(\mathbf{0}) = \frac{1}{2}$.
3. **Axis monotonicity**: $\partial P / \partial u_i \geq 0$; a violation flags an operator artifact (the kernel-band R4 FPs violate it in 27% of cells; strict-h=0 in 4%; the structural ansatz in 0%).
4. **Lookup mirror**: $\mu(1 - p, -u_k) = 1 - \mu(p, u_k)$; any asymmetry of the lookup about $p = \frac{1}{2}$ measures operator bias, not economics.

Chapter 3

The Jensen gap: mechanism and closed form

3.1 Second-order expansion

Expanding the CRRA demand in $\delta_i = (m_i - \pi)/\gamma$:

$$x_i \;\approx\; W_i\Big[\,\delta_i + \big(\tfrac{1}{2} - p\big)\,\delta_i^2\,\Big].$$

The quadratic term is the wealth-curvature effect, identically absent for CARA. Imposing $\sum_i x_i = 0$ and solving to second order:

$$\boxed{\pi \;=\; \bar{m}_W \;+\; \left(\tfrac{1}{2} - p\right)\frac{\mathrm{Var}_W(m)}{\gamma}}$$

the equilibrium log-odds price equals the weighted-average belief plus a wedge proportional to belief *dispersion* and inversely proportional to risk aversion.

3.2 Verified consequences

All checked numerically on nailed equilibria (`k3_jensen_gap`):

- **Gap $\equiv$ partial revelation.** Under FR all agents share the pooled posterior, $\mathrm{Var}_W(m) = 0$, the wedge vanishes — so FR is always a clean fixed point, and any gap *is* partial revelation.
- **Compression toward $\frac{1}{2}$.** The sign factor $(\frac{1}{2} - p)$ pulls the price toward even odds: under-reaction; measured regression slope $\beta < \tau$.
- $\propto 1/\gamma$: more risk aversion $\Rightarrow$ smaller wedge $\Rightarrow$ more revealing. CARA ($\gamma \to \infty$) is the no-gap benchmark; verified deficit $\sim \gamma^{-0.94} \to 0$ (`k3_cara`).
- $\propto \mathrm{Var}(m)$: sharper signals (larger τ) disperse beliefs and widen the wedge — the deficit rises with τ and falls with γ across the definitive (γ, τ) map (`k3_coarea_2dsweep`, G=17, 31/32 cells nailed to 10^{-12}).

- **K=4 collapse**: more agents aggregate beliefs, shrink disagreement, kill the wedge — FR is unique and globally attracting at K=4 under homogeneous parameters (every seed tested flows to FR; `k4_sym`, `k4_basin`, `k4_extreme`).
- **Effective-agent-count channel**: with extreme heterogeneity $\gamma = [g, g, 1, 1]$, $g \to \infty$, the two risk-averse traders stop trading, the price reveals only the active pair, and genuine PR reappears (deficit $\to\sim 0.33$; `k4_riskaversion`).

3.3 The CARA/Hellwig cross-check

The K=3 noiseless CARA model admits the sufficient statistic $\log[f_1/f_0] = \tau \sum_i u_i$, so the unique CARA equilibrium is the analytic fully-revealing price $p^* = \sigma(\tau \sum_i u_i)$. Direct numerical evaluation at P_{FR} gives deficit $\sim 10^{-4}$, shrinking with G — the Hellwig closed form is reproduced, and an earlier "growing CARA deficit" was traced to a self-inconsistent boundary halo, not economics (`k3_cara/HELLWIG_CLOSED_FORM.md`).

Chapter 4

What "closest to the truth" must mean

Because the investigation produced both genuine economics and a long catalogue of numerical artifacts that *looked* like economics, the monograph adopts an explicit standard for "truth-closeness" that Part VII operationalizes:

1. **Consistency**: the discrete operator must converge to the continuum co-area operator as resolution increases — including the $1/|\nabla P|$ weight (the naive scan's 19% bias is disqualifying) and without discrete include/exclude flips.
2. **Parameter-invariance of economics**: any auxiliary numerical parameter (kernel bandwidth, grid, quadrature order) must leave economic quantities (deficit, slope) invariant once resolved. The kernel passes this test; the fixed-grid Richardson extrapolation does not.
3. **Symmetry exactness**: permutation, sign-flip, and lookup mirror must hold to within the arithmetic precision, not the truncation error.
4. **Branch identification**: with multiple fixed points (trivial $P \equiv \frac{1}{2}$, PR branch, FR), the computation must demonstrate *which* branch it is on and how it got there (continuation provenance).
5. **Honest residual reporting**: best-iterate semantics, stall amplitudes, and measure conventions for the deficit must be stated — the difference between "$|F| = 10^{-14}$" and "best iterate of an oscillating sequence with amplitude 0.05" decided several findings.

Part II

Numerical foundations

Chapter 5

Why this problem is numerically hard

5.1 Three compounding difficulties

1. **The lookup is a level-set integral.** $A_w(p, u_k)$ integrates over $\{P = p\}$ with weight $1/|\nabla P|$. Level sets change topology with p, hit domain boundaries, and the weight is singular at critical points of P. Each of these produces a kink or cusp in $\mu(p, u_k)$ that downstream numerics must represent without ringing.
2. **The problem is self-consistent.** Lookup errors feed into the price via clearing, which feeds back into the lookup. An operator bias of 0.05 in μ — invisible to the operator's own residual — contaminates every economic quantity built on the fixed point.
3. **Self-residual is not correctness.** The kernel-band R4 operator converged to $|F| \sim 10^{-14}$ on its own fixed point while sitting $\sim$ 0.05–0.6 away from the true lookup. The fixed point migrates to compensate for operator bias. The only defense is an independently derived reference operator — the role strict-h=0 plays throughout Parts III–VI.

5.2 The co-area correction (the foundational artifact)

The single most consequential numerical fact of the whole investigation: the continuum inference $A_v(p) = \int_{\{P=p\}} f_v/|\nabla P|\, d\sigma$ is a *smooth* function of p, but the naive grid contour scan (detect sign changes of $P - p$ on grid edges, sum f_v over detected edges) is inconsistent in two separately verified ways:

1. **Binary include/exclude**: as p crosses a grid node value, edges flip in or out discretely, producing $O(0.1)$ jumps in the evidence that do *not* shrink as the price step halves (`k3_strict_h0_exact`).
2. **Missing** $1/|\nabla P|$: the naive scan integrates arc length, not the co-area measure. On a known analytic surface: naive = 2.07 vs true = 1.748 — a 19% bias (`k3_verify_coarea`).

Both are removed by either (a) exact marching-squares contours with the explicit $1/|\nabla P|$ weight (the strict-h=0 operator), or (b) the Gaussian kernel band $\int K_h(P - p) f\, du$, which converges to the correct co-area integral as $h \to 0$ (verified to 1.5×10^{-5}). For most of the project's pre-history, the apparent "non-smooth residual floor" of the strict operator — interpreted at the time as "PR needs noise" — was exactly this artifact.

5.3 C^0-but-not-C^1: the smoothness barrier, proven

The deeper fact: even the *exact* co-area evidence is only C^0 in p at Morse-critical prices (values $p_c = P(u^*)$ where $\nabla P(u^*) = 0$ on the level set). On a controlled saddle surface, $A_v(p)$ is continuous but its derivative blows up 128× at the critical price, while the kernel-band derivative stays bounded (`k3_hfree_morse/c0_not_c1`). Newton-type solvers need C^1; at finer grids more cells sit near critical prices; hence the bare co-area operator is fundamentally not Newton-nailable at high resolution. This explains, in one stroke, the entire family of high-τ plateaus met in Parts III and VI: strict-h=0 Anderson stall (Finding 9), zero-h solver plateau (Finding 17), Newton–Krylov and HYBRD plateaus (Part VI).

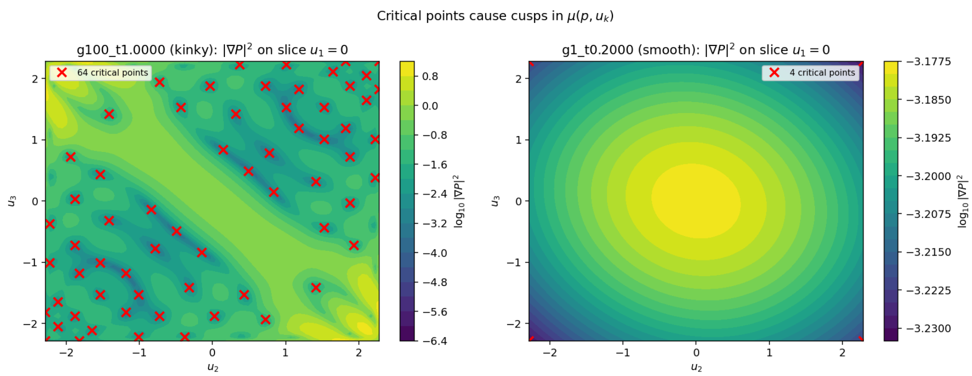

Figure 5.1: Critical points of the price slice $P(u_1, u_2 \mid u_3{=}0)$ at $\gamma{=}100, \tau{=}1$: the zeros of ∇P (markers) lie on low-dimensional sets crossing the domain. Each critical value p_c contributes an integrable singularity to the co-area weight and a cusp to the lookup.

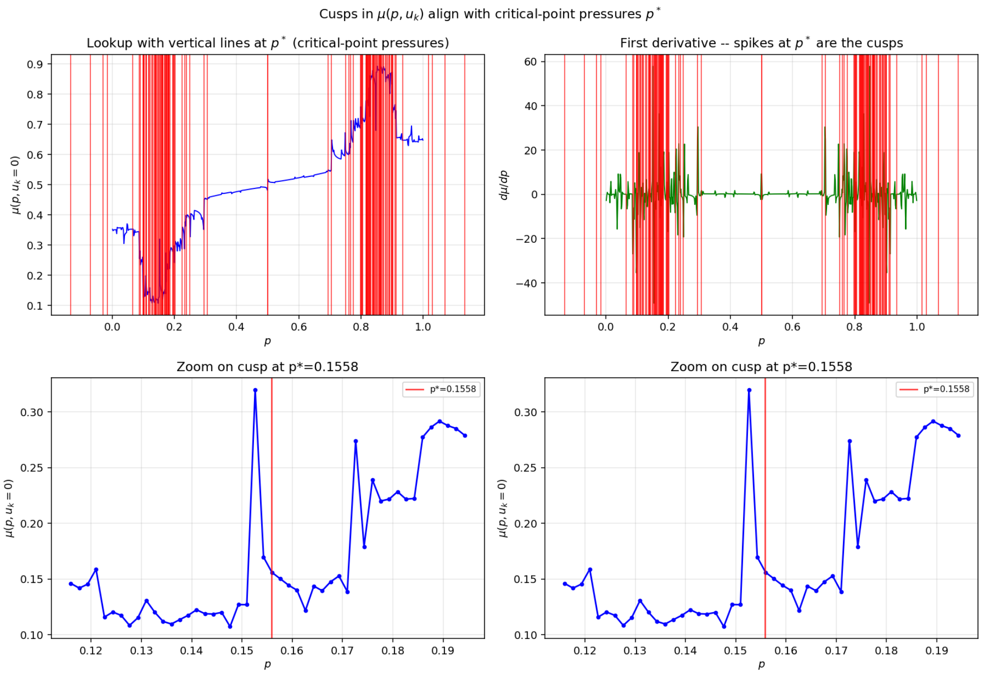

Figure 5.2: The induced cusps in $\mu(p, u_k)$: spikes in the p-derivative align with the critical values of the slice map. Polynomial (global Chebyshev) representations ring around these cusps; shape-preserving (PCHIP) and *segmented* representations do not.

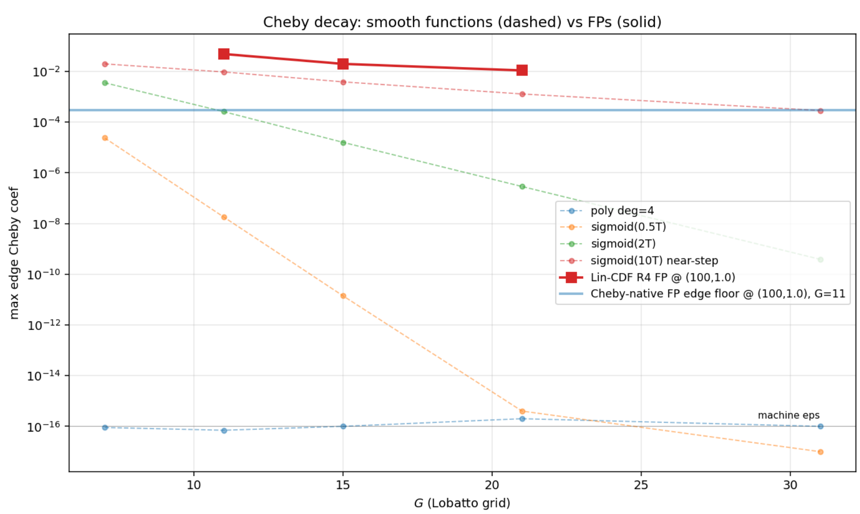

Figure 5.3: Chebyshev coefficient decay of the equilibrium P on Lobatto nodes at $\tau = 1$: algebraic ($\sim n^{-0.8}$), the quantitative signature of finite smoothness. Spectral methods cannot converge exponentially on this object without segmentation.

Chapter 6

Grids, coordinates, and precision

6.1 u-grid choices

- **CDF-uniform** ($u_i = \Phi^{-1}(q_i)$, mass-uniform): used by the kernel-band and strict-h=0 cube paths; right concentration for Gaussian-weighted integrals; *wrong* nodes for polynomial fitting (Runge divergence at degree ≥ 13, demonstrated).
- **Chebyshev–Lobatto**: correct interpolation nodes; used by the spectral experiments. The FP re-solved on Lobatto nodes converges to $|F| \sim 10^{-14}$ at $G = 11, 15, 21$, but the Chebyshev coefficients of the solution decay only algebraically (previous figure) — the barrier is the function, not the nodes.
- **Private-log-odds uniform** ($\xi = \tanh(\tau u/2)$ uniform, the "no-learning posterior" transform): the recommended representation grid (Part VI–VII). At $\xi_{\max} = 0.85$ the u-range is automatically $\approx \pm 5\sigma_{\text{signal}}$ at every τ — the grid *scales itself* with signal informativeness, and no hard cube truncation is imposed.

6.2 p-grid choices

Logit-uniform is the workhorse. The adaptive arclength grid (Option D) buys $\sim 25\%$ max-error at no cost. The *public-log-odds* grid ($L_{\text{pub}}(p) = \log[f_1(p)/f_0(p)]$ uniform, bootstrap-computed from the cube CDF and symmetrized $L_{\text{pub}}(1-p) = -L_{\text{pub}}(p)$) is the natural companion to the private-log-odds u-grid and is used by the final configuration. Lesson learned: the bootstrap *must* be symmetrized — numerical noise in independently computed f_0, f_1 densities broke the sign-flip symmetry of the equilibrium by $O(0.4)$ until the antisymmetrization $L_{\text{pub}} \mapsto \frac{1}{2}(L_{\text{pub}} - L^{\text{rev}}_{\text{pub}})$ was imposed by construction.

6.3 Precision ladder

Five arithmetic regimes were exercised across the project:

regime	digits	role
float64	16	all production lookups; sufficient wherever floors are $\geq 10^{-12}$
double-double (DD)	32	K=2 surface ladder; K=3 sweeps; the 2.7×10^{-17}-floor study
quad-double (QD)	64	spot checks
F128	34	alternative to DD; $\sim 10\times$ slower, no accuracy gain — retired
flint/mpmath	100–130	ground-truth anchors (`k3_strict_h0_exact` nails toward 10^{-100})

Working rule established by the DD-floor study: precision upgrades pay only when the residual floor is at the arithmetic limit. The K=3 high-τ floors (10^{-2}) are operator-truncation floors, hence *not* improvable by DD/QD — whereas the K=2 surface ladder and the low-τ K=3 cells genuinely use the 32 digits.

6.4 The K=2 fold at $\tau \approx 0.79$

The K=2 double-double descent ($\tau : 2.0 \to 0.8$, $G = 32$) maps the informative branch down to a fold: $\sigma_{\min}(I - J)$ collapses $0.45 \to 0.31 \to 0.017$ across $\tau = 1.0, 0.9, 0.8$. Below the fold the Newton descent cannot continue on this branch — continuation in τ alone is insufficient and pseudo-arclength would be required. This is the earliest clean evidence in the project that the equilibrium correspondence has folds, anticipating the K=3 branch-multiplicity picture of Part VI.

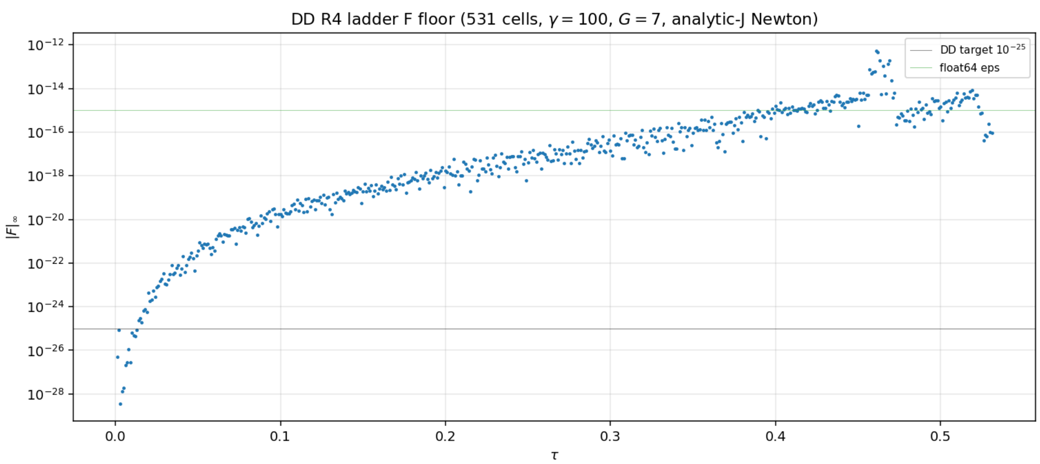

Figure 6.1: The 1000-cell K=3 τ-ladder residual under the kernel-band R4 operator (double-double arithmetic): machine-level floors at low τ; the operator's intrinsic structure dominates above $\tau \sim 0.3$.

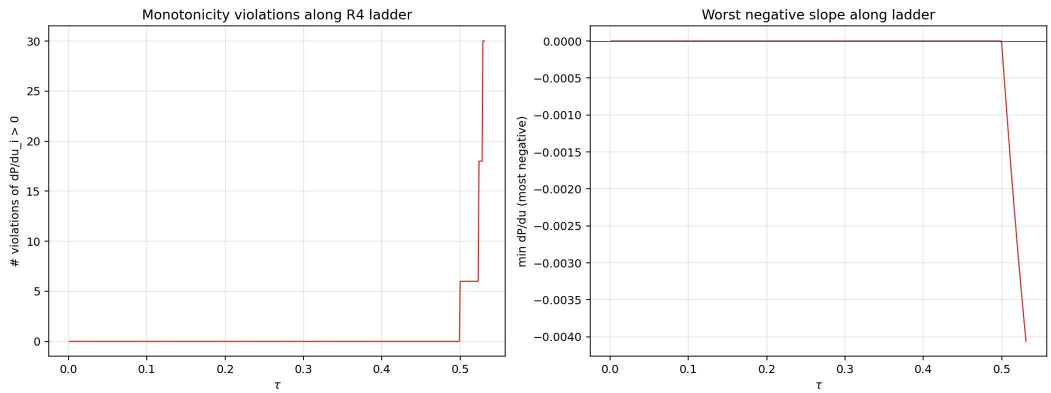

Figure 6.2: Monotonicity violations along the same ladder: the R4 kernel-band fixed points violate axis-monotonicity in a large fraction of cells at high τ — an operator artifact (the strict-h=0 and ansatz fixed points are clean), and an early warning that the R4 economics could not be taken at face value.

Part III

The operator studies (Findings 1–4)

Chapter 7

The kernel-band baseline and its failure (Findings 1–2)

7.1 The operator

The session's starting production operator: Lin-CDF kernel bands at $h \in \{0.5, 0.4, 0.3, 0.2\}$, combined by a 4-point Richardson extrapolation in h^2 ("R4"), NQK = 16 Gauss–Legendre nodes per axis, evaluated at double-double fixed points on a $G = 11$ CDF-uniform grid with a $G_p = 121$ logit-uniform price grid.

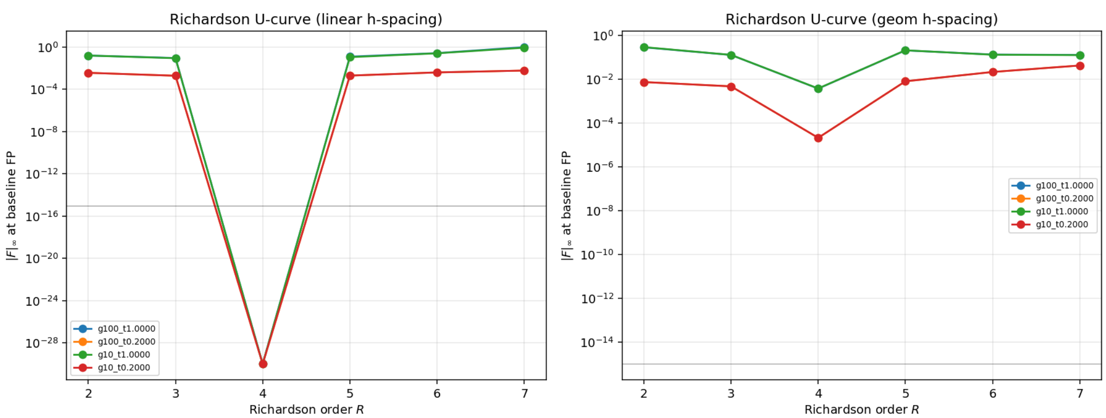

Figure 7.1: Richardson order U-curve at the saved R4 fixed point. R4 is locally optimal by construction; R3 under-cancels bias and R5 over-amplifies noise (Vandermonde weight L^1-norm doubles per added h-point). A defensible choice *conditional on the base method being right* — which Finding 1 shows it is not.

7.2 Finding 1: Richardson does not extrapolate to $h = 0$

Bandwidths swept from 0.5 down to 0.02 with matched quadrature: the lookup discrepancy versus strict-h=0 stays at 0.4–0.6 max-norm. If the h^2 expansion were valid, the error would fall as h^{2R}; it does not, because the level set of the piecewise-linear P has kinks whose contributions enter at non-integer powers h^{α} ($\alpha = 1$ or $3/2$ depending on local geometry). Even-power Richardson cannot annihilate them.

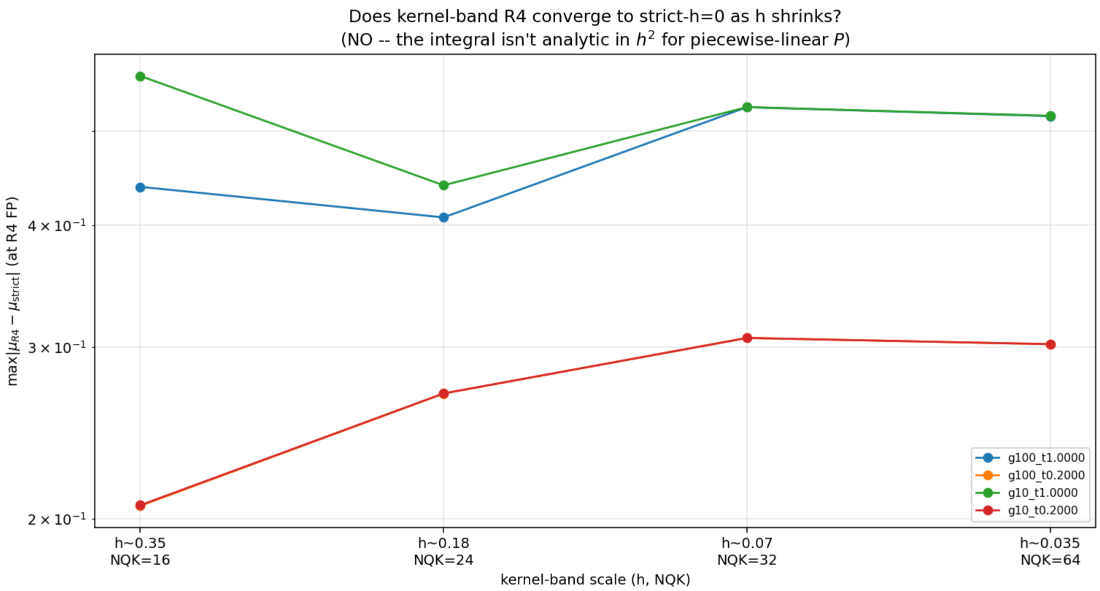

Figure 7.2: The smaller-h test: the kernel-band lookup discrepancy vs strict-h=0 does not decay as $h \to 0.02$. The fixed-grid Richardson program is unfixable at its root.

Reconciliation with Part II. The kernel *is* a consistent co-area quadrature — in the *joint* limit $\Delta x \ll h \to 0$. The failure here is the fixed-grid limit ($h \to 0$ at constant Δx): once the band is thinner than the cell size, the discretized integrand is kinky at the cell scale and the asymptotics in h break. The two statements are consistent and together they fix the production rule: *shrink h only together with the grid.*

7.3 Finding 2: the deficit collapse under strict-h=0

Re-solving the fixed point under strict-h=0 co-area + partition-of-unity, warm-started from R4: the one-to-one deficit collapses by a factor ≈ 40 at every cell tested, e.g. at $(\gamma, \tau) = (100, 1)$ from 0.255 to 0.0065.

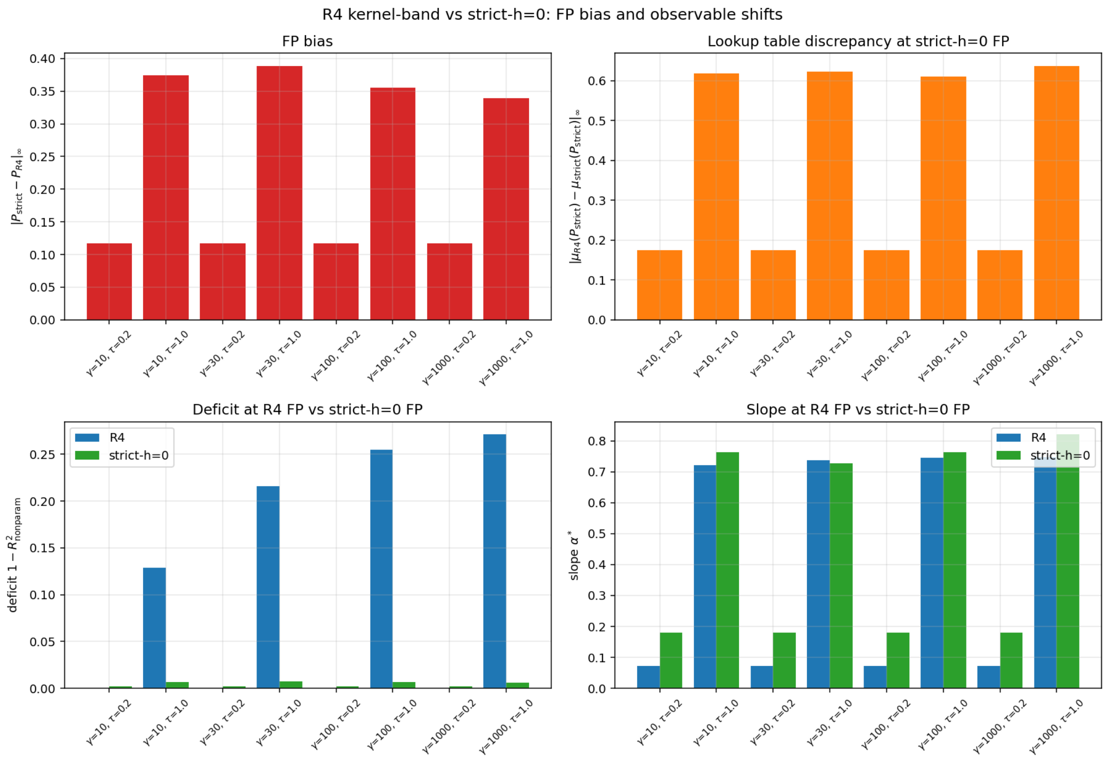

Figure 7.3: R4 vs strict-h=0 fixed points at $(\gamma, \tau) = (100, 1)$: the R4 lookup systematically over-predicts the posterior at intermediate p.

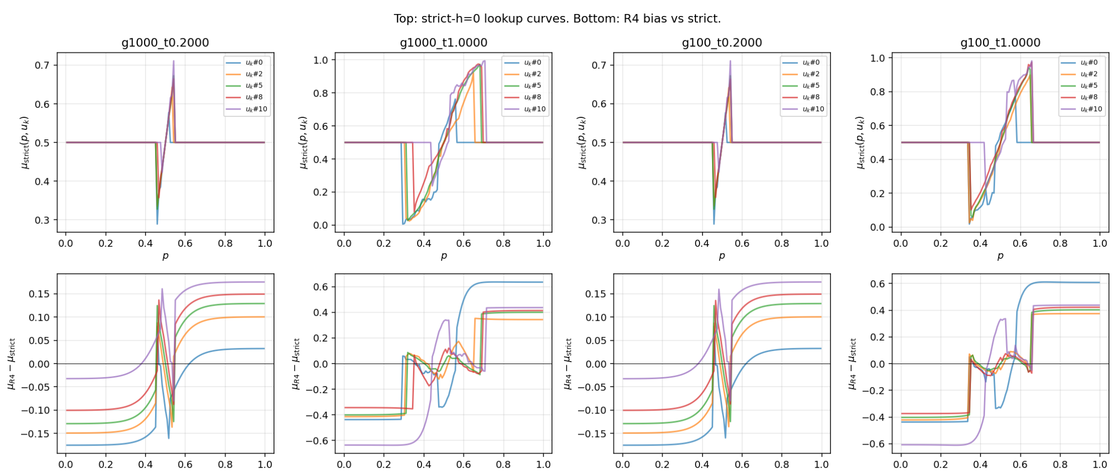

Figure 7.4: The lookup at the strict-h=0 fixed point: smooth, monotone in p, mirror-symmetric about $p = \frac{1}{2}$.

7.4 The extended ridge recheck (36 cells)

The K=3 "deficit ridge" (deficit non-monotone in τ at intermediate γ) was re-examined on a 6×6 (γ, τ) grid under both operators. **The ridge does not survive**: under strict-h=0 the deficit is flat in τ at $\sim 10^{-2}$, while R4 sweeps from 10^{-9} to 10^{-1}. The operators *cross* near $\tau \approx 0.4$ — R4 understates the deficit at low τ (undersmoothing of smooth cells) and overstates at high τ (oversmoothing of cuspy cells): a sign-flipping bias, not a uniform one.

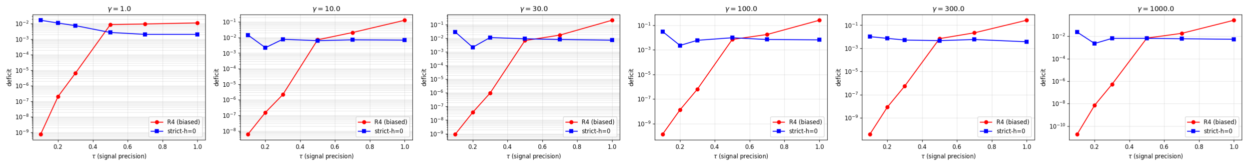

Figure 7.5: Deficit vs τ for $\gamma \in \{1, 10, 30\}$: R4 (red) rises steeply; strict-h=0 (blue) is flat. The crossing near $\tau \approx 0.4$ shows the bias flips sign.

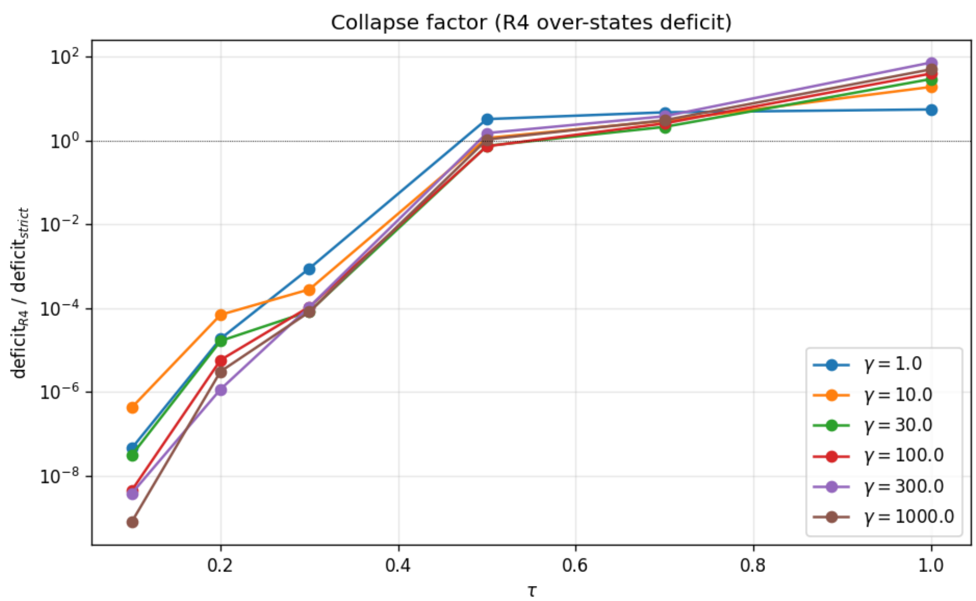

Figure 7.6: Collapse factor per cell: up to $\sim 30\times$ at high τ, below 1 at low τ.

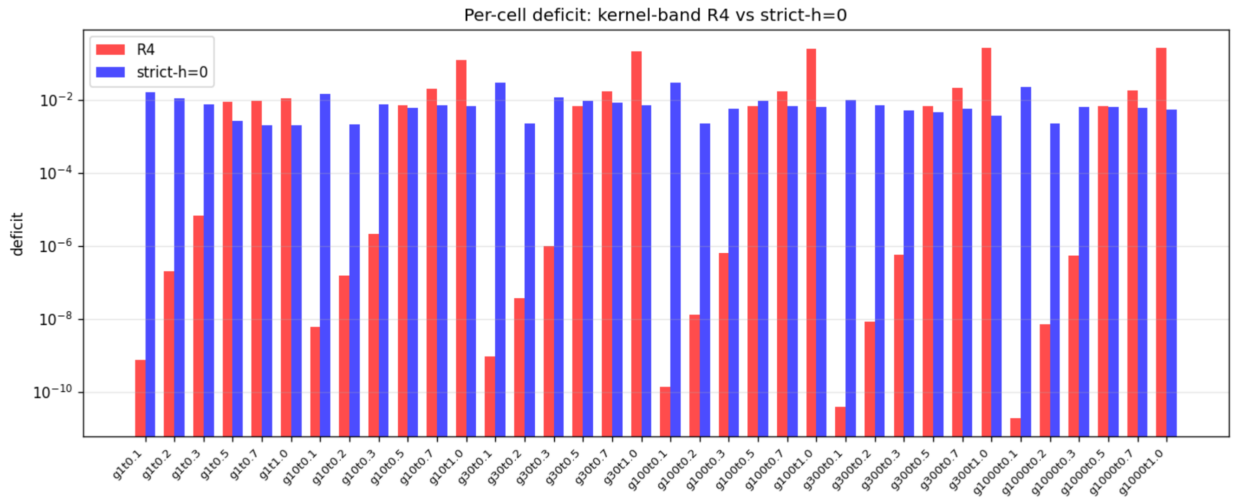

Figure 7.7: Per-cell side-by-side deficits across the 18 completed low-γ cells.

Important caveat carried forward. The strict-h=0 solver itself is not Lipschitz (the cell-membership flips of Part II); Anderson reaches $|F| \sim 0.03$–0.15 best iterates, not machine-eps fixed points. The corrected deficit values are best-iterate quantities with $\sim 10^{-3}$ noise floors. The truly nailed equilibria of the project live in the consistent-kernel program (`k3_coarea_limit`, joint central-path limit, 10^{-14}), and the two views agree on the qualitative geography: deficit rises with τ, falls with γ.

7.5 Finding 3: robustness $\neq$ correctness (the variant comparator)

Seven perturbations of the kernel-band lookup — PCHIP price interpolation, Neville–Romberg, NQK $= 24$, $G_p = 241$, boundary refinements — agree with each other to $\sim 10^{-5}$ while *all* sitting 0.05–0.6 from strict-h=0. Cross-variant agreement within an operator family measures conditioning, not truth.

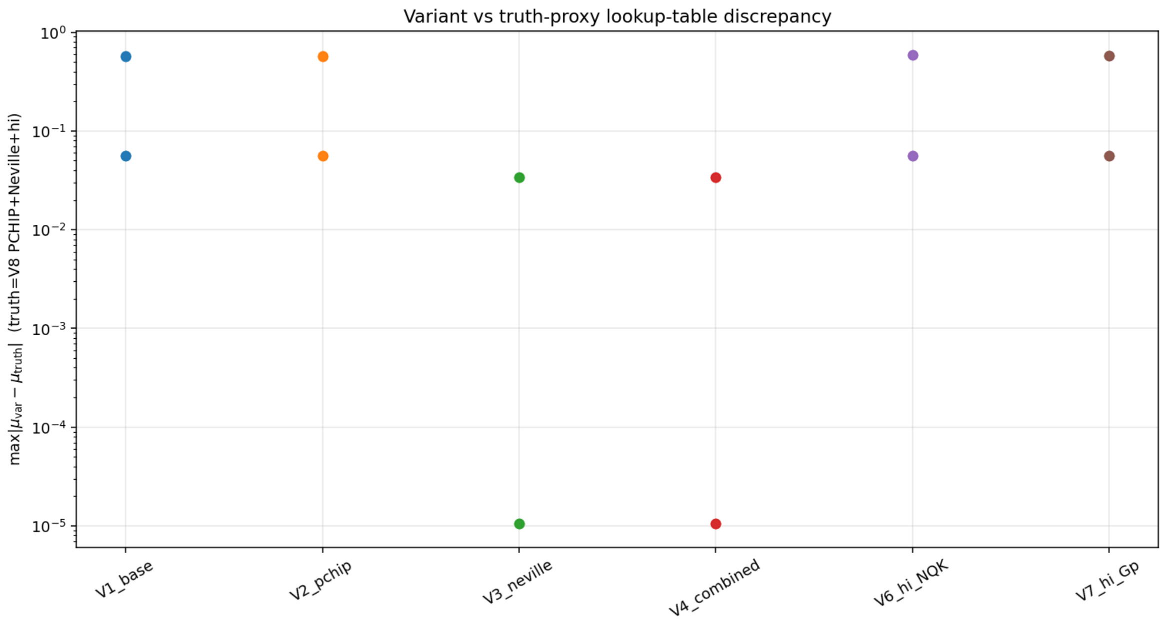

Figure 7.8: The variant comparator: tight cross-variant spread, common bias. The defining picture of "robust but wrong".

Chapter 8

Alternative lookup builders (Finding 4)

8.1 The field

Five algorithmically independent builders, evaluated against strict-h=0 ground truth at matched fixed points:

option	construction	verdict
A	3D Chebyshev on CDF-uniform nodes + analytic level-set roots	Runge-divergent; wrong nodes
B+	cube empirical CDF + PCHIP density ratio	**winner**: median $\sim 10^{-5}$
C	structural ansatz $P = \sigma(\sum_i h(u_i))$	exact at $\tau \to 0$; Part IV
D	adaptive p-grid (arclength equidistribution)	-25% max error, free
E	Hermite-cube derivative matching	no improvement

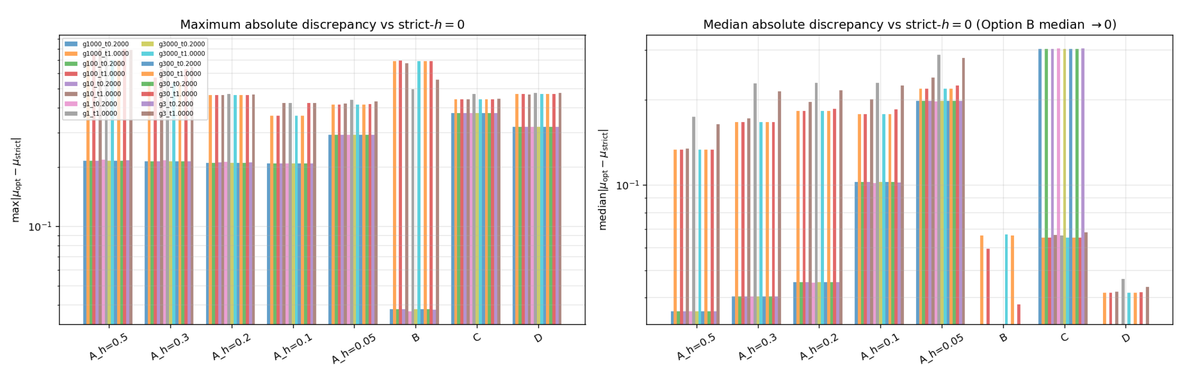

Figure 8.1: Max and median lookup error across builders and (γ, τ) cells: Option B+ improves the median by four decades over kernel-band R4.

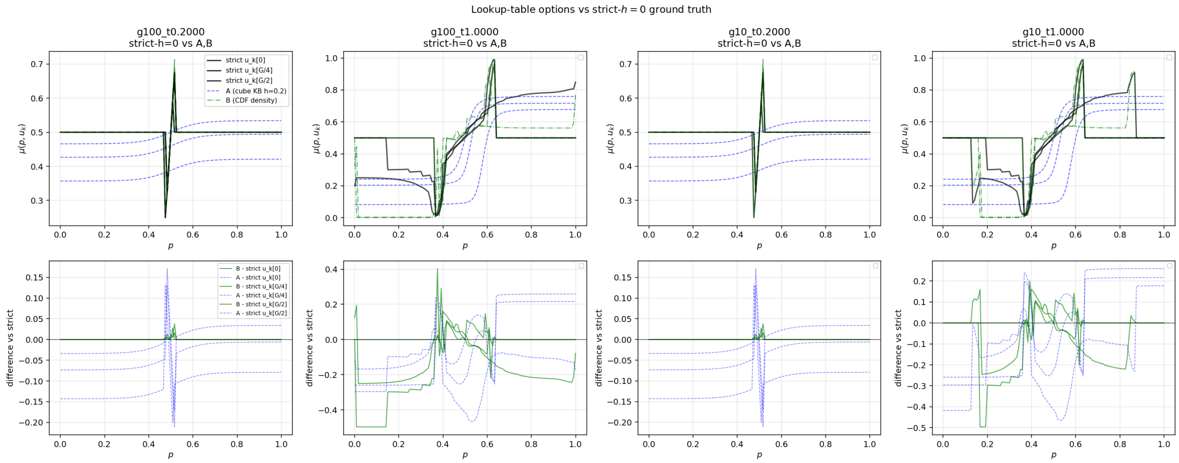

Figure 8.2: Visual comparison of the builders' $\mu(p, u_k)$ against strict-h=0: B+ tracks; A oscillates in the tails; C is exact at low τ but underfits at $\tau = 1$.

8.2 Option B+ anatomy

The construction is geometric, not extrapolative: accumulate Gaussian-weighted cube-cell mass into an empirical CDF in p per state, differentiate with shape-preserving PCHIP, take the density ratio. PCHIP's C^1-not-C^∞ regularity is exactly matched to the lookup's cusp structure — it represents kinks without ringing.

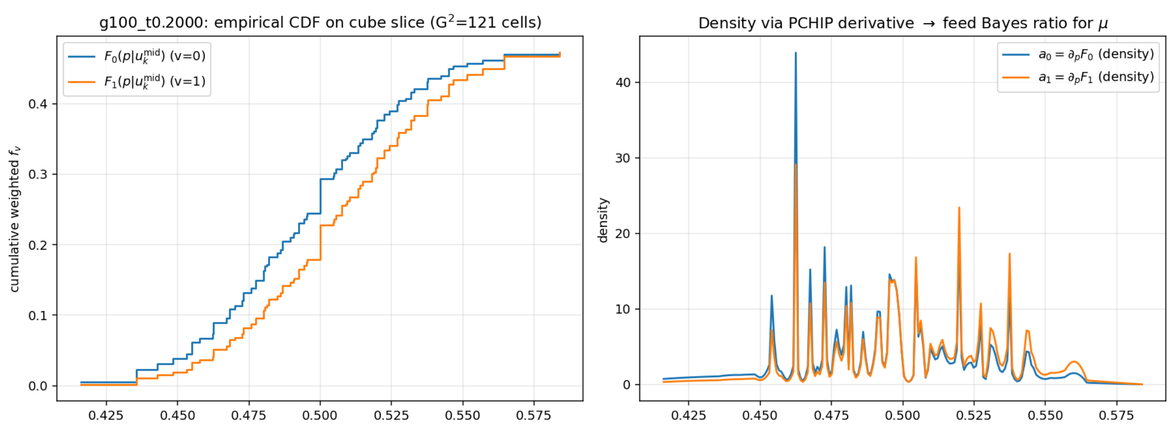

Figure 8.3: Option B+ construction: cube CDF, PCHIP density, resulting lookup vs strict-h=0.

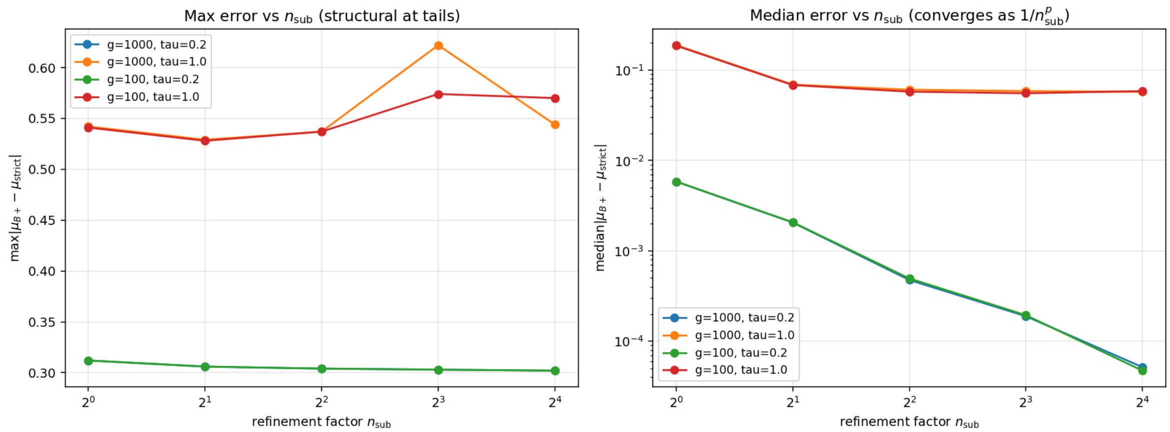

Figure 8.4: Sub-grid refinement: a 4× refinement inside each bin buys roughly a decade of median accuracy at trivial cost (numba).

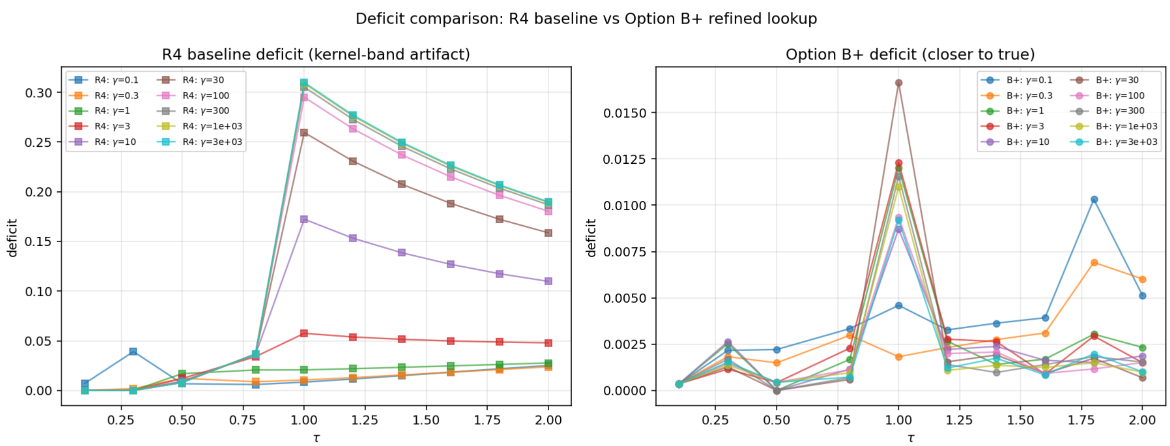

Figure 8.5: The economic consequence: deficit maps under kernel-band R4 vs Option B+ over the (γ, τ) region — the spurious ridge versus the true pattern.

B+ is an evaluator, not a solver. Used as the inner operator of a Picard loop it destabilizes the outer iteration — it lacks the contraction structure of the co-area operator. The division of labor that survives to the final recommendation: co-area (kernel or strict) *solves*; B+/per-segment-Chebyshev *evaluates*.

8.3 Option D: the adaptive price grid

Four monitor functions tested for p-grid equidistribution; arclength of $\mu(p)$ wins (most noise-robust), reduces max interpolation error by $\sim 25\%$ at $< 5\%$ cost, stacking multiplicatively with B+.

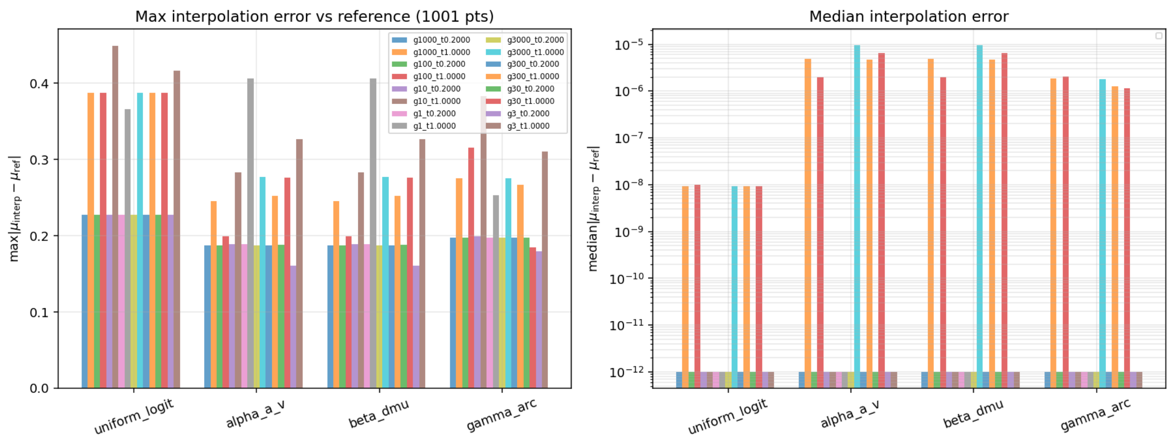

Figure 8.6: Monitor-function comparison for the adaptive p-grid.

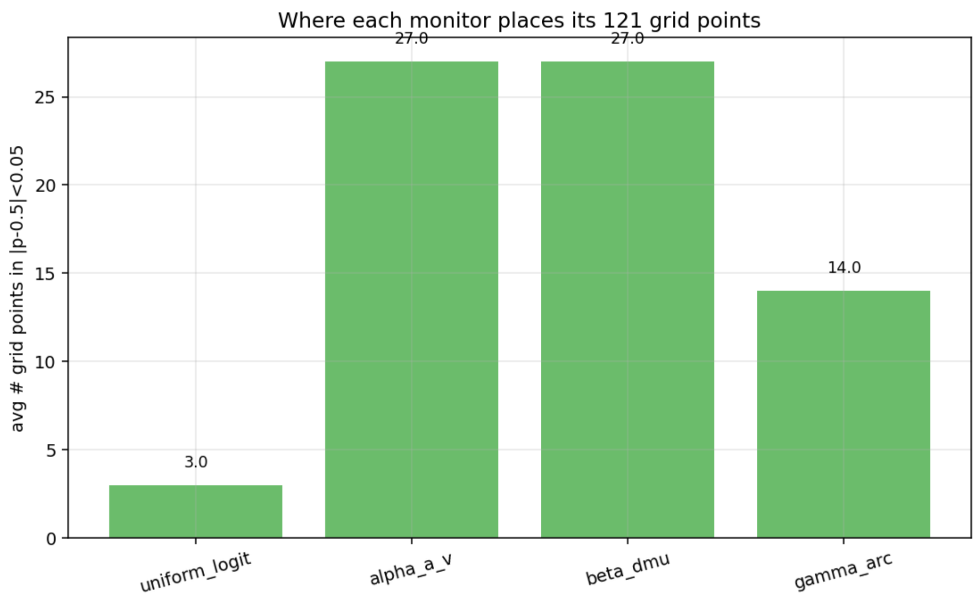

Figure 8.7: Resulting node-density profiles: the arclength monitor puts $\sim 23\times$ more nodes where the lookup actually bends.

Chapter 9

The spectral program and its limit

9.1 Chebyshev–Lobatto + analytic contours

The elegant idea: fit a global 3D Chebyshev polynomial to P, then the level set is an algebraic variety — companion-matrix root-finding gives exact contours and the lookup becomes clean 1D quadrature. Two results:

1. On CDF-uniform nodes the fit diverges (Runge; edge coefficients $> 10^6$ at degree 15). Resolving the FP on true Lobatto nodes fixes conditioning ($|F| \sim 10^{-14}$ at $G = 11, 15, 21$).
2. But convergence of the *lookup* is algebraic only: top-edge coefficient $\sim G^{-0.8}$, lookup median $\sim G^{-1.6}$. Machine epsilon requires effectively infinite G. The equilibrium is not analytic at high τ — consistent with the C^0-not-C^1 co-area theorem of Part II.

G	FP $\lvert F\rvert_\infty$	top edge coef	lookup max	lookup median
11	6.8e-14	0.0491	0.434	0.251
15	2.4e-14	0.0198	0.407	0.127
21	4.3e-14	0.0110	0.380	0.081

9.2 Ruling out the operator: the Chebyshev-native fixed point

To exclude the possibility that the algebraic rate came from mixing the spectral fit with the kernel-band operator, a Chebyshev-native FP solver was built (lookup evaluated by analytic contours at the exact cell price; no kernel, no Richardson, no bilinear interpolation). Same algebraic decay. **The slow convergence is a property of the equilibrium, not of any operator** — the single most load-bearing negative result of the spectral program, and the direct motivation for the *segmented* spectral representation that finally succeeds in Part V.

Part IV

Structural ansatz and coordinate transformations (Findings 5–7)

Chapter 10

Method C v2: the additive sigmoid ansatz

10.1 Motivation and form

In the $\tau \to 0$ linear-Gaussian limit, signals enter posterior log-odds additively and the (CARA/Hellwig) price is exactly $\sigma(\tau \sum_i u_i)$. The structural ansatz preserves this skeleton with a learned one-dimensional profile:

$$P(\mathbf{u}) = \sigma\big(h(u_1) + h(u_2) + h(u_3)\big), \qquad h(u) = \int_0^u s^2(t)\,dt, \qquad s(t) = \sum_{n=0}^{4} c_n\, T_{2n}(t/U_{\max}),$$

five free coefficients. By construction: permutation-symmetric, sign-flip symmetric, axis-monotone (zero violations across every run), and equipped with a *closed-form 1D lookup*: under the ansatz the level set in (u_a, u_b) at fixed u_k collapses, after $v = h(u)$, to the affine line $v_a + v_b = \operatorname{logit} p - h(u_k)$, making $\mu(p, u_k)$ a single Gauss–Legendre integral ($\sim 10\,\mu$s per call — four orders of magnitude faster than any cube-based builder).

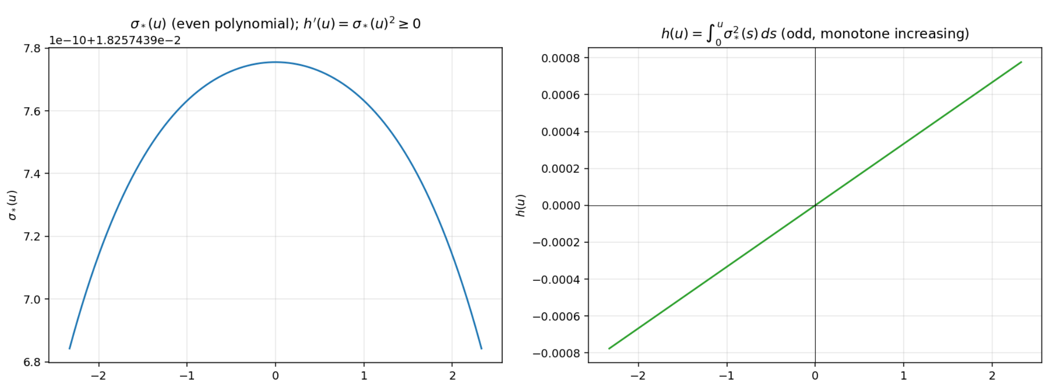

Figure 10.1: The fitted profile at $\tau = 0.001$: $s^2(\cdot)$ and the resulting odd-monotone $h(u)$, matching the analytic linear-Gaussian limit $h(u) = \tau u + O(\tau^3 u^3)$ to plotting accuracy.

10.2 The grids and ladders

Trust-region least squares against strict-h=0 truth solves a cell in ~ 1 second. Two production sweeps:

- **100-cell** (γ, τ) **grid** (2 minutes): the residual is essentially a function of τ alone; at $\tau = 0.001$ the fit is exact to 10^{-10}, at $\tau = 1$ it floors at $\|F\|_\infty \approx 0.08$–$0.10$.
- **1000-cell** τ**-ladder** (16.5 minutes, chained warm-start, zero monotonicity violations): the residual scales as $\sim \tau^4$ at small τ (consistent with the $O(\tau^3 u^3)$ truncation of h feeding an $O(\tau^4)$ fixed-point error) and saturates at the non-additive plateau above $\tau \approx 0.3$–0.4.

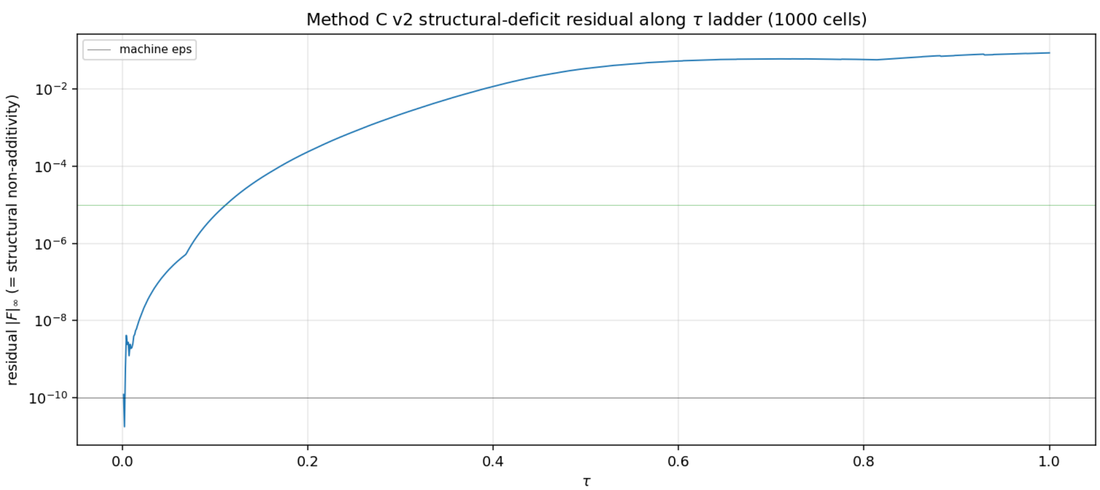

Figure 10.2: Method C v2 1000-cell ladder: residual vs τ. The sharp transition near $\tau \approx 0.3$ separates "additive ansatz exact" from "non-additive correction required".

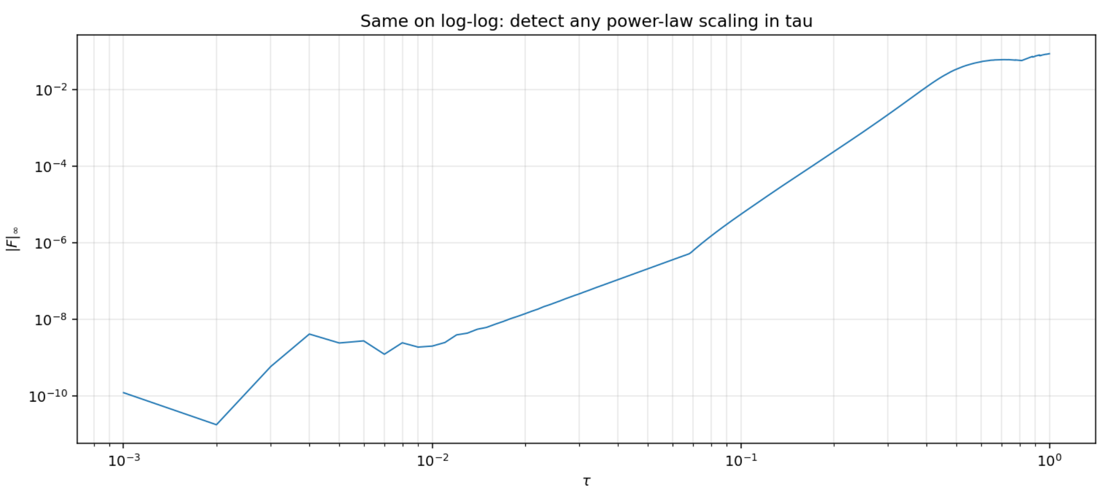

Figure 10.3: Same data, log-log: the $\sim \tau^4$ scaling at small τ and the saturation plateau.

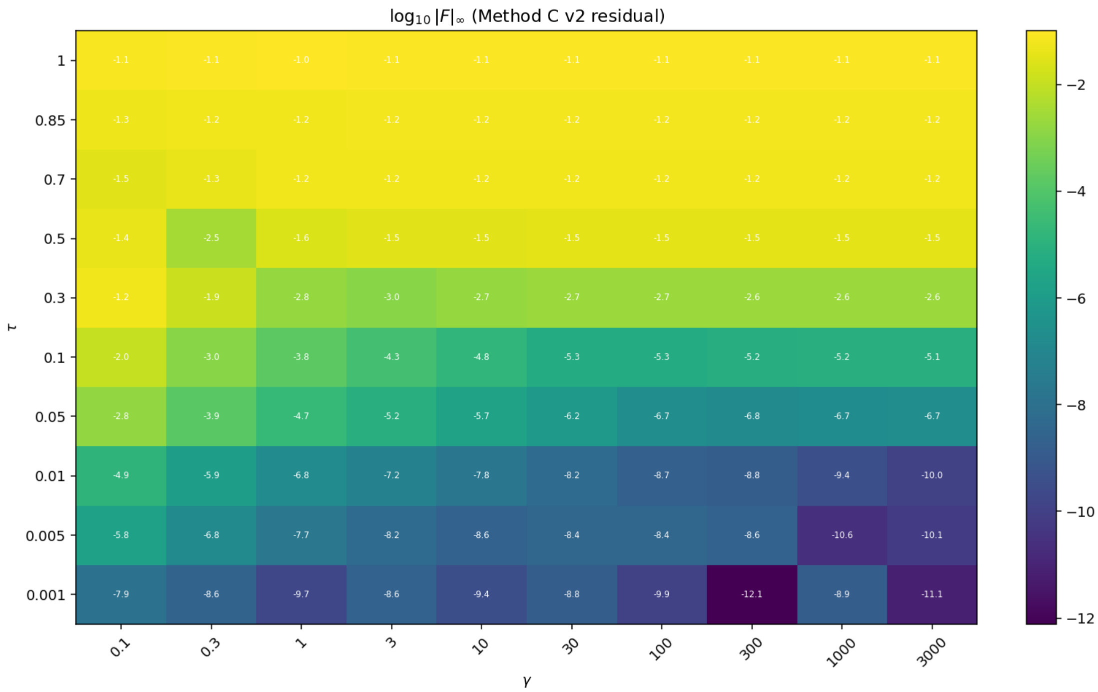

Figure 10.4: The 100-cell (γ, τ) grid heat map: residual depends almost entirely on τ, not γ — the additive structure is τ-driven.

10.3 Two control experiments

- **Logit-space loss**: refitting v2 with an L-space loss leaves the converged residual unchanged (median 1.7×10^{-3} in both) — the bottleneck is ansatz expressiveness, not loss geometry.
- **Strict-operator inner loop**: v2 fit with strict-h=0 as the inner operator at $\tau = 0.001$ converges to $P \equiv \frac{1}{2}$ exactly — the first clean sighting of the trivial branch that Part VI later confirms is an exact fixed point at all (γ, τ).

Chapter 11

Finding 5: the $(L_{\text{priv}}, L_{\text{pub}})$ coordinate test

11.1 Hypothesis

Bayes suggests $\operatorname{logit}\mu(p, u_k) = L_{\text{priv}}(u_k) + L_{\text{pub}}(p) + \delta(p, u_k)$ with $L_{\text{priv}} = \tau u_k$ (closed form), $L_{\text{pub}} = \log[f_1(p)/f_0(p)]$, and δ a small smooth K=3 conditional-overlap correction. If δ were small *and* smooth, Chebyshev in the transformed coordinates would converge spectrally with a handful of coefficients.

11.2 Result: bulk absorbed, cusps preserved

The optimal additive (ANOVA) decomposition removes $\sim 50\%$ of the RMS magnitude — but the Chebyshev tail-decay of δ is *indistinguishable* from that of the raw lookup, and the SVD of δ is as full-rank as the original. The cusps live on the algebraic variety $\{|\nabla P| = 0\}$, which no global change of coordinates can remove.

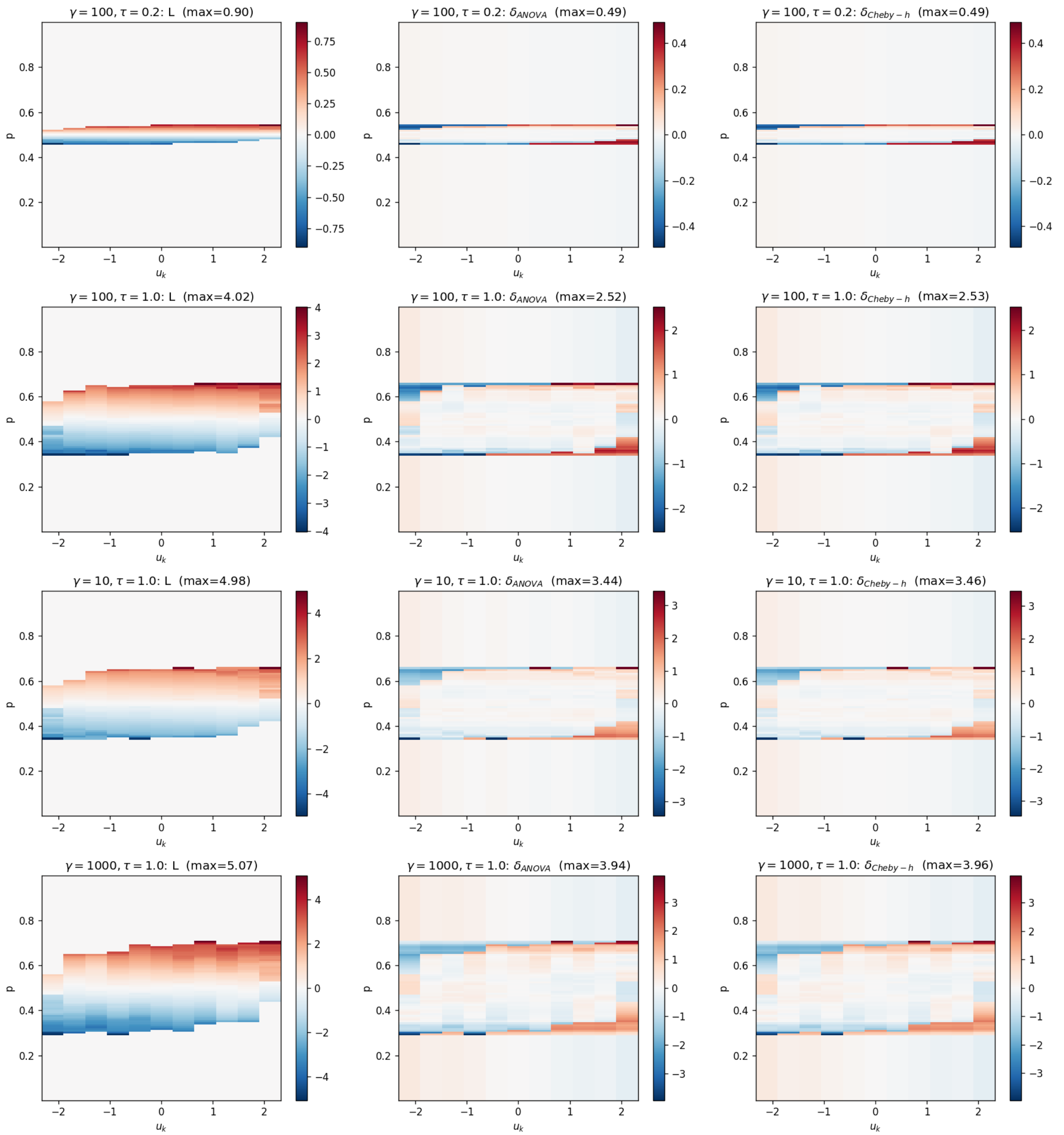

Figure 11.1: L, δ_{ANOVA}, $\delta_{\text{Cheby-h}}$ heat maps across four cells: the smooth vertical gradient is removed; the sharp horizontal cusp bands survive untouched.

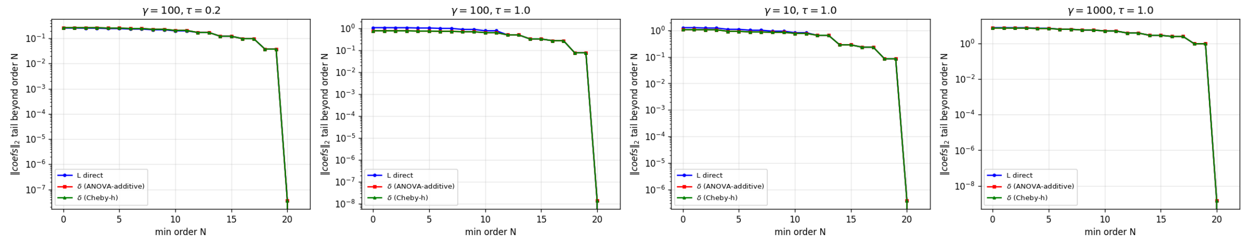

Figure 11.2: Chebyshev tail-norm decay: the three curves (raw L, ANOVA residual, Cheby-h residual) coincide. The coordinate transformation does not improve the convergence rate.

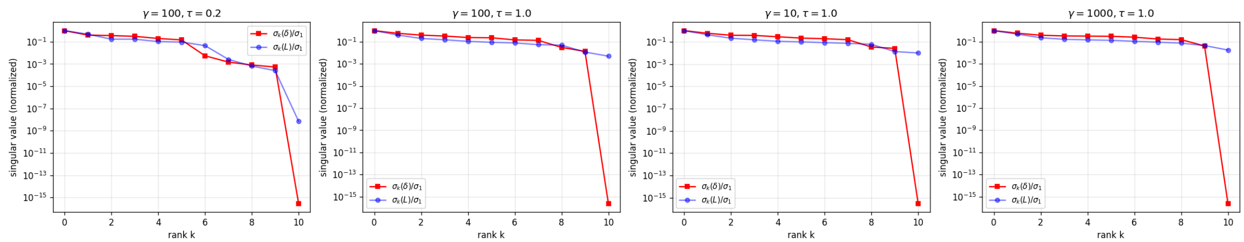

Figure 11.3: SVD spectra: δ is essentially full rank; rank-1 captures only 30–45% of its Frobenius norm. No low-rank multiplicative correction exists either.

11.3 The economic by-product

The fitted u_k-effect $b(u_k)$ is roughly $10\times$ smaller than the textbook τu_k: at $(\gamma, \tau) = (100, 1)$ the closed form predicts ± 2.33 where the data give ± 0.22. Conditional on the price, the agent's own signal adds little — the K=3 price already does most of the aggregation. This is the lookup-level reflection of the Jensen-gap economics of Part I: the price is a powerful but *biased* (compressed) aggregator.

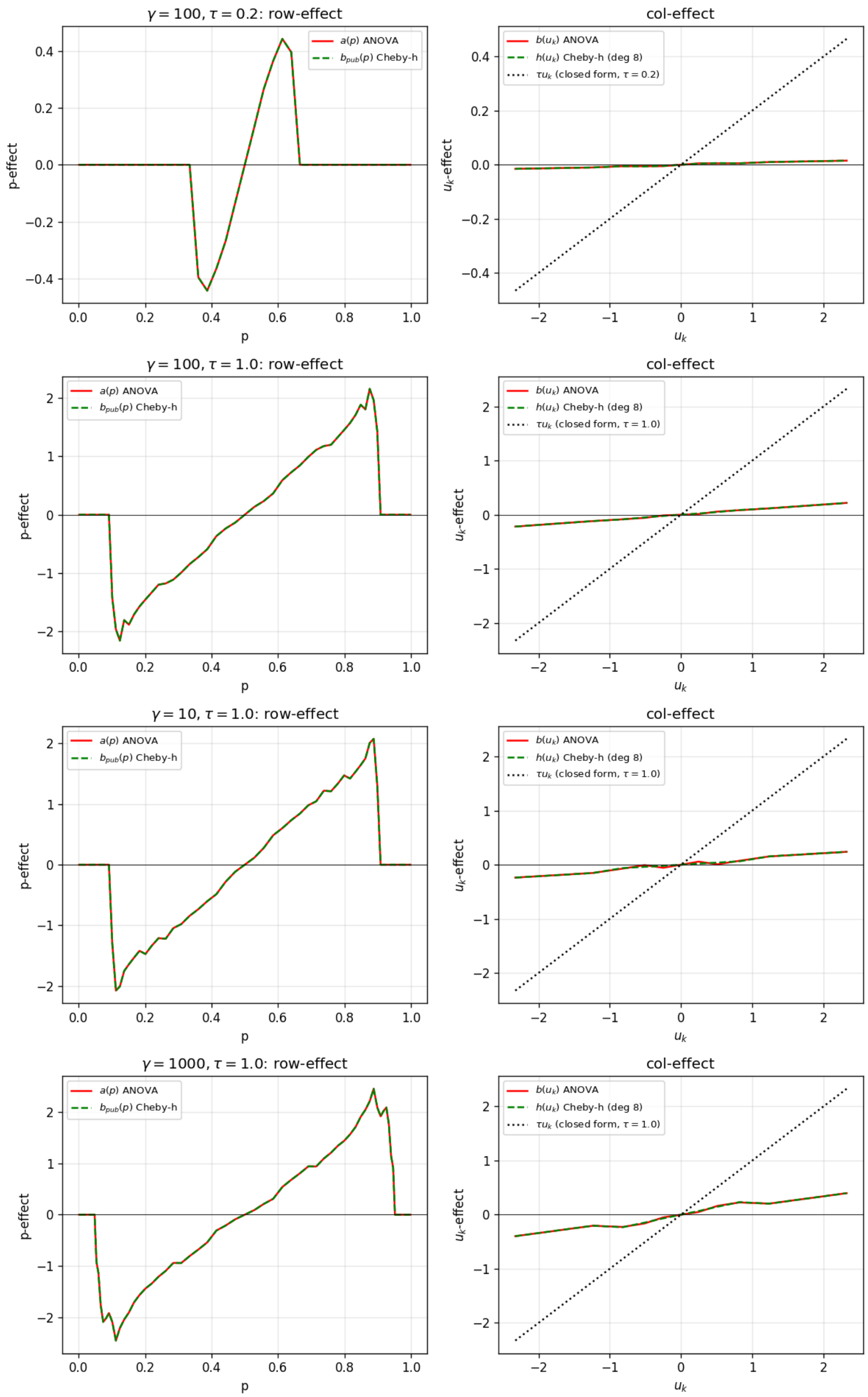

Figure 11.4: Row and column effects of the additive decomposition; dotted line: the naive τu_k prediction, an order of magnitude too steep.

Chapter 12

Findings 6–7: the limits of finite-parameter structure

12.1 Finding 7: no low-degree polynomial closes the gap

At $(\gamma, \tau) = (100, 1)$, the *best achievable* additive fit to the strict-h=0 log-odds surface leaves $|R|_\infty = 3.5$ (larger than $|L|_\infty = 2.56$ itself), flat across 3–11 profile parameters; and no symmetric polynomial correction captures the residual:

candidate g	best κ	share of $\|R\|^2$ captured
$u_1 u_2 u_3$	-0.103	7.8%
$\sum_{i<j}(u_i - u_j)^2$	≈ 0	0.0%
$(u_1+u_2+u_3)^3$	-0.009	22.8%
$\sum u_i^3$, $\sum u_i^5$	≈ 0	0.0%
six-term LSQ combination	—	63.3%

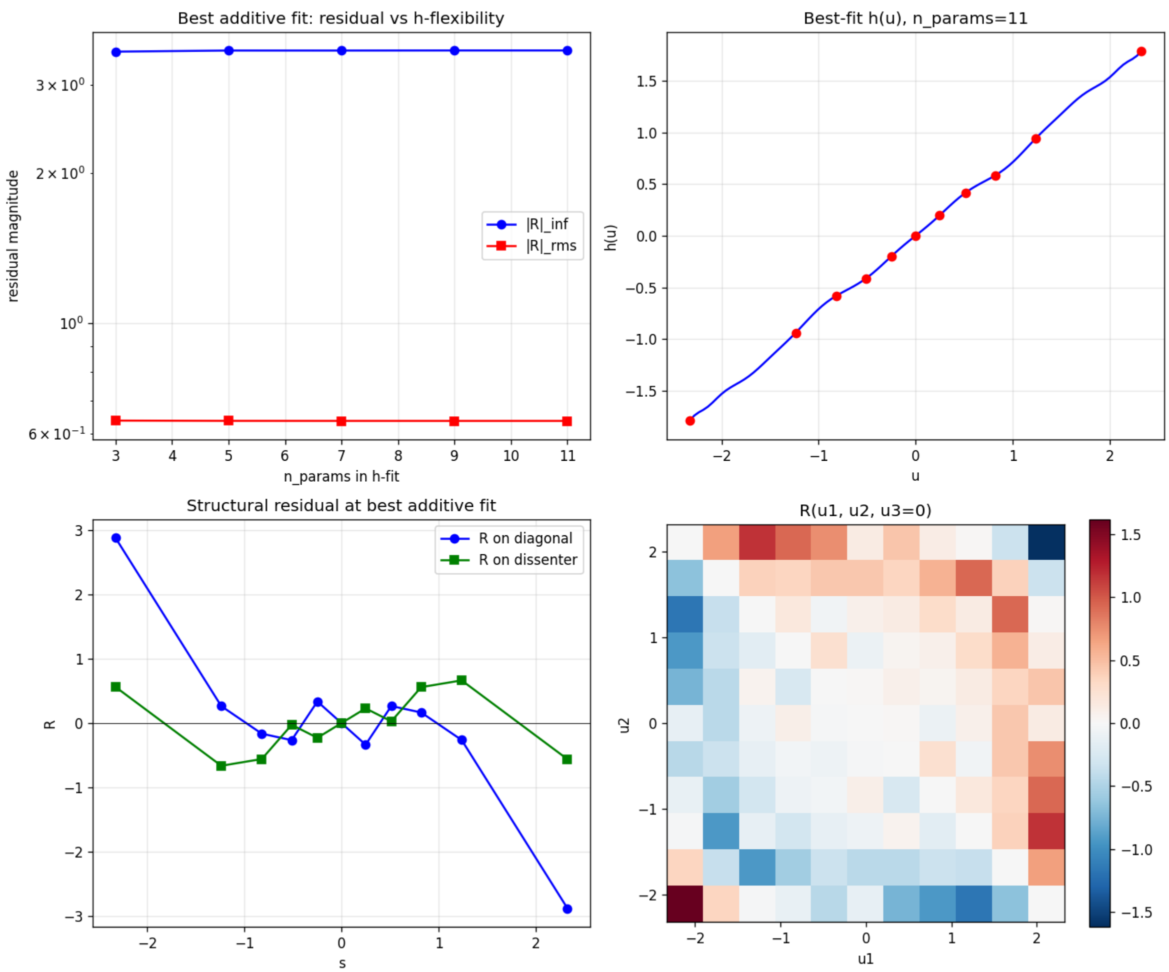

Figure 12.1: The fingerprint test: residual flat in profile flexibility (top left); the residual heat map (bottom right) is a high-rank checkerboard concentrated at corners — not any smooth low-degree polynomial.

12.2 Interpretation

Findings 5 and 7 are the same fact seen from two angles: **the non-additive structure of the K=3 equilibrium at high τ is high-rank and cusp-bearing**. Neither clever coordinates nor finite-parameter corrections reach 10^{-4}; the viable routes are (i) segmented representations that put the cusps on segment boundaries (Part V — this works), and (ii) brute resolution with a consistent smooth operator (Part VII recommendation for the surface itself).

Part V

The machine-epsilon lookup (Findings 8–16)

Chapter 13

Cusp detection (Findings 8 and 13)

13.1 The 3D detector — and why it is not enough

A per-cube-cell critical-point finder (trilinear roots of $\nabla\hat{P} = 0$, quadratic-Hermite fallback) enumerates the interior critical points of the discretized price cube. At the four reference strict-h=0 fixed points it finds 40–140 points (16–42 unique critical values p_c), all with $|\nabla\hat{P}|$ at machine epsilon, symmetric about $p = \frac{1}{2}$, broadening with τ.

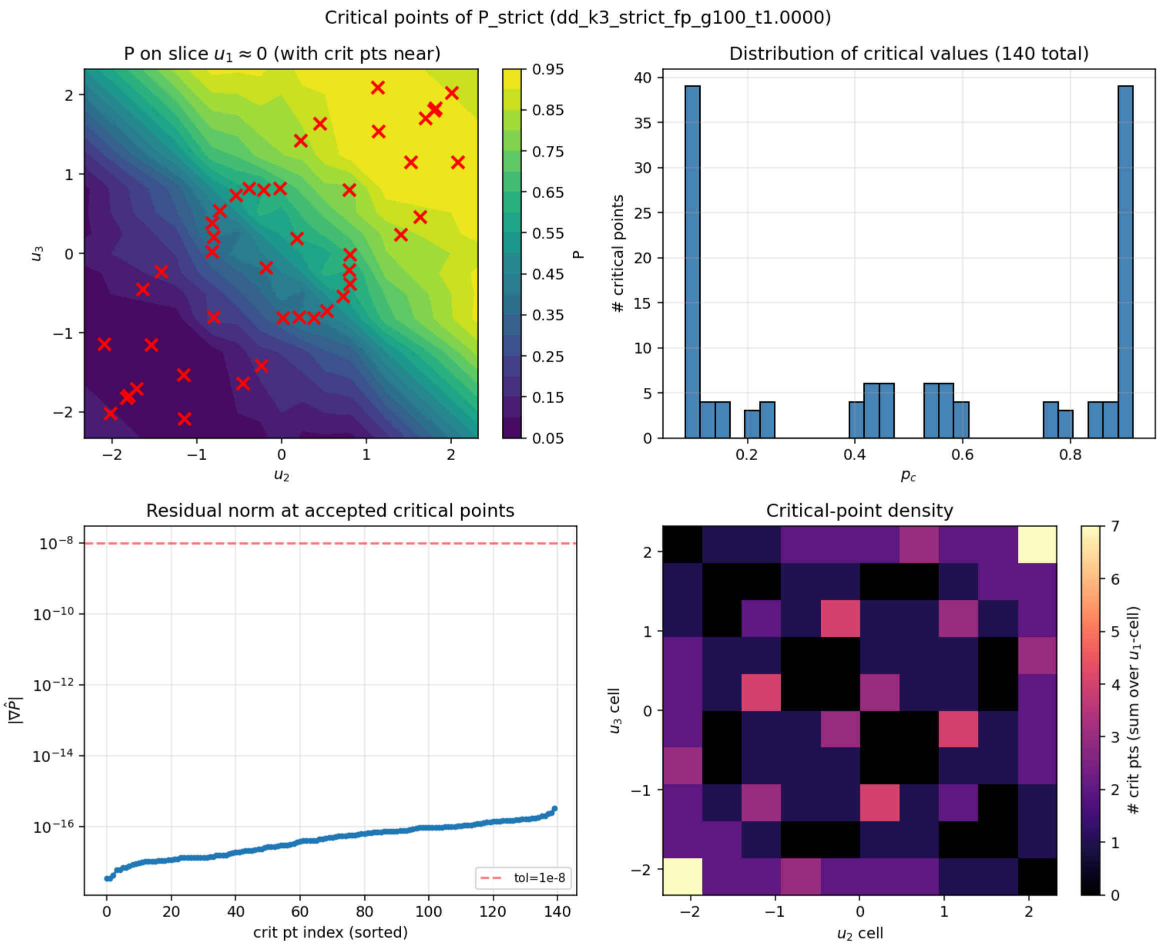

Figure 13.1: The 3D critical-point detector at $(\gamma, \tau) = (100, 1)$: 140 interior critical points; bimodal distribution of critical values; residuals at machine epsilon.

Finding 13 (the correction that unlocked everything): the cusps of the *one-dimensional* lookup $\mu(p, u_k)$ at fixed u_k are critical values of the *slice* map $P(\cdot, \cdot, u_k)$ — not of the full 3D map. Knotting segments at the 3D critical values leaves a 3–9% error floor at $\tau = 1$; detecting the cusps directly from μ itself (second-difference spikes plus first-difference jumps, pooled across u_k, deduplicated in logit distance) finds the correct knots and collapses the floor to machine epsilon. The lesson generalizes: *locate singularities in the object you are approximating, not in its generator.*

13.2 Multi-element OptB+ (Finding 8): a half-step

Inserting the (3D) critical values as forced PCHIP knots drives the *median* error to zero at the knots by construction but leaves the max error unchanged — because the max error was never at the bulk cusps; it was in the tails ($p \to 0, 1$), where the empirical CDF runs out of support. This diagnosis redirected the program to the tail treatment of Findings 11–14.

Chapter 14

The segmented spectral representation (Finding 10)

14.1 Per-segment Chebyshev between cusp knots

Between adjacent cusps the lookup is C^∞; a degree-d Chebyshev fit per segment therefore converges exponentially *per segment*. With the correct (internal, μ-based) knots:

cell	degree	in-segment max	in-segment median
$(100, 0.2)$	4	6.7e-16	—
$(100, 1.0)$	3	3.4e-15	3.3e-16
$(1000, 0.2)$	4	6.7e-16	2.8e-16
$(1000, 1.0)$	6	1.6e-15	2.2e-16

Machine epsilon at degree 3–6, all four reference cells — the cusps are honest piecewise-C^k discontinuities at discrete p_c, not smoothable curvature bumps. Notably, 67–92% of segments contain ≤ 2 samples (cusps cluster), forcing degree 0–1 on them; they reach machine epsilon anyway because they are tiny in p-width.

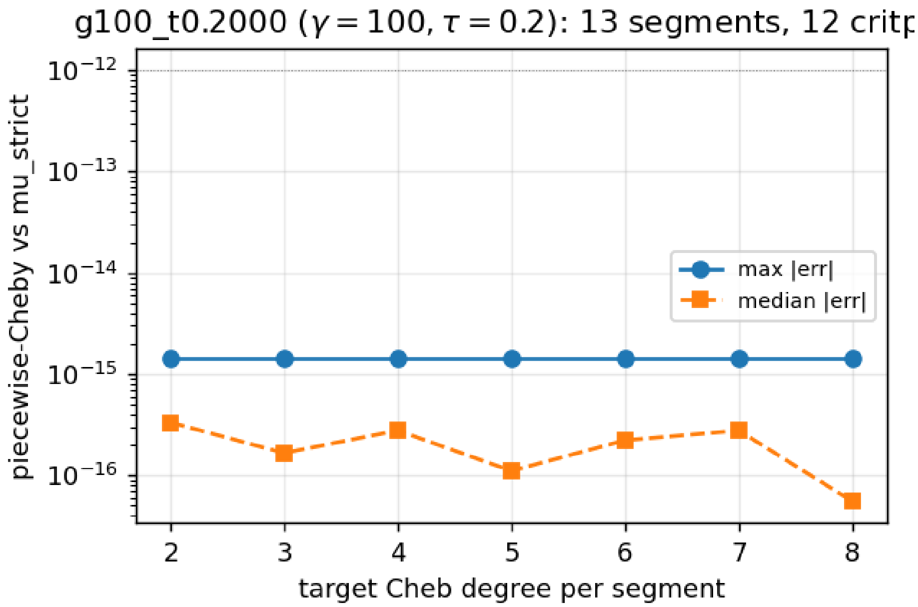

Figure 14.1: Per-segment convergence at $(\gamma, \tau) = (100, 0.2)$: machine epsilon at any degree once knots are correct.

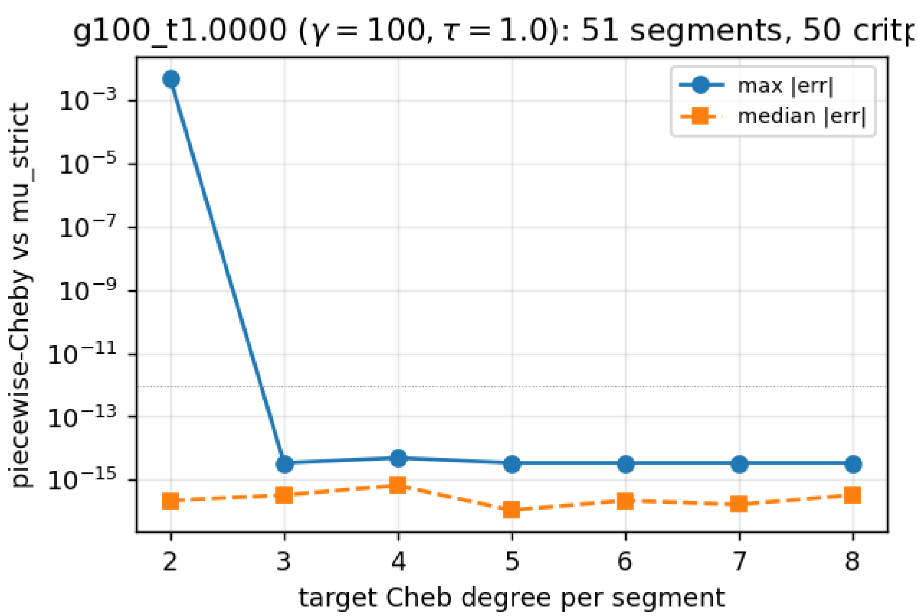

Figure 14.2: $(\gamma, \tau) = (100, 1.0)$: the in-segment median drops to 10^{-16} from degree 3; the blue max curve is dominated by out-of-support points (the fallback artifact addressed in Finding 14).

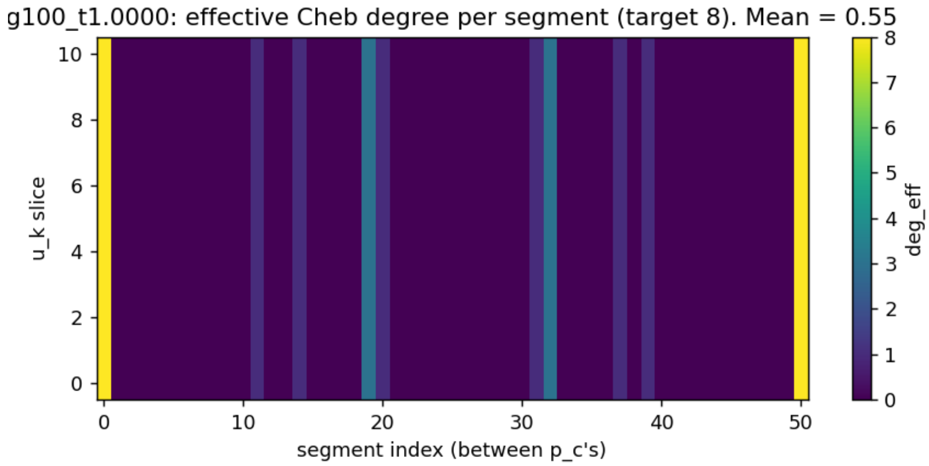

Figure 14.3: Effective degree per $(u_k, \text{segment})$: large central segments support high degree; clustered cusp segments are forced low — harmlessly.

Chapter 15

Negative control experiments (Findings 9, 11, 12, 17)

15.1 Finding 9: smoothing the strict operator has no sweet spot

Replacing the discrete crossing test $ab < 0$ with the smooth weight $\sigma(-ab/\varepsilon^2)$ makes the operator Lipschitz and Anderson converges to machine epsilon for $\varepsilon \geq 0.1$ — but the resulting fixed point is biased 0.13–0.49 against the original operator; and as $\varepsilon \to 0$ the convergence failure returns before the bias disappears. There is no ε at which both residuals are small. The diagnostic that motivated this: *starting Picard at the saved "strict fixed point" diverges* (from $|F| = 0.06$ to 0.5) — the saved object is a best iterate, not a fixed point.

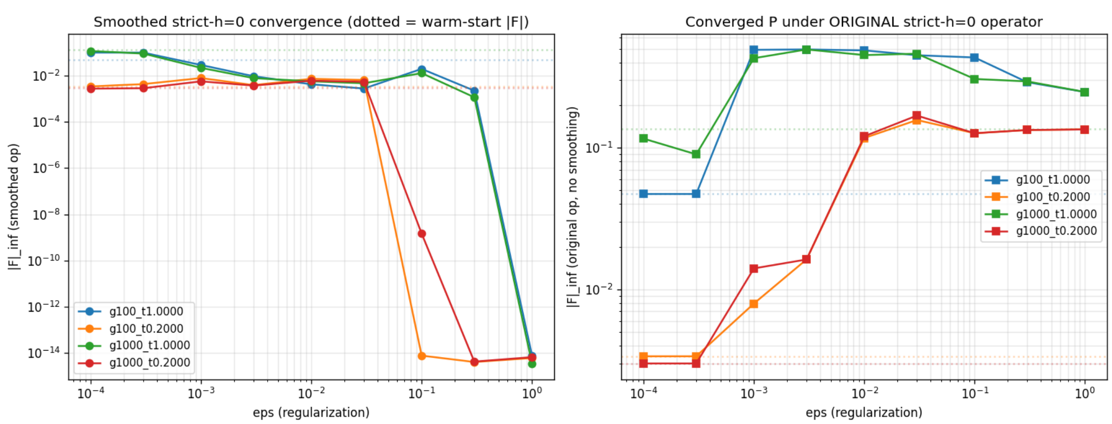

Figure 15.1: The ε-sweep: clean convergence under the smoothed operator (left) coexists with large bias under the original operator (right) at every ε.

15.2 Findings 11–12: tail BCs and exact cell mass are not the levers

Boundary-augmented PCHIP, logit-space CDF fitting, and Beta-CDF parametric tails all leave the max error *identical* to baseline (the boundary error is interpolation failure, not missing BCs); exact per-cell Gauss–Legendre mass integration halves the in-support median but worsens the max (a smoother CDF extrapolates worse outside support). Useful contributors, not regime changers.

15.3 Finding 17: a smooth lookup does not fix the FP iteration

The decisive control: using the machine-epsilon segmented lookup as the *inner operator* of the FP iteration, with cusp knots *frozen* (eliminating cusp drift, the conjectured source of non-smoothness) still plateaus at $|F| \approx 1.1 \times 10^{-2}$ — essentially the same wall as the recompute mode (1.5e-2) and only 5× better than the kernel-band baseline (5.7e-2). The wall is the $G = 11$ truncation error of the co-area evidence, not operator roughness. PCHIP segments in the inner loop diverge outright.

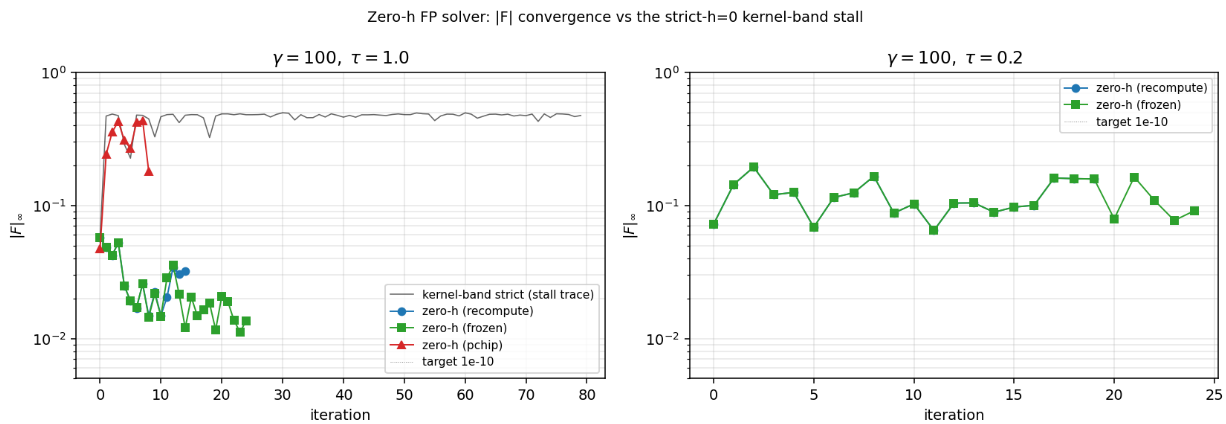

Figure 15.2: The zero-h FP solver: recompute and frozen-knot modes plateau together at $\sim 10^{-2}$ while the kernel-band trace (grey) diverges. Smoothness of the lookup was not the binding constraint.

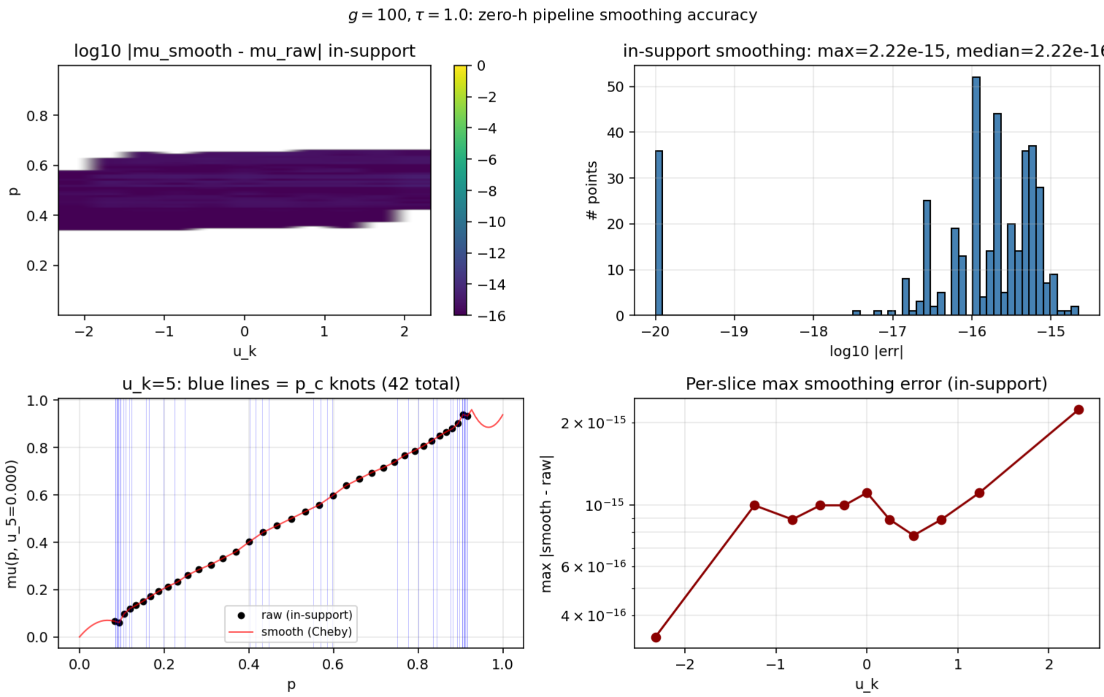

Figure 15.3: Verification that the inner lookup is exact: the segmented pipeline reproduces all 369 in-support μ values at the warm-start to 2.2×10^{-15}.

Chapter 16

The production pipeline (Findings 14–16)

16.1 The fallback fix

The raw strict-h=0 lookup returns $\mu = 0.5$ wherever the co-area integral finds no roots (p outside the cube's empirical price range), producing -0.5 jumps and 0/11 monotone slices at high τ. The fix (`fix_fallback`) replaces the fallback region by PCHIP extrapolation through the in-support values pinned to the Bayesian boundary conditions $\mu(\epsilon) = 0$, $\mu(1-\epsilon) = 1$: afterwards, 11/11 slices monotone with correct tails at every cell tested, and the spurious $|\mu''|$ spikes drop by half.

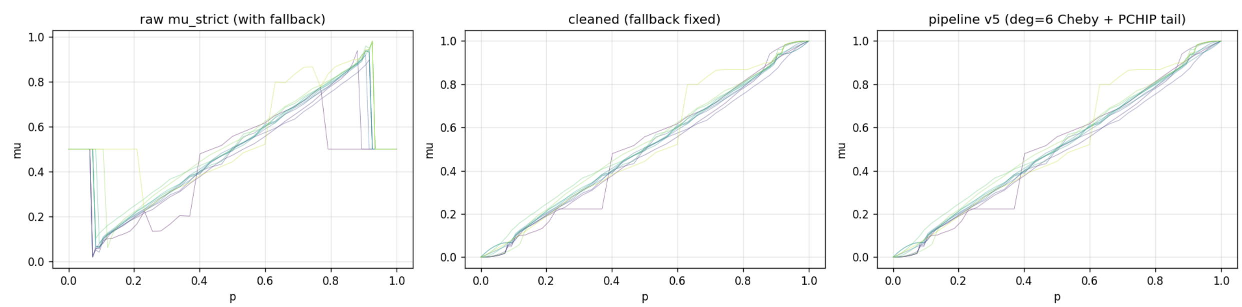

Figure 16.1: The before/after of the production pipeline at $(\gamma, \tau) = (100, 1)$: raw μ_{strict} with fallback discontinuities (left); cleaned (middle); pipeline v5 output — monotone, smooth, boundary-correct (right).

16.2 Pipeline v5 and the 36-cell certification

v5 = `fix_fallback` (clean) $\to$ internal cusp detector (knots) $\to$ degree-6 Chebyshev per segment (in-support) $\to$ PCHIP shape-preserving tails (out-of-support). Applied to the full 6×6 (γ, τ) ridge grid:

metric	result
cells processed	36/36, 7 seconds total
median error vs cleaned truth	1.11×10^{-16} at every cell
max error at machine eps	31/36 cells (10^{-15}–10^{-14})
outlier cells	5, max 10^{-2}–9×10^{-2} (corner cells, fallback-smoothing artifacts)
monotone slices (dense 501-pt check)	11/11 at every cell
boundary conditions	satisfied at every cell

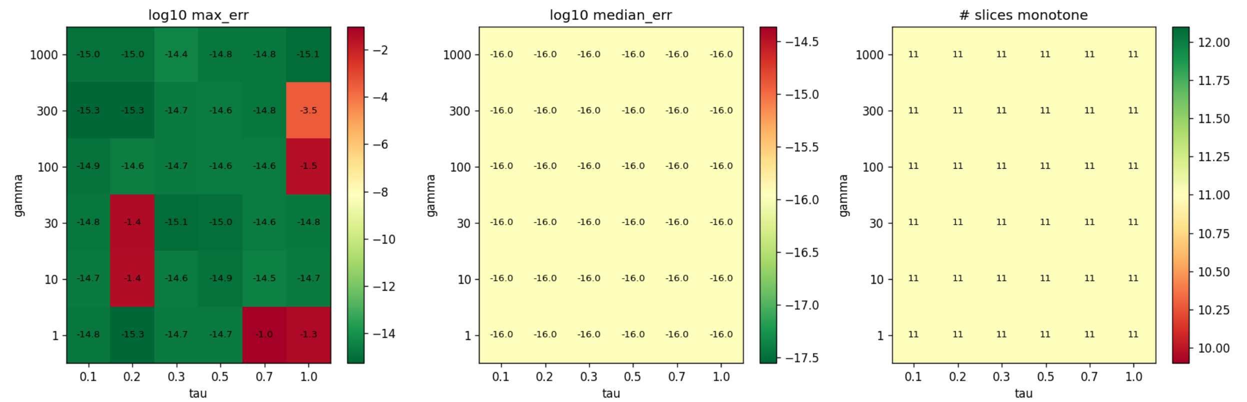

Figure 16.2: The certification heat maps: $\log_{10}$ max error (left), $\log_{10}$ median error — uniformly -16 (middle), and monotone slice count — uniformly 11/11 (right), across the (γ, τ) grid.

16.3 What is and is not claimed

The lookup problem — evaluating the Bayes-correct conditional posterior given a price surface — is solved to machine epsilon in-support, with controlled monotone Bayesian extrapolation out-of-support, at one second per cell. The claim is conditional on the input surface: it inherits whatever distance the underlying (best-iterate) strict-h=0 surface has from the true equilibrium. The surface question is Part VI's subject and drives the final recommendation.

Part VI

The equilibrium surface: log-odds coordinates, branches, and solvers

Chapter 17

The log-odds configuration

17.1 Specification

The final coordinate system, run at the user's direction:

- u-**grid**: uniform in the no-learning private posterior log-odds transform $\xi = \tanh(\tau u/2)$, $\xi \in [-0.85, 0.85]$, mapped back by $u = (2/\tau)\,\mathrm{atanh}(\xi)$. No bounded cube: the grid self-scales to $\approx \pm 5\sigma_{\text{signal}}$ at every τ (at $\tau = 0.005$ the cube spans ± 70 in u; at $\tau = 1$, ± 2.4).
- p-**grid**: uniform in public-only posterior log-odds $L_{\text{pub}}(p)$, bootstrap-computed from the cube empirical CDF and *antisymmetrized by construction* ($L_{\text{pub}} \mapsto \frac{1}{2}(L_{\text{pub}} - L_{\text{pub}}^{\text{rev}})$) after the unsymmetrized bootstrap was caught breaking the equilibrium's sign-flip symmetry at $O(0.4)$.
- **Operator**: strict-h=0 co-area + POU (zero kernel).
- **Clearing**: exact CRRA bisection (the linearized clearing bug and its repair are recorded in Part I).
- **Residual**: both raw $|F|_\infty$ and the density-weighted $F_w = \max|\Phi(P) - P| \cdot W_3$, $W_3 \propto \prod_i \bar{f}(u_i)$ — the raw max over an unbounded cube is dominated by corner cells of vanishing probability and is not the economically meaningful residual.
- **Symmetry projection** (permutation + sign-flip) every iteration; live per-iteration reporting.

17.2 The G-ladder and γ-sweep

At $(\gamma, \tau) = (100, 0.1)$: F_w best of 1.3×10^{-2} (G=7) down to 1.6×10^{-3} (G=9), non-monotone across $G \in \{7..15\}$ — the strict-h=0 stall manifests differently per grid. The γ-sweep at $G = 11$ spans $F_w = 1.8 \times 10^{-2}$–2.6×10^{-2} across $\gamma \in [1, 1000]$: genuinely γ-dependent (after the clearing fix), all cells with sign-flip symmetry at 1.1×10^{-16} and $P(\mathbf{0}) = \frac{1}{2}$ exactly.

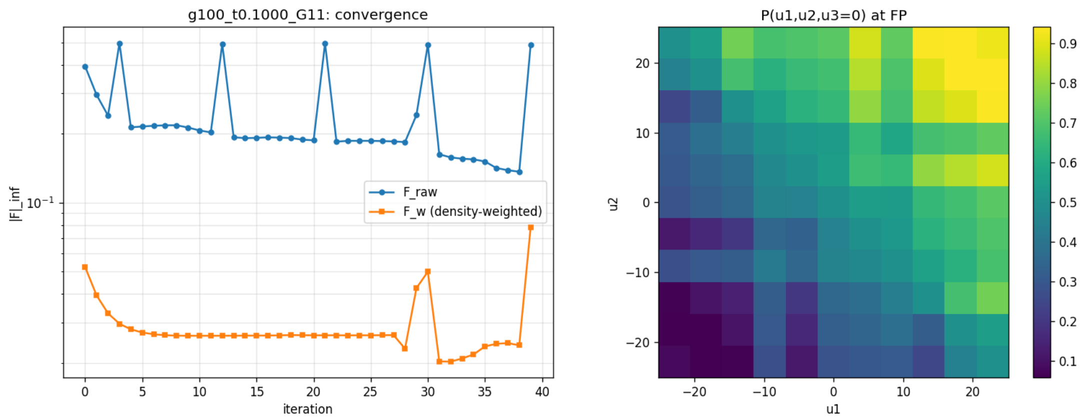

Figure 17.1: The log-odds configuration at $(\gamma, \tau, G) = (100, 0.1, 11)$: raw vs density-weighted residual traces (left) and the equilibrium slice (right). The raw residual is dominated by zero-density corners; the weighted residual is the meaningful one.

17.3 The 50-cell τ-ladder

Fifty log-spaced $\tau \in [0.001, 1.2]$, chained warm-start, 16 seconds total. Three regimes:

$$\underbrace{\tau \leq 0.005}_{F_w \approx 2.5\times 10^{-15}} \quad \longrightarrow \quad \underbrace{0.005 \lesssim \tau \lesssim 0.02}_{F_w : 10^{-7} \to 10^{-3}} \quad \longrightarrow \quad \underbrace{\tau \gtrsim 0.05}_{F_w \approx 10^{-2}\ \text{(stall)}}$$

with perfect sign-flip symmetry everywhere and the price range opening monotonically with τ.

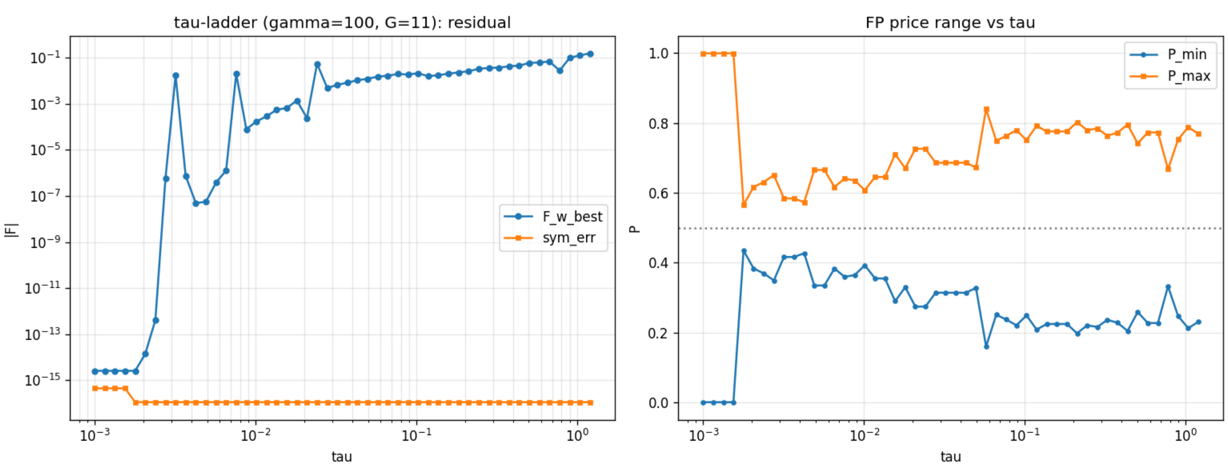

Figure 17.2: The 50-cell Picard τ-ladder: residual regimes (left) and equilibrium price range vs τ (right).

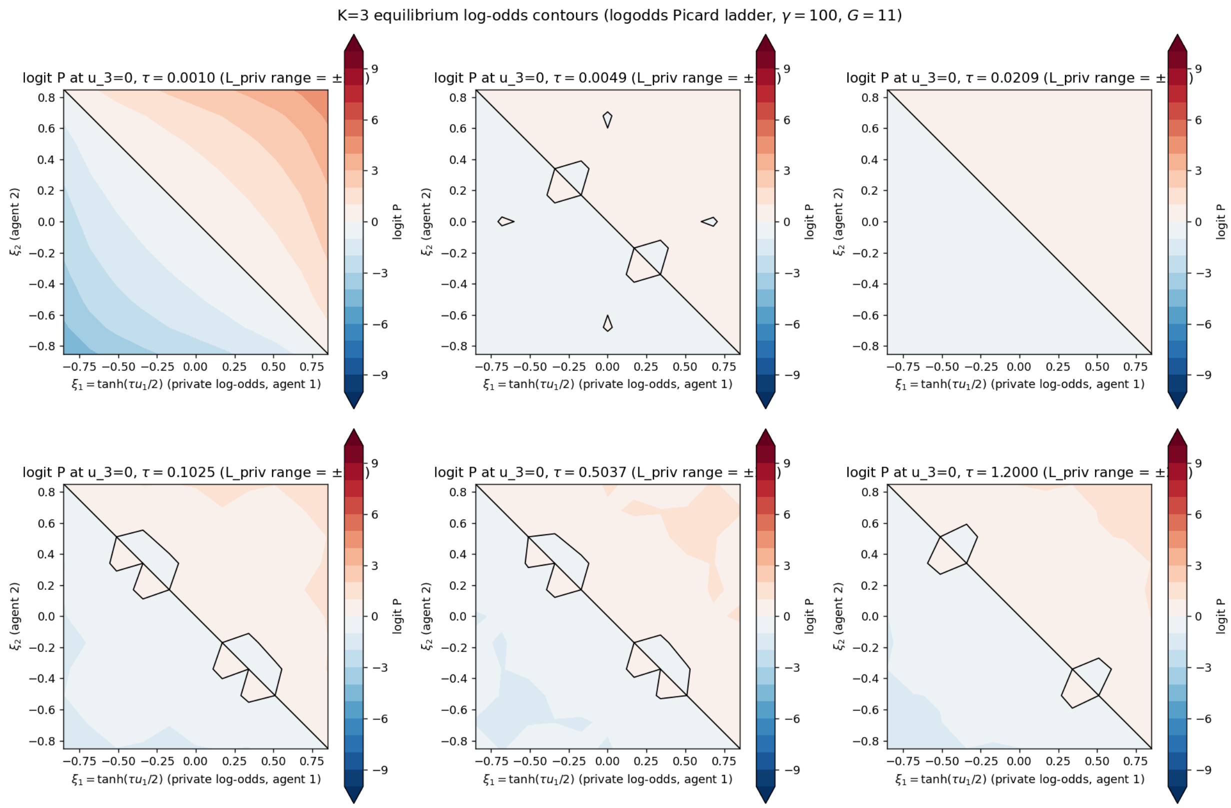

Figure 17.3: Equilibrium log-odds contours logit $P(u_1, u_2 \mid u_3{=}0)$ on private-log-odds axes across six τ values: a clean diagonal zero contour at low τ deforming into the kinked high-τ structure whose critical values generate the lookup cusps.

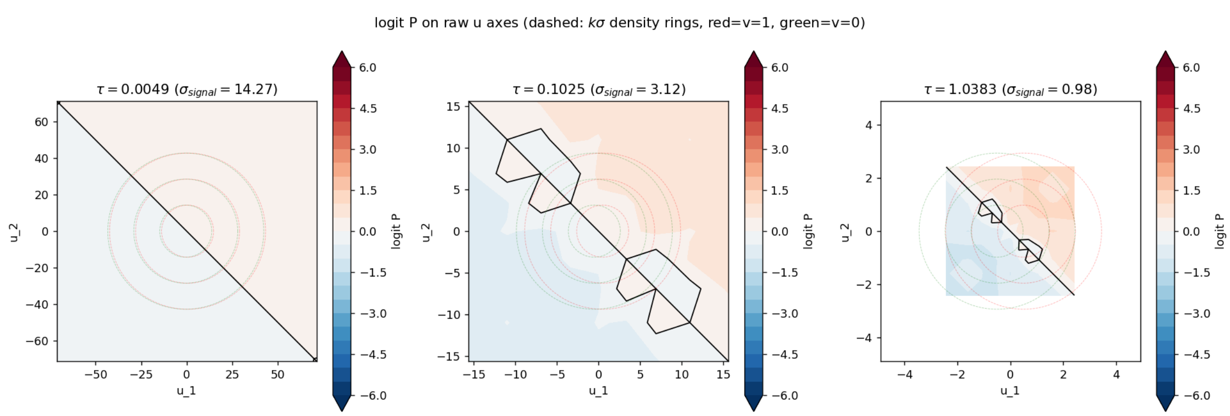

Figure 17.4: The same surfaces on raw u axes with $k\sigma$ density rings: the unbounded-cube grid keeps the Gaussian mass interior at every τ.

17.4 Anatomy of one lookup evaluation

For each (p, u_k) query the strict-h=0 builder consumes: all G^2 slice vertex values (linear interpolation backbone), 2×16 Gauss–Legendre sweep nodes, and the level-set roots found per sweep line — typically crossing 25–56 of the $(G-1)^2$ cells. One full table is $G_p \times G = 891$ such queries.

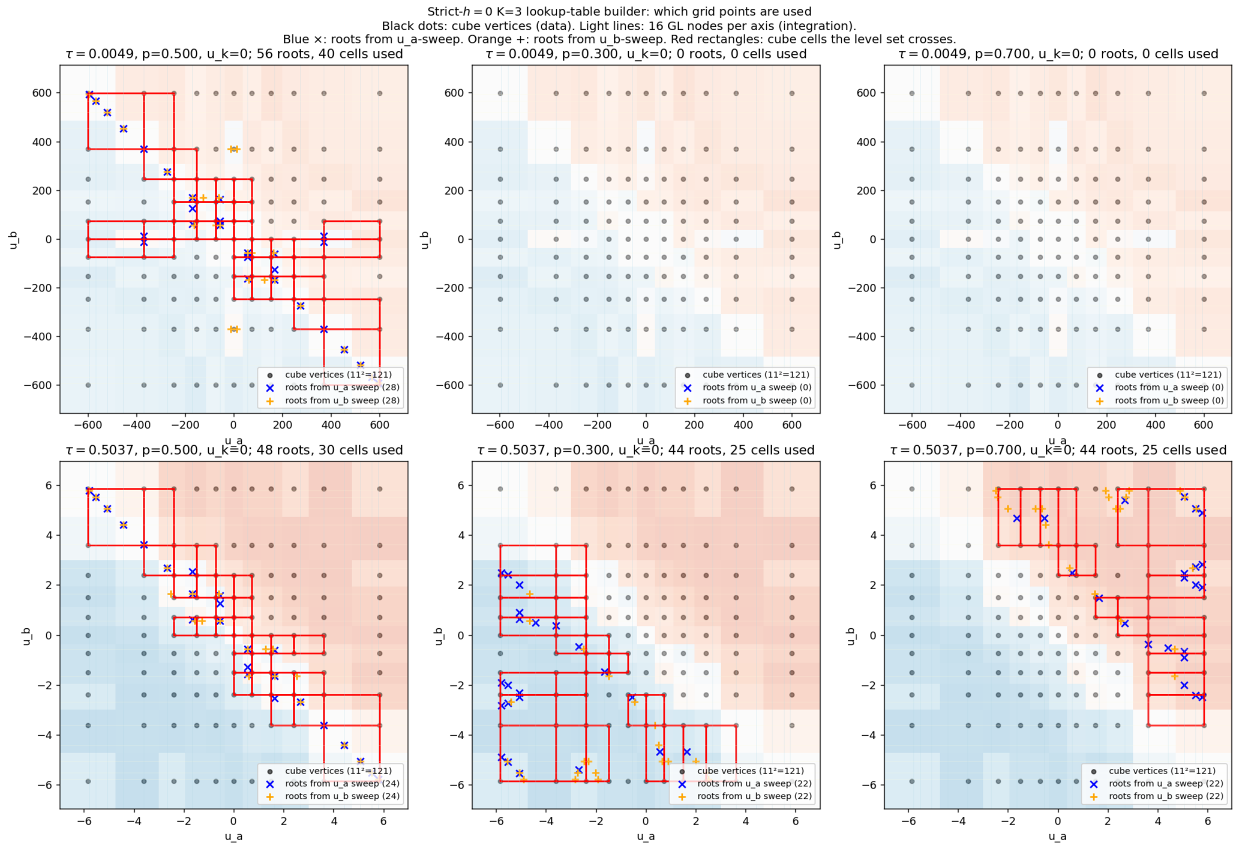

Figure 17.5: Which grid points the lookup uses: cube vertices (black), GL sweep lines (light), level-set roots from each sweep (blue × / orange +), and the cells crossed by the level set (red). Top row: $\tau = 0.005$ — the level set hugs the antidiagonal and off-median prices find no roots (the fallback region of Finding 14). Bottom: $\tau = 0.5$ — well-resolved level sets across the cube.

Chapter 18

Branches and Newton-type solvers

18.1 The trivial branch is exact

$P \equiv \frac{1}{2}$ satisfies the K=3 CRRA clearing identically (all posteriors equal the prior, all $R_i = 1$, zero demand): cold-started solvers of every type converge to it instantly, to $F = 0$ exactly. The equilibrium correspondence is multi-valued everywhere; *branch identification is part of the computation's burden of proof.* The informative branch is reached only by continuation (warm-starting along the τ-ladder from the analytic low-τ solution).

18.2 Newton–Krylov via the implicit function theorem

The IFT Newton direction $\delta P = -(\partial\Phi/\partial P - I)^{-1}(\Phi(P) - P)$, implemented matrix-free by LGMRES with finite-difference Jacobian-vector products and Wolfe line search, warm-started from the Picard ladder:

- **Low** τ: weighted residuals driven as deep as 10^{-25}–10^{-50} on cells where the basin is locally smooth — far below the Picard floor.
- **High** τ: plateaus at $F_w \sim 10^{-3}$–10^{-2} with intermittent symmetry breaks — the finite-difference Jacobian inherits the operator's C^0-not-C^1 roughness.
- **Cost**: 200–1400 F-evaluations per cell (50–100× Picard); total ladder 207 s vs 16 s.

MINPACK HYBRD (explicit numerical Jacobian + Broyden updates) was also tried: in logit space it is captured by the trivial branch even from informative warm starts; in P-space it exits the unit cube (one cell reached $P \in [-1590, 1591]$) even with bound penalties. Newton–Krylov with continuation warm starts is the only stable Newton variant found.

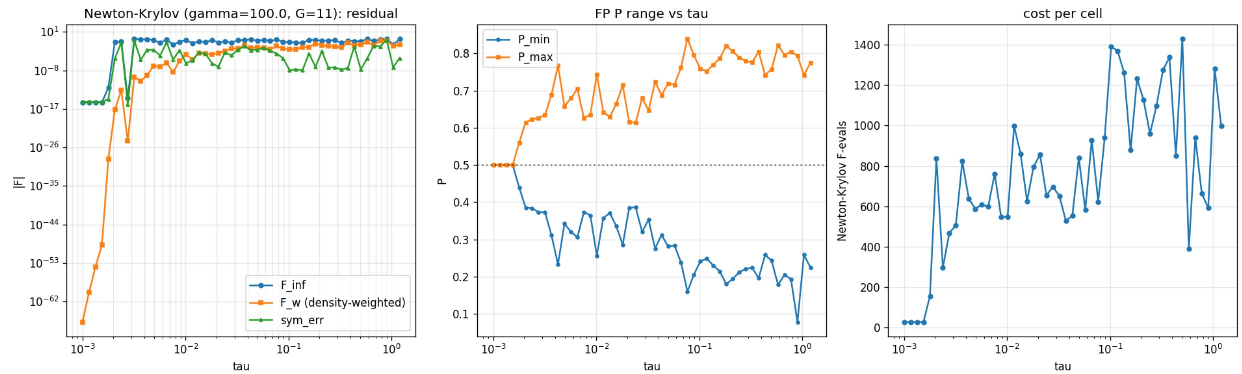

Figure 18.1: The IFT-Newton ladder vs Picard: deeper convergence at low τ, the same operator wall at high τ, at 50–100× the cost. Newton refines; it does not rescue.

18.3 Synthesis of the surface-side evidence

Across Picard, Anderson, Newton–Krylov, HYBRD, smoothed-ε operators, frozen-knot inner loops, and the structural ansatz: every method that respects the strict-h=0 operator's discretization plateaus at the same G-dependent wall at high τ, and every method that smooths past the wall pays in bias. Combined with the C^0-not-C^1 theorem and the success of the consistent-kernel joint limit at 10^{-14} (`k3_coarea_limit`), the conclusion is structural: **the high-resolution surface computation must use a consistent C^∞ operator (kernel co-area in the central-path limit), with the bare strict-h=0 operator reserved for the role of unbiased cross-check at matched tolerance**. That division is the spine of the recommendation in Part VII.

Part VII

The recommendation: a reproducible strict-$h=0$-anchored gold standard

Chapter 19

Design principles

The recommendation reconciles two facts that pull in opposite directions:

1. The *bare* strict-h=0 operator is the unbiased definition of the equilibrium (no auxiliary parameter at all) — but it is C^0-not-C^1 at Morse-critical prices and therefore cannot be nailed at high resolution by any Newton-type method (Part II theorem; Findings 9, 17; Part VI plateaus).
2. The *consistent kernel* co-area operator is C^∞ and nails the equilibrium to 10^{-14} — but carries the bandwidth h, which must be demonstrated (not assumed) to be a pure convergence knob.

The resolution: **solve with the kernel in the joint central-path limit; certify against strict-h=0 at matched tolerance; report the h-ladder as part of the result**. The kernel is then never trusted — it is verified, cell by cell.

Chapter 20

The protocol, step by step

20.1 Step 0: coordinates and grids

- u-grid: uniform private log-odds $\xi = \tanh(\tau u/2)$, $\xi_{\max} = 0.85$ (auto-scaling $\pm 5\sigma$; no hard truncation). G per the accuracy target below.
- p-grid: uniform public log-odds L_{pub}, bootstrap from the cube CDF, antisymmetrized by construction; refreshed during warm-up only (iteration 5), frozen afterwards.
- Weights: trapezoid in u; density weight $W_3 = \prod_i \bar{f}(u_i)$ cached for the residual metric.

20.2 Step 1: continuation ladder (the branch certificate)

Start at $\tau_0 = 10^{-3}$ from the analytic additive solution $P = \sigma(\tau \sum_i u_i)$ (Method C v2 closed form; exact there to 10^{-10}). March τ upward in log steps of $\leq 5\%$, each cell warm-started from its predecessor, recording per cell: F_w best, symmetry errors, price range, wall time. This ladder *is* the proof of branch identity: a connected continuation path from the analytic limit to the target cell. Never accept a cold-started high-τ fixed point — the trivial branch is exact and attracting.

20.3 Step 2: solve each cell with the consistent kernel, joint limit

At each (γ, τ) target:

1. Solve at (G_1, h_1) with the Gaussian-band co-area operator, Anderson or Picard, to its floor; require the floor to be at arithmetic level for the precision in use.
2. Refine along the central path: $(G_{k+1}, h_{k+1}) = (\lceil 1.4 G_k \rceil,\ h_k/1.4)$ keeping $h \gtrsim 3\Delta\xi$ always, warm-starting each rung by interpolation. Three rungs minimum.
3. Convergence claim: economic quantities (deficit, slope, P-range) must agree across the last two rungs to the target tolerance, demonstrating h-invariance. *The ladder table, not a single number, is the result.*
4. Symmetry projection (permutation + sign-flip) every iteration; symmetry errors reported at arithmetic level.

5. Precision: float64 until the rung floor reaches 10^{-13}; double-double beyond. (QD/F128 retired: no benefit at these floors.)

20.4 Step 3: the strict-h=0 cross-check (the unbiasedness certificate)

At the finest rung, evaluate the *bare* strict-h=0 residual at the kernel solution. Accept if $|F_{\text{strict}}|_w \leq C \cdot$ (kernel-rung truncation estimate) with $C \approx 3$. This certifies that the kernel solution is a strict-h=0 near-fixed-point to the accuracy the bare operator can measure — the kernel never substitutes its own notion of equilibrium.

20.5 Step 4: extract the lookup with pipeline v5

On the certified surface: `fix_fallback` $\rightarrow$ internal μ-based cusp detector $\rightarrow$ degree-6 per-segment Chebyshev $\rightarrow$ PCHIP monotone tails with $\mu(\epsilon) = 0$, $\mu(1 - \epsilon) = 1$. Certify per cell: in-support max error vs the cleaned co-area lookup at machine epsilon; 11/11 monotone slices on a dense grid; boundary conditions met. (Demonstrated at 36/36 cells, 7 seconds total.)

20.6 Step 5: the validation battery (all five, every cell)

1. **Symmetry**: permutation, sign-flip, lookup mirror at arithmetic level.
2. **Monotonicity**: P axis-monotone; μ monotone in p per slice (dense check).
3. **Boundary conditions**: lookup tails; $P(\mathbf{0}) = \frac{1}{2}$ exactly.
4. **Operator cross-check**: Step 3 inequality, plus the kernel h-ladder invariance table.
5. **Measure robustness**: report the deficit under at least two state weightings (uniform and signal-weighted); the ~ 0.12 measure spread is part of the honest result, the price *surface* being the invariant object.

20.7 Step 6: reporting standard

Every published cell carries: the continuation provenance (ladder path), the final rung $(G, h, \text{precision})$, the kernel-ladder invariance table, the strict-h=0 cross-check value, best-iterate vs converged status, the symmetry residuals, and the deficit with its measure convention. Nothing else counts as a result.

Chapter 21

Accuracy targets and cost

target	grid/rungs	precision	cost/cell (order)
survey (10^{-3} econ)	G=11, 1 rung + v5 lookup	float64	seconds
publication (10^{-6} econ)	G=17−25, 3 rungs	float64	minutes
anchor (10^{-12+})	$G \geq 31$, 4 rungs	DD (& flint spot checks)	hours

The existing anchors: `k3_coarea_limit` (10^{-14}, joint limit), `k3_strict_h0_exact` (flint, toward 10^{-100} at coarse G), and the G=9 no-kernel nail at 8×10^{-15} — three independent operator families agreeing on the same PR equilibrium is the strongest truth-closeness evidence the project owns.

Chapter 22

Open problems

1. **Analytic Jacobian through the kernel co-area operator** (~ 200 LOC chain rule): would replace finite-difference Newton noise and likely extend deep convergence to higher τ at fixed cost.
2. **Pseudo-arclength continuation** past the K=2 fold at $\tau \approx 0.79$ and a systematic K=3 fold map: the branch structure above $\tau \sim 1$ is mapped only along ladders, not in the (γ, τ) plane.
3. **The 5 corner outlier cells** of the v5 certification (10^{-2} max): localized fallback-smoothing artifacts at extreme parameter corners; a targeted G_p refinement there is unfinished.
4. **K=4 extreme-heterogeneity PR** (`k4_pr_nail` floors at 2.6×10^{-3}): whether it nails fully under the consistent co-area operator is open — the natural next economics result.
5. **A cusp-tracking surface representation**: the lookup-side segmentation idea lifted to the surface itself (multi-element cube patches bounded by the critical manifold) — the principled route to exponential convergence of P, beyond the scope of the present cycle.

Appendices

Appendix A

Per-cell result tables

A.1 The 50-cell log-odds Picard ladder ($\gamma = 100$, $G = 11$)

idx	τ	F_w best	$P_{\min}$	$P_{\max}$	sym err
0	0.00100	2.55e-15	0.0005	0.9995	4.4e-16
1	0.00116	2.55e-15	0.0005	0.9995	4.4e-16
2	0.00134	2.55e-15	0.0005	0.9995	4.4e-16
3	0.00154	2.55e-15	0.0005	0.9995	4.4e-16
4	0.00178	2.55e-15	0.4347	0.5653	1.1e-16
5	0.00206	1.38e-14	0.3832	0.6168	1.1e-16
6	0.00238	4.08e-13	0.3693	0.6307	1.1e-16
7	0.00275	5.69e-07	0.3491	0.6509	1.1e-16
8	0.00318	1.76e-02	0.4162	0.5838	1.1e-16
9	0.00368	7.37e-07	0.4162	0.5838	1.1e-16
10	0.00425	4.91e-08	0.4270	0.5730	1.1e-16
11	0.00491	5.57e-08	0.3346	0.6654	1.1e-16
12	0.00568	3.77e-07	0.3346	0.6654	1.1e-16
13	0.00656	1.27e-06	0.3833	0.6167	1.1e-16
14	0.00758	1.98e-02	0.3593	0.6407	1.1e-16
15	0.00876	7.89e-05	0.3646	0.6354	1.1e-16
16	0.01013	1.71e-04	0.3925	0.6075	1.1e-16
17	0.01170	2.86e-04	0.3552	0.6448	1.1e-16
18	0.01352	5.53e-04	0.3543	0.6457	1.1e-16
19	0.01563	6.53e-04	0.2897	0.7103	1.1e-16
20	0.01806	1.39e-03	0.3298	0.6702	1.1e-16
21	0.02088	2.41e-04	0.2741	0.7259	1.1e-16
22	0.02413	5.53e-02	0.2741	0.7259	1.1e-16
23	0.02788	4.91e-03	0.3139	0.6861	1.1e-16
24	0.03222	6.53e-03	0.3139	0.6861	1.1e-16
25	0.03724	8.35e-03	0.3139	0.6861	1.1e-16
26	0.04304	1.06e-02	0.3139	0.6861	1.1e-16
27	0.04974	1.22e-02	0.3276	0.6724	1.1e-16
28	0.05748	1.52e-02	0.1599	0.8401	1.1e-16
29	0.06643	1.66e-02	0.2509	0.7491	1.1e-16
30	0.07677	1.98e-02	0.2370	0.7630	1.1e-16
31	0.08873	1.86e-02	0.2205	0.7795	1.1e-16
32	0.10254	2.13e-02	0.2491	0.7509	1.1e-16

idx	τ	F_w best	$P_{\min}$	$P_{\max}$	sym err
33	0.11850	1.58e-02	0.2083	0.7917	1.1e-16
34	0.13695	1.74e-02	0.2244	0.7756	1.1e-16
35	0.15828	2.04e-02	0.2244	0.7756	1.1e-16
36	0.18292	2.31e-02	0.2244	0.7756	1.1e-16
37	0.21140	2.59e-02	0.1977	0.8023	1.1e-16
38	0.24431	3.31e-02	0.2207	0.7793	1.1e-16
39	0.28234	3.58e-02	0.2159	0.7841	1.1e-16
40	0.32630	3.71e-02	0.2362	0.7638	1.1e-16
41	0.37710	4.28e-02	0.2280	0.7720	1.1e-16
42	0.43581	4.54e-02	0.2046	0.7954	1.1e-16
43	0.50366	5.95e-02	0.2587	0.7413	1.1e-16
44	0.58208	6.19e-02	0.2271	0.7729	1.1e-16
45	0.67270	6.87e-02	0.2271	0.7729	1.1e-16
46	0.77743	2.75e-02	0.3324	0.6676	1.1e-16
47	0.89846	9.99e-02	0.2471	0.7529	1.1e-16
48	1.03834	1.26e-01	0.2121	0.7879	1.1e-16
49	1.20000	1.56e-01	0.2308	0.7692	1.1e-16

A.2 The 50-cell IFT-Newton ladder (warm-started from Picard)

idx	τ	F_∞	F_w	$P_{\min}$	$P_{\max}$	nfev
0	0.00100	3.33e-16	1.46e-67	0.5000	0.5000	27
1	0.00116	3.33e-16	1.15e-60	0.5000	0.5000	27
2	0.00134	3.33e-16	1.07e-54	0.5000	0.5000	27
3	0.00154	3.33e-16	1.56e-49	0.5000	0.5000	27
4	0.00178	9.59e-13	2.02e-29	0.4398	0.5602	156
5	0.00206	4.80e-02	6.31e-18	0.3855	0.6145	837
6	0.00238	6.22e-02	2.37e-13	0.3838	0.6231	297
7	0.00275	3.22e-15	3.13e-25	0.3735	0.6265	468
8	0.00318	2.84e-01	2.66e-10	0.3731	0.6345	507
9	0.00368	1.54e-01	3.15e-11	0.3111	0.6889	823
10	0.00425	2.02e-01	6.87e-10	0.2330	0.7668	639
11	0.00491	1.27e-01	1.12e-07	0.3425	0.6578	587
12	0.00568	2.70e-02	8.30e-08	0.3205	0.6795	609
13	0.00656	1.61e-01	5.03e-07	0.3064	0.7042	598
14	0.00758	1.14e-02	4.05e-09	0.3733	0.6267	761
15	0.00876	4.56e-02	1.50e-06	0.3648	0.6352	548
16	0.01013	1.41e-01	4.97e-05	0.2567	0.7433	548
17	0.01170	2.15e-02	3.57e-06	0.3578	0.6422	998
18	0.01352	6.86e-02	1.24e-04	0.3708	0.6292	861
19	0.01563	3.87e-02	7.32e-05	0.3361	0.6639	626
20	0.01806	1.19e-01	4.97e-05	0.2859	0.7141	794
21	0.02088	3.61e-02	1.31e-04	0.3847	0.6153	857
22	0.02413	5.69e-02	3.10e-04	0.3866	0.6134	654
23	0.02788	3.33e-02	1.34e-03	0.3202	0.6798	697
24	0.03222	1.13e-01	1.21e-03	0.3534	0.6464	651
25	0.03724	1.38e-01	6.84e-03	0.2757	0.7235	528
26	0.04304	1.86e-01	1.67e-03	0.3122	0.6878	554
27	0.04974	9.35e-02	2.74e-03	0.2818	0.7180	841

idx	τ	F_∞	F_w	$P_{\min}$	$P_{\max}$	nfev
28	0.05748	1.40e-01	1.93e-03	0.2839	0.7157	582
29	0.06643	7.64e-02	7.74e-04	0.2387	0.7614	927
30	0.07677	1.71e-01	4.89e-03	0.1604	0.8395	623
31	0.08873	4.54e-02	1.28e-03	0.2056	0.7944	939
32	0.10254	4.57e-02	8.03e-04	0.2418	0.7582	1392
33	0.11850	2.93e-02	9.77e-04	0.2481	0.7519	1369
34	0.13695	2.83e-02	2.52e-03	0.2303	0.7697	1262
35	0.15828	1.45e-01	2.75e-03	0.2142	0.7858	878
36	0.18292	1.26e-01	1.79e-02	0.1800	0.8200	1232
37	0.21140	8.21e-02	8.63e-03	0.1943	0.8057	1127
38	0.24431	9.06e-02	4.80e-03	0.2119	0.7881	959
39	0.28234	1.17e-01	3.95e-03	0.2210	0.7790	1098
40	0.32630	2.83e-02	2.35e-03	0.2246	0.7754	1274
41	0.37710	8.25e-02	2.65e-02	0.1959	0.8041	1338
42	0.43581	7.24e-02	8.82e-03	0.2587	0.7418	851
43	0.50366	6.59e-02	1.12e-02	0.2430	0.7570	1431
44	0.58208	2.13e-01	3.94e-02	0.1786	0.8214	390
45	0.67270	7.29e-02	8.38e-03	0.2052	0.7950	940
46	0.77743	2.01e-01	8.57e-02	0.1936	0.8046	664
47	0.89846	2.56e-01	8.64e-02	0.0782	0.7935	592
48	1.03834	1.03e-02	5.04e-03	0.2586	0.7414	1282
49	1.20000	2.09e-01	1.02e-02	0.2252	0.7748	997

A.3 Pipeline v5 certification, 36 ridge cells

γ	τ	cusps	max err	median err	monotone	min diff
1	0.1	83	1.55e-15	1.11e-16	11/11	5.4e-04
1	0.2	119	5.55e-16	1.11e-16	11/11	0.0e+00
1	0.3	56	2.11e-15	1.11e-16	11/11	0.0e+00
1	0.5	44	2.11e-15	1.11e-16	11/11	0.0e+00
1	0.7	117	9.37e-02	1.11e-16	11/11	-1.1e-16
1	1.0	45	4.82e-02	1.11e-16	11/11	-1.4e-17
10	0.1	88	2.11e-15	1.11e-16	11/11	7.7e-04
10	0.2	55	3.72e-02	1.11e-16	11/11	1.2e-05
10	0.3	55	2.55e-15	1.11e-16	11/11	1.5e-05
10	0.5	41	1.22e-15	1.11e-16	11/11	0.0e+00
10	0.7	51	3.22e-15	1.11e-16	11/11	-1.1e-16
10	1.0	62	2.22e-15	1.11e-16	11/11	-2.2e-16
30	0.1	52	1.55e-15	1.11e-16	11/11	1.7e-03
30	0.2	67	3.84e-02	1.11e-16	11/11	1.2e-05
30	0.3	55	8.88e-16	1.11e-16	11/11	0.0e+00
30	0.5	52	9.99e-16	1.11e-16	11/11	0.0e+00
30	0.7	44	2.55e-15	1.11e-16	11/11	-1.1e-16
30	1.0	54	1.78e-15	1.11e-16	11/11	0.0e+00
100	0.1	49	1.33e-15	1.11e-16	11/11	7.4e-04
100	0.2	47	2.66e-15	1.11e-16	11/11	1.2e-05
100	0.3	60	2.11e-15	1.11e-16	11/11	9.0e-06
100	0.5	51	2.44e-15	1.11e-16	11/11	0.0e+00
100	0.7	42	2.44e-15	1.11e-16	11/11	-1.1e-16

γ	τ	cusps	max err	median err	monotone	min diff
100	1.0	61	3.22e-02	1.11e-16	11/11	0.0e+00
300	0.1	119	5.55e-16	1.11e-16	11/11	5.0e-04
300	0.2	79	5.55e-16	1.11e-16	11/11	1.4e-05
300	0.3	70	2.00e-15	1.11e-16	11/11	0.0e+00
300	0.5	39	2.33e-15	1.11e-16	11/11	0.0e+00
300	0.7	47	1.67e-15	1.11e-16	11/11	-1.1e-16
300	1.0	50	3.17e-04	1.11e-16	11/11	0.0e+00
1000	0.1	52	1.11e-15	1.11e-16	11/11	9.7e-04
1000	0.2	53	1.11e-15	1.11e-16	11/11	1.2e-05
1000	0.3	53	4.11e-15	1.11e-16	11/11	0.0e+00
1000	0.5	45	1.55e-15	1.11e-16	11/11	5.5e-06
1000	0.7	46	1.78e-15	1.11e-16	11/11	0.0e+00
1000	1.0	52	8.88e-16	1.11e-16	11/11	-1.1e-16

A.4 Method C v2: the 100-cell (γ, τ) grid

γ	τ	$\|F\|_\infty$	slope	$P_{\min}$	$P_{\max}$
0.1	0.001	1.24e-08	0.0003	0.4994	0.5006
0.1	0.005	1.72e-06	0.0017	0.4971	0.5029
0.1	0.01	1.37e-05	0.0033	0.4942	0.5058
0.1	0.05	1.64e-03	0.0167	0.4710	0.5290
0.1	0.1	1.05e-02	0.0337	0.4424	0.5576
0.1	0.3	5.91e-02	0.1123	0.3241	0.6759
0.1	0.5	4.16e-02	0.2406	0.1812	0.8188
0.1	0.7	3.27e-02	0.3627	0.0966	0.9034
0.1	0.85	5.43e-02	0.6891	0.0075	0.9925
0.1	1	7.90e-02	0.9228	0.0010	0.9990
0.3	0.001	2.73e-09	0.0003	0.4994	0.5006
0.3	0.005	1.41e-07	0.0017	0.4971	0.5029
0.3	0.01	1.14e-06	0.0033	0.4942	0.5058
0.3	0.05	1.39e-04	0.0167	0.4709	0.5291
0.3	0.1	1.04e-03	0.0338	0.4414	0.5586
0.3	0.3	1.28e-02	0.1149	0.3135	0.6865
0.3	0.5	3.32e-03	0.2676	0.1418	0.8582
0.3	0.7	4.55e-02	0.5272	0.0246	0.9754
0.3	0.85	5.81e-02	0.7410	0.0052	0.9948
0.3	1	8.83e-02	0.9479	0.0011	0.9989
1	0.001	1.78e-10	0.0003	0.4994	0.5006
1	0.005	1.88e-08	0.0017	0.4971	0.5029
1	0.01	1.72e-07	0.0033	0.4942	0.5058
1	0.05	2.14e-05	0.0168	0.4708	0.5292
1	0.1	1.64e-04	0.0340	0.4410	0.5590
1	0.3	1.76e-03	0.1200	0.3029	0.6971
1	0.5	2.47e-02	0.2989	0.1116	0.8884
1	0.7	5.68e-02	0.5738	0.0169	0.9831
1	0.85	5.89e-02	0.7966	0.0036	0.9964
1	1	1.02e-01	1.0132	0.0007	0.9993
3	0.001	2.33e-09	0.0003	0.4994	0.5006
3	0.005	5.94e-09	0.0017	0.4971	0.5029

γ	τ	$\|F\|_\infty$	slope	$P_{\min}$	$P_{\max}$
3	0.01	5.76e-08	0.0033	0.4942	0.5058
3	0.05	7.02e-06	0.0168	0.4708	0.5292
3	0.1	5.01e-05	0.0341	0.4409	0.5591
3	0.3	1.12e-03	0.1220	0.2994	0.7006
3	0.5	3.12e-02	0.3086	0.1039	0.8961
3	0.7	5.93e-02	0.5879	0.0153	0.9847
3	0.85	6.35e-02	0.8126	0.0032	0.9968
3	1	8.58e-02	1.0335	0.0009	0.9991
10	0.001	3.55e-10	0.0003	0.4994	0.5006
10	0.005	2.39e-09	0.0017	0.4971	0.5029
10	0.01	1.71e-08	0.0033	0.4942	0.5058
10	0.05	1.98e-06	0.0168	0.4708	0.5292
10	0.1	1.57e-05	0.0341	0.4408	0.5592
10	0.3	1.86e-03	0.1227	0.2982	0.7018
10	0.5	3.34e-02	0.3120	0.1014	0.8986
10	0.7	6.03e-02	0.5927	0.0147	0.9853
10	0.85	6.51e-02	0.8177	0.0031	0.9969
10	1	8.41e-02	1.0475	0.0008	0.9992
30	0.001	1.47e-09	0.0003	0.4994	0.5006
30	0.005	3.71e-09	0.0017	0.4971	0.5029
30	0.01	6.01e-09	0.0033	0.4942	0.5058
30	0.05	7.00e-07	0.0168	0.4708	0.5292
30	0.1	5.44e-06	0.0341	0.4408	0.5592
30	0.3	2.13e-03	0.1230	0.2978	0.7022
30	0.5	3.40e-02	0.3130	0.1007	0.8993
30	0.7	6.06e-02	0.5941	0.0145	0.9855
30	0.85	6.65e-02	0.8234	0.0029	0.9971
30	1	8.45e-02	1.0500	0.0007	0.9993
100	0.001	1.21e-10	0.0003	0.4994	0.5006
100	0.005	3.73e-09	0.0017	0.4971	0.5029
100	0.01	2.00e-09	0.0033	0.4942	0.5058
100	0.05	2.10e-07	0.0168	0.4708	0.5292
100	0.1	5.51e-06	0.0341	0.4408	0.5592
100	0.3	2.22e-03	0.1230	0.2977	0.7023
100	0.5	3.42e-02	0.3133	0.1004	0.8996
100	0.7	6.06e-02	0.5946	0.0145	0.9855
100	0.85	6.66e-02	0.8239	0.0029	0.9971
100	1	8.46e-02	1.0510	0.0007	0.9993
300	0.001	7.69e-13	0.0003	0.4994	0.5006
300	0.005	2.31e-09	0.0017	0.4971	0.5029
300	0.01	1.48e-09	0.0033	0.4942	0.5058
300	0.05	1.53e-07	0.0168	0.4708	0.5292
300	0.1	6.66e-06	0.0341	0.4408	0.5592
300	0.3	2.25e-03	0.1231	0.2976	0.7024
300	0.5	3.43e-02	0.3134	0.1004	0.8996
300	0.7	6.07e-02	0.5947	0.0145	0.9855
300	0.85	6.67e-02	0.8241	0.0029	0.9971
300	1	8.46e-02	1.0512	0.0007	0.9993
1000	0.001	1.40e-09	0.0003	0.4994	0.5006
1000	0.005	2.34e-11	0.0017	0.4971	0.5029
1000	0.01	3.91e-10	0.0033	0.4942	0.5058
1000	0.05	2.03e-07	0.0168	0.4708	0.5292

γ	τ	$\|F\|_\infty$	slope	$P_{\min}$	$P_{\max}$
1000	0.1	7.06e-06	0.0341	0.4408	0.5592
1000	0.3	2.26e-03	0.1231	0.2976	0.7024
1000	0.5	3.43e-02	0.3135	0.1003	0.8997
1000	0.7	6.07e-02	0.5948	0.0144	0.9856
1000	0.85	6.67e-02	0.8242	0.0029	0.9971
1000	1	8.46e-02	1.0513	0.0007	0.9993
3000	0.001	7.44e-12	0.0003	0.4994	0.5006
3000	0.005	7.39e-11	0.0017	0.4971	0.5029
3000	0.01	1.12e-10	0.0033	0.4942	0.5058
3000	0.05	2.17e-07	0.0168	0.4708	0.5292
3000	0.1	7.18e-06	0.0341	0.4408	0.5592
3000	0.3	2.26e-03	0.1231	0.2976	0.7024
3000	0.5	3.43e-02	0.3135	0.1003	0.8997
3000	0.7	6.07e-02	0.5948	0.0144	0.9856
3000	0.85	6.67e-02	0.8242	0.0029	0.9971
3000	1	8.46e-02	1.0513	0.0007	0.9993

A.5 The strict-ridge deficit recheck (completed cells)

γ	τ	F_{strict}	deficit (R4)	deficit (strict)	ratio
1	0.1	4.33e-03	7.61e-10	1.68e-02	0.0
1	0.2	7.33e-03	2.06e-07	1.11e-02	0.0
1	0.3	1.45e-02	6.66e-06	7.54e-03	0.0
1	0.5	3.27e-02	8.82e-03	2.74e-03	3.2
1	0.7	1.49e-01	9.70e-03	2.07e-03	4.7
1	1	1.16e-01	1.14e-02	2.07e-03	5.5
10	0.1	9.02e-03	6.16e-09	1.45e-02	0.0
10	0.2	3.24e-03	1.54e-07	2.23e-03	0.0
10	0.3	2.49e-02	2.15e-06	7.72e-03	0.0
10	0.5	7.58e-02	7.18e-03	6.30e-03	1.1
10	0.7	5.35e-02	2.09e-02	7.13e-03	2.9
10	1	3.86e-02	1.29e-01	6.83e-03	18.8
30	0.1	4.57e-03	9.19e-10	2.96e-02	0.0
30	0.2	3.15e-03	3.67e-08	2.23e-03	0.0
30	0.3	3.09e-02	9.68e-07	1.16e-02	0.0
30	0.5	7.20e-02	6.98e-03	9.39e-03	0.7
30	0.7	2.82e-02	1.75e-02	8.34e-03	2.1
30	1	6.39e-02	2.16e-01	7.43e-03	29.0
100	0.1	6.47e-03	1.38e-10	3.07e-02	0.0
100	0.2	3.37e-03	1.32e-08	2.28e-03	0.0
100	0.3	1.12e-02	6.41e-07	5.99e-03	0.0
100	0.5	4.77e-02	7.00e-03	9.66e-03	0.7
100	0.7	5.53e-02	1.77e-02	6.95e-03	2.5
100	1	4.72e-02	2.55e-01	6.50e-03	39.2
300	0.1	6.87e-03	3.93e-11	1.03e-02	0.0
300	0.2	1.64e-02	8.49e-09	7.34e-03	0.0
300	0.3	1.03e-02	5.58e-07	5.25e-03	0.0
300	0.5	4.88e-02	6.98e-03	4.67e-03	1.5
300	0.7	3.98e-02	2.20e-02	5.84e-03	3.8

γ	τ	F_{strict}	deficit (R4)	deficit (strict)	ratio
300	1	6.39e-02	2.67e-01	3.72e-03	71.8
1000	0.1	5.08e-03	1.88e-11	2.36e-02	0.0
1000	0.2	3.00e-03	7.08e-09	2.30e-03	0.0
1000	0.3	1.18e-02	5.30e-07	6.56e-03	0.0
1000	0.5	8.69e-02	6.97e-03	6.62e-03	1.1
1000	0.7	8.59e-02	1.84e-02	6.10e-03	3.0
1000	1	4.15e-02	2.71e-01	5.48e-03	49.5

Appendix B

Repository map

B.1 Solver code (projects/REZN/solver_code/highprec_dd_qd/)

file	role
dd_ops.py, dd_k3_ops.py	double-double primitives; exact CRRA clearing bisection
dd_solver.py	K=2 DD surface solver (the τ-fold study)
dd_k3_solver.py, dd_k3_sweep.py	K=3 kernel-band R4 production (superseded for economics)
dd_k3_strict_solve.py	strict-h=0 Anderson + deficit metrics
dd_k3_compare.py, dd_k3_rconverge.py, dd_k3_small_h.py	Findings 1, 3 diagnostics
dd_k3_lookup_alts.py, dd_k3_optB_numba.py, dd_k3_optB_refined.py	Options A–E; B+ production
dd_k3_optD_adaptive_p.py	adaptive p-grid
dd_k3_cheby_h0.py, dd_k3_cheby_lobatto.py	spectral program
dd_k3_method_c_v2*.py	structural ansatz: solver, grid, ladder, logit-loss, strict variants
dd_k3_mu_analytic.py	closed-form 1D lookup under the ansatz
dd_k3_v3_fingerprint.py, dd_k3_v3_refit.py	Finding 7
dd_k3_lpub_lpriv_v2.py, dd_k3_delta_svd.py	Finding 5
dd_k3_critpts.py	3D critical-point detector (Finding 8)
dd_k3_optB_pieceCheby.py	per-segment Chebyshev + the internal cusp detector (Findings 10, 13)
dd_k3_optB_tail.py, dd_k3_optB_exact.py, dd_k3_optB_multielem.py	Findings 8, 11, 12
dd_k3_strict_smoothed*.py, dd_k3_stall_trace.py	Finding 9
dd_k3_strict_fixed_fallback.py	the fallback fix (Finding 14)
dd_k3_zero_h_pipeline_v2/v3/v4.py, dd_k3_pipeline_v5_final.py	pipeline v2–v5
dd_k3_v5_grid.py	36-cell certification (Finding 16)
dd_k3_zero_h_solver.py	Finding 17
dd_k3_logodds.py, dd_k3_logodds_ladder.py	the final coordinate configuration + 50-cell ladder
dd_k3_logodds_newton.py, dd_k3_logodds_ift.py	IFT Newton (Krylov; HYBRD experiments)

B.2 Pre-K=3 foundations (`projects/REZN/solved_fixed_points/`)

directory	contribution
`methodology/`	9pp method paper + runnable reference operator — start here
`k3_coarea_limit/`	the key anchor: joint $h \to 0$ ladder, PR nailed to 10^{-14}, deficit $\to 0.28$
`k3_verify_coarea/`	the 19% naive-scan bias, proven on an analytic surface
`k3_strict_h0_exact/`	flint marching-squares strict-h=0: continuity of the bare operator; nails toward 10^{-100}
`k3_hfree_morse/`	the C^0-not-C^1 proof (128× derivative blowup at a Morse price)
`k3_coarea_2dsweep/`	the definitive (γ, τ) deficit map (31/32 cells at 10^{-12})
`k3_jensen_gap/`	the closed-form wedge verified on nailed equilibria
`k3_cara/`	CARA = $\gamma \to \infty$ benchmark; Hellwig closed-form cross-check
`k4_*`	K=4: FR globally attracting; PR via extreme-γ dropout
`highprec_dd/`	K=2 DD ladder; the $\tau \approx 0.79$ fold
`flint_*`, `k3_plateau_*`	superseded naive-scan era (kept as the artifact record)
`FINDINGS.md`	the human-readable index of all of the above

Appendix C

Glossary

REE rational-expectations equilibrium: price function such that demands conditioned on (own signal, price) clear at every signal profile.

FR / PR fully / partially revealing price function.

deficit $1 - R^2$ of logit P on the pooled statistic (with stated state-measure); 0 under FR.

co-area integral $\int_{\{P=p\}} f/|\nabla P|\, d\sigma$ — the correct level-set evidence measure.

strict-h=0 the bare co-area operator: explicit contours + $1/|\nabla P|$, no kernel.

kernel band Gaussian-smoothed indicator $\int K_h(P-p) f\, du$; consistent quadrature of the co-area integral as $h \to 0$.

central path / joint limit $h \to 0$ together with $\Delta x \to 0$, $\Delta x \ll h$ — the regime where the kernel is both consistent and smooth.

lookup $\mu(p, u_k)$: Bayes posterior of $v = 1$ given own signal and the price level set.

POU partition of unity across the two sweep directions of the level-set integral.

Morse-critical price $p_c = P(u^*)$ with $\nabla P(u^*) = 0$: location of the C^0-not-C^1 kink in the evidence and of the lookup cusps.

F_w density-weighted residual $\max |\Phi(P) - P| \cdot \prod_i \bar{f}(u_i)$ — the economically meaningful convergence metric on unbounded grids.

trivial branch $P \equiv \frac{1}{2}$: an exact fixed point at all parameters; the uninformative pooled-prior equilibrium.

best iterate the lowest-residual point of a non-converging iteration — reported as such, never as a fixed point.

DD / QD double-double (32 digits) / quad-double (64 digits) arithmetic.

Appendix D

Chronology of the seventeen findings

#	verdict	one-line content
1	negative	fixed-grid kernel-band Richardson does not extrapolate to $h = 0$ (kink powers)
2	positive	strict-h=0 re-solve collapses the R4 deficit $\sim 40\times$; ridge is an artifact
3	negative	cross-variant agreement $\neq$ correctness (robust-but-wrong comparator)
4	positive	Option B+ (cube CDF + PCHIP) wins the builder bake-off; median 10^{-5}
5	mixed	$(L_{\text{priv}}, L_{\text{pub}})$ transform absorbs bulk, preserves cusps; $b(u_k) \approx \tau u_k/10$
6	positive	adaptive arclength p-grid: -25% max error, free
7	negative	no finite-parameter polynomial correction reaches 10^{-4} (high-rank residual)
8	mixed	3D critical-point detector works; knots fix the median only
9	negative	ε-smoothed strict-h=0: no sweet spot (bias vs convergence)
10	breakthrough	per-segment Chebyshev between cusps: machine eps in-segment
11	negative	tail BCs alone do not move the max error
12	mixed	exact cell mass: halves the median, worsens the max
13	correction	cusps of μ are *slice*-critical values; internal detector is the right one
14	positive	`fix_fallback`: monotone, BC-correct cleaned lookup
15	positive	pipeline v5 assembled: the production zero-h lookup
16	certification	36/36 cells: median 10^{-16}, 11/11 monotone, 7 s total
17	negative	smooth lookup inside the FP loop does not break the G-truncation wall

Appendix E

Extended figure gallery

This appendix collects the remaining result figures referenced by the findings but not embedded in the main text, with interpretive captions. Together with the in-text figures they constitute the complete graphic record of the investigation.

E.1 Kernel-band era and the full ladder

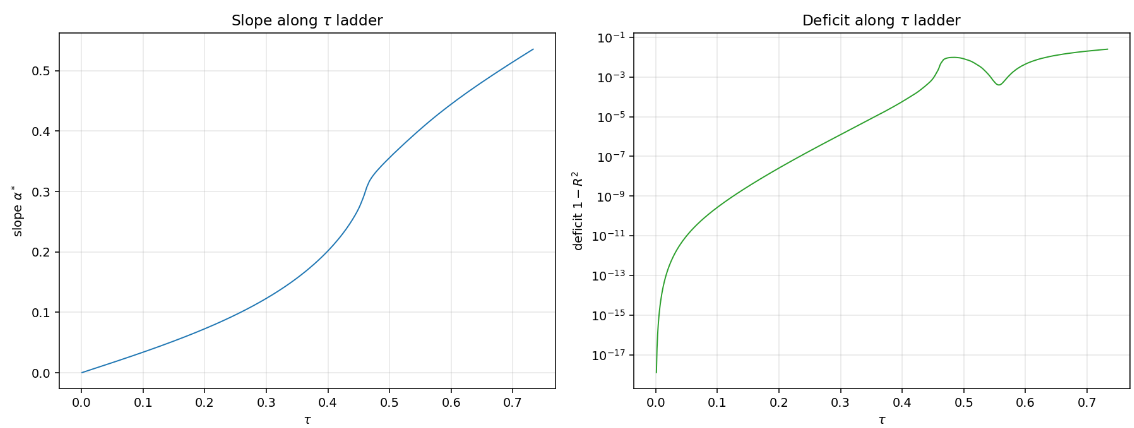

Figure E.1: Slope and deficit along the 1000-cell R4 DD ladder. These are the quantities later revised by the strict-h=0 recheck: the high-τ deficit rise is predominantly operator bias.

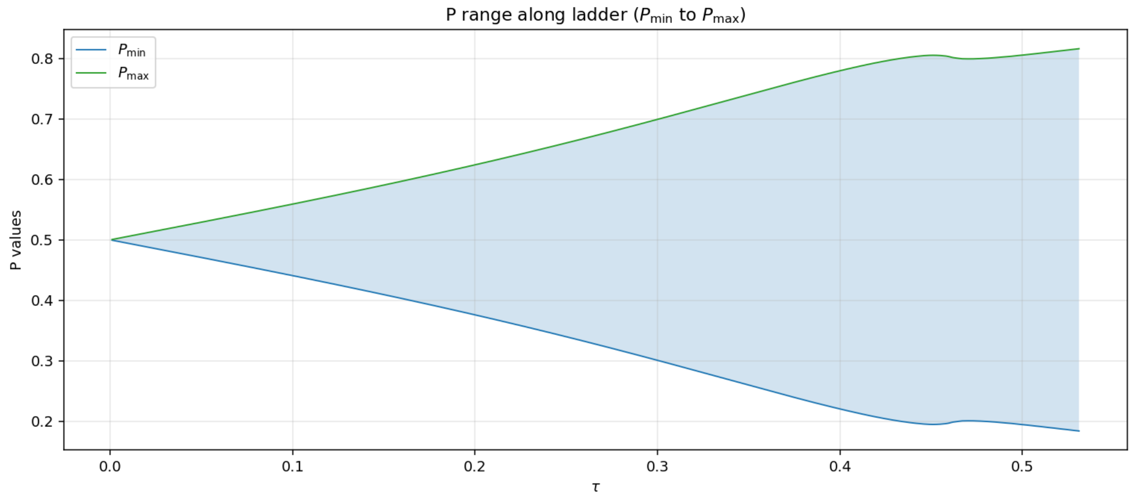

Figure E.2: Price range along the R4 ladder: the equilibrium sharpens with τ — qualitatively robust across operators even where the deficit numbers are not.

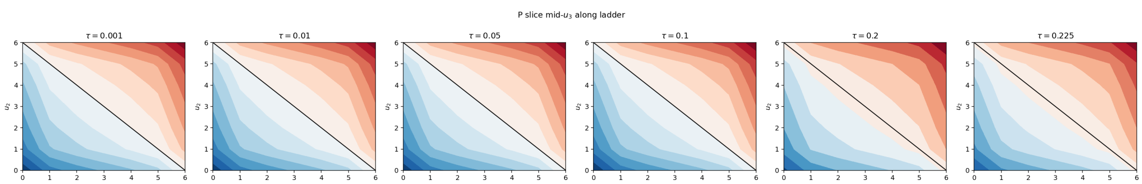

Figure E.3: R4 equilibrium contours across the ladder: the early visual evidence of the kink structure that the spectral and cusp analyses later formalized.

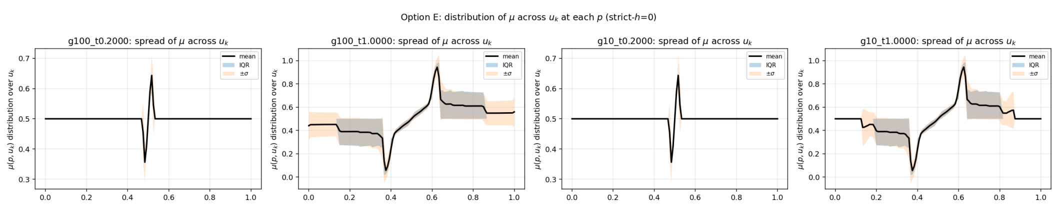

Figure E.4: Error distributions across the five lookup builders: Option B+ concentrates its mass two decades left of all kernel-band variants.

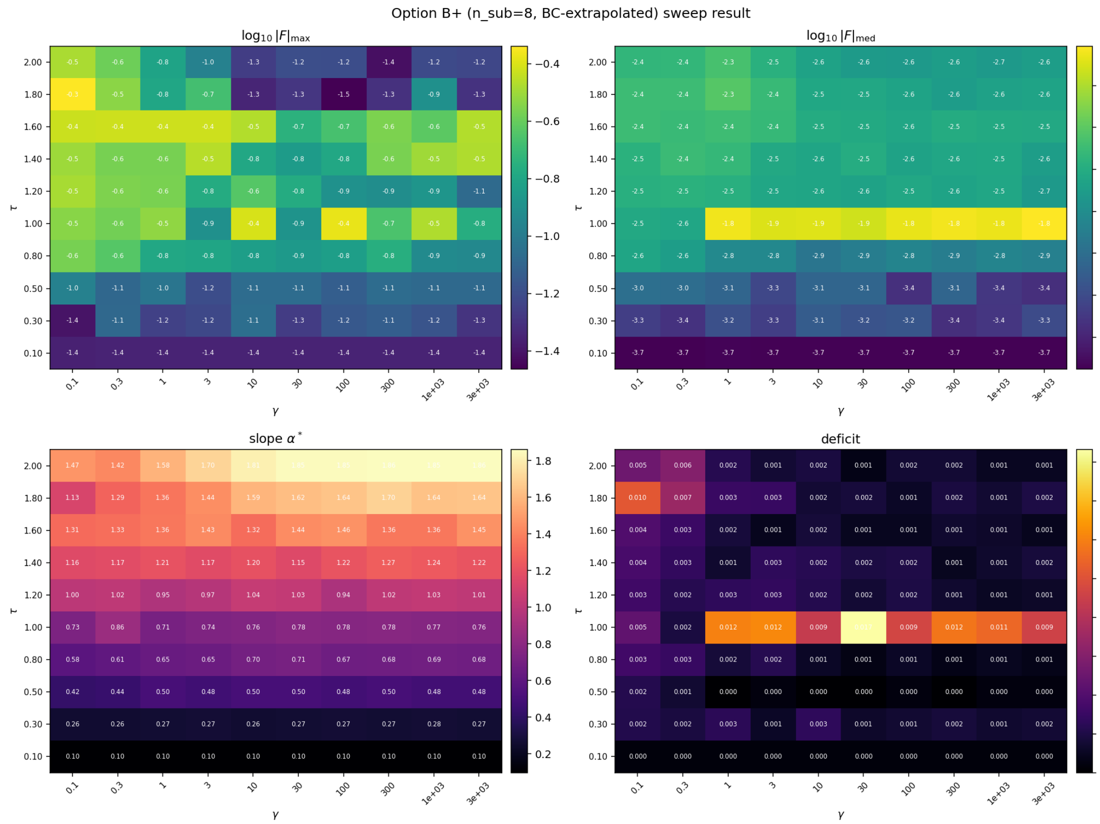

Figure E.5: Option B+ accuracy across (γ, τ): uniform in γ, mildly degrading at high τ as the cube CDF's tails thin.

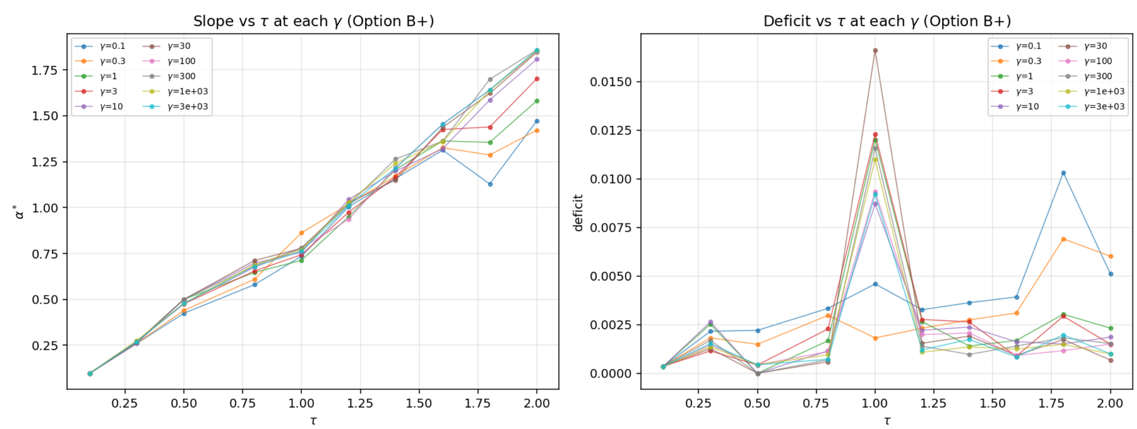

Figure E.6: Cross-sectional view: B+ tracks strict-h=0 to plotting accuracy at a representative cell.

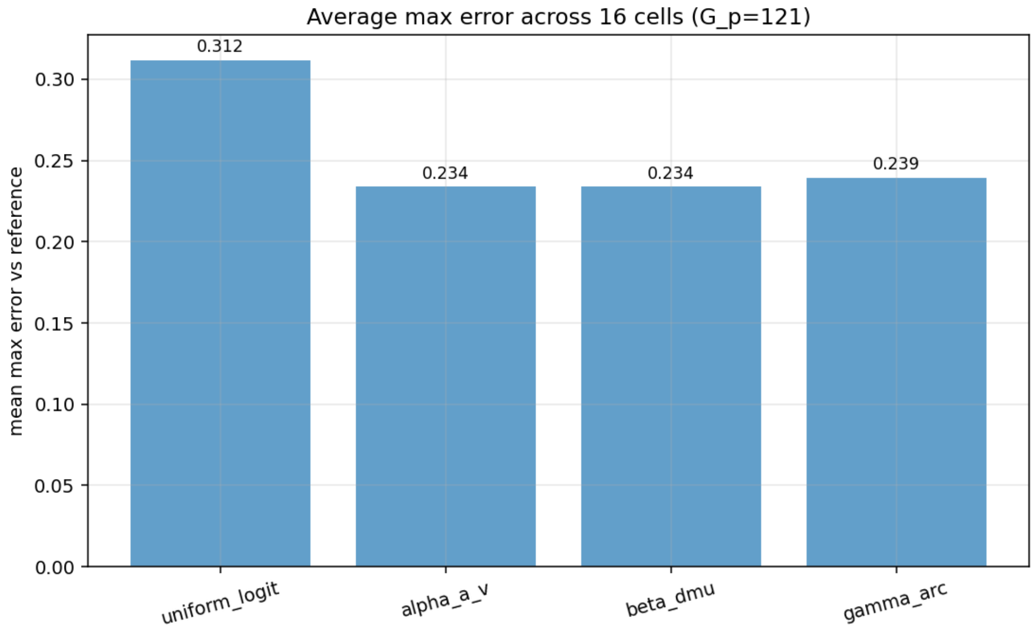

Figure E.7: Adaptive p-grid: average max-error by monitor function — arclength wins consistently.

E.2 Cusp geometry, continued

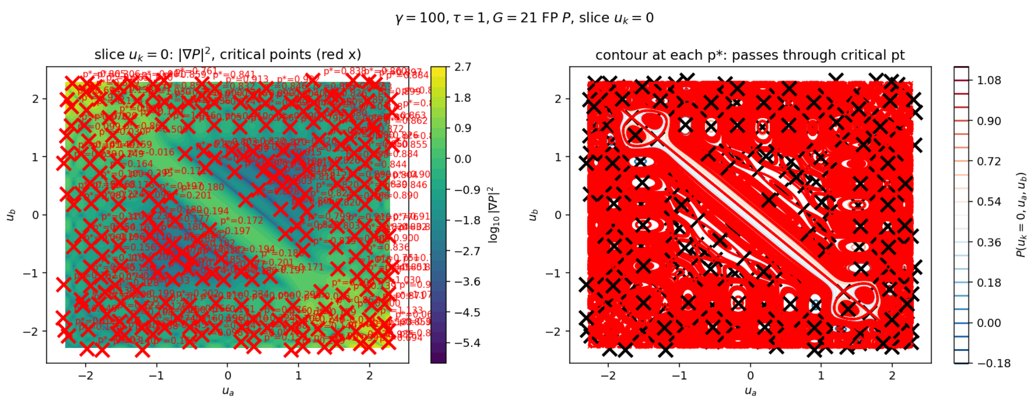

Figure E.8: Critical points on a second slice/regime, complementing the main-text figure: the critical manifold moves with u_k — the geometric root of Finding 13 (slice criticality, not 3D criticality, governs the lookup cusps).

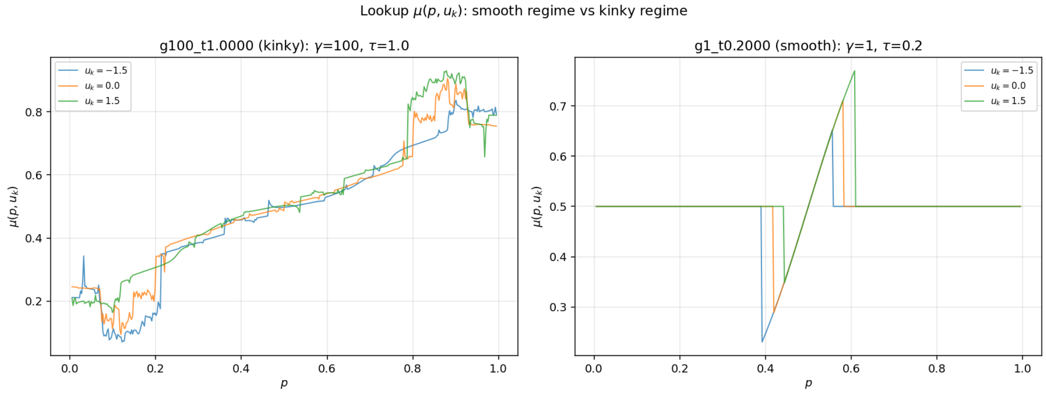

Figure E.9: Lookup cusps for the slice of the previous figure.

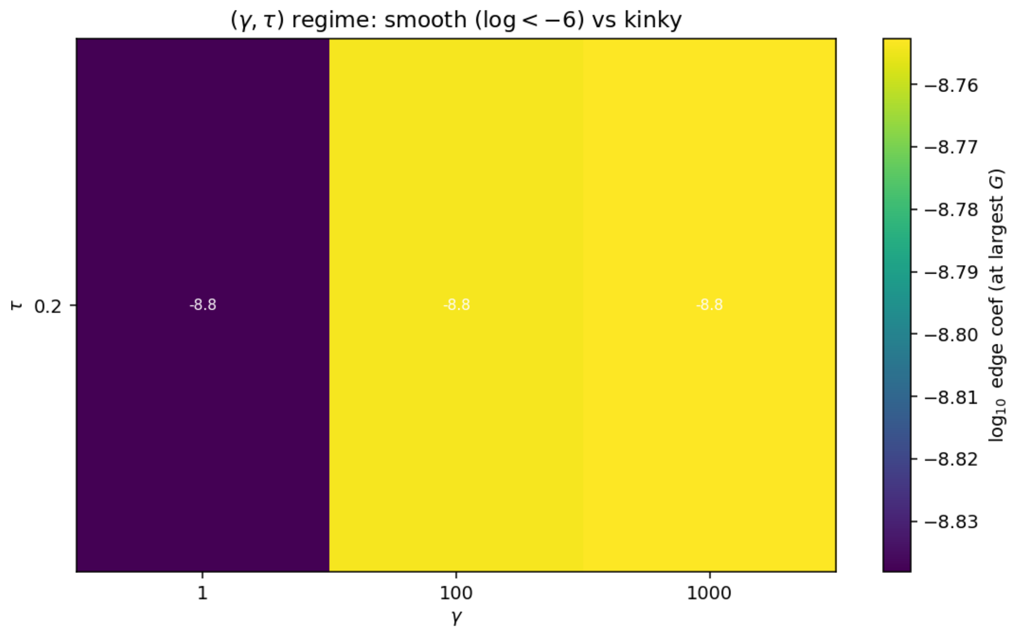

Figure E.10: The smooth/cuspy regime boundary in (γ, τ): interior critical points (and hence lookup cusps) emerge with rising τ; below the boundary global spectral methods would suffice.

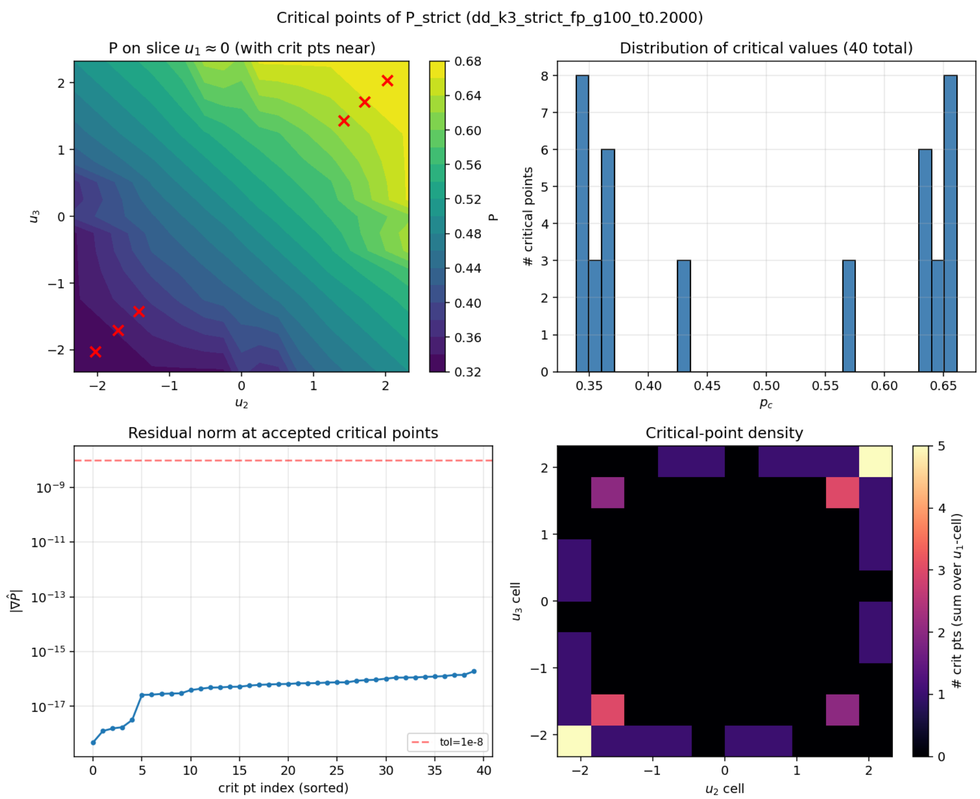

Figure E.11: 3D critical points at $(\gamma, \tau) = (100, 0.2)$: 40 points, 16 unique critical values confined to $p \in [0.34, 0.66]$ — the narrower cusp band of the low-τ regime.

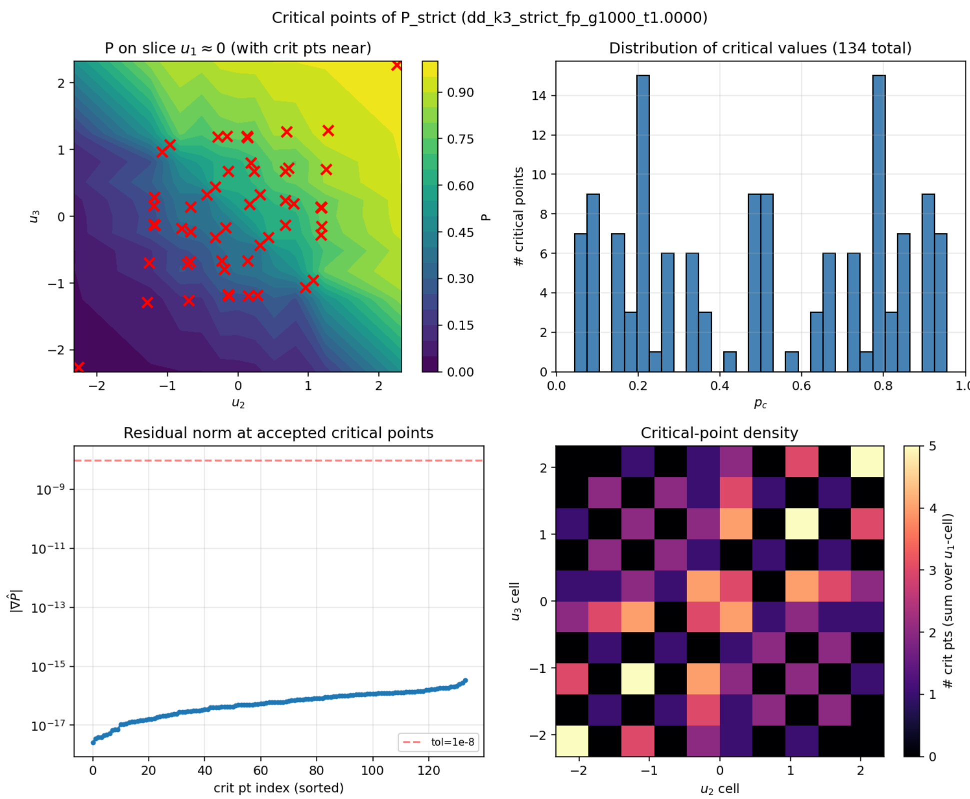

Figure E.12: 3D critical points at $(\gamma, \tau) = (1000, 1)$: the high-$\gamma$, high-$\tau$ corner; 134 points spanning $p \in [0.05, 0.96]$.

E.3 Structural ansatz, continued

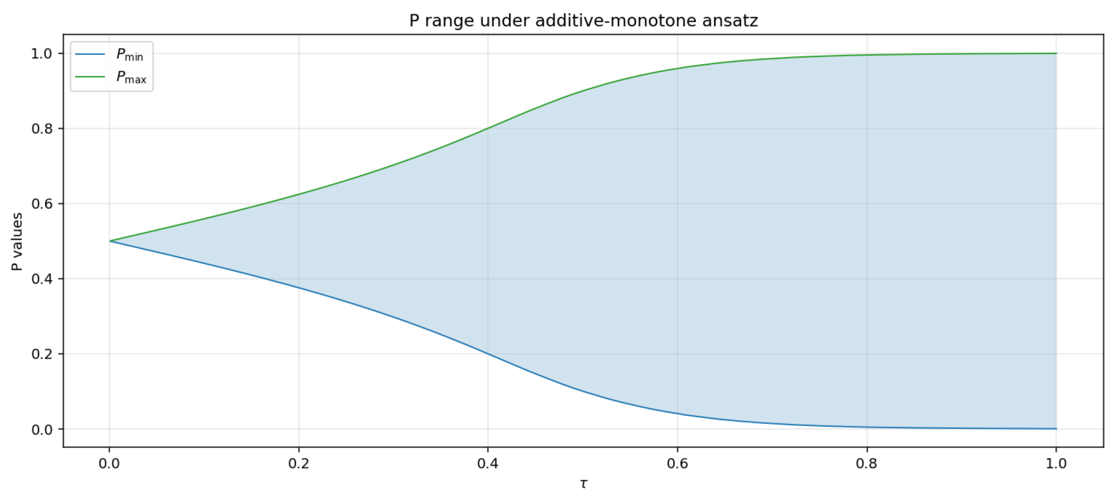

Figure E.13: Method C v2 ladder: price range vs τ under the additive ansatz — smooth and monotone by construction.

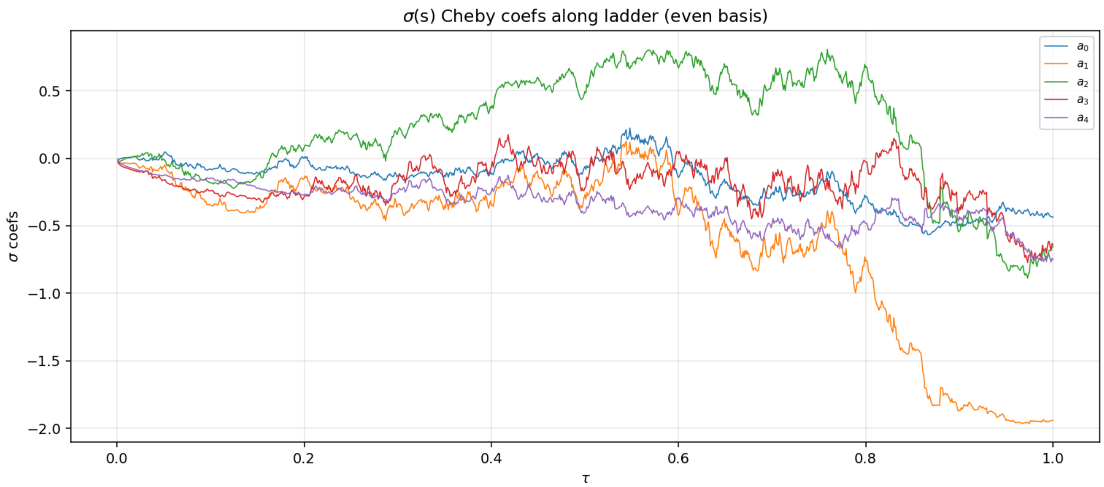

Figure E.14: Evolution of the five ansatz coefficients along the τ-ladder: smooth trajectories, suitable for warm-starting and for interpolating the ansatz across τ.

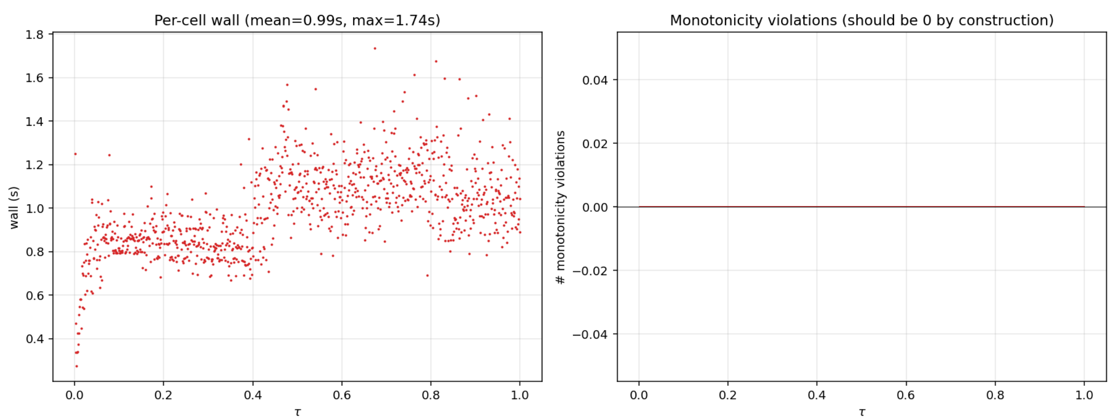

Figure E.15: Ansatz diagnostics along the ladder.

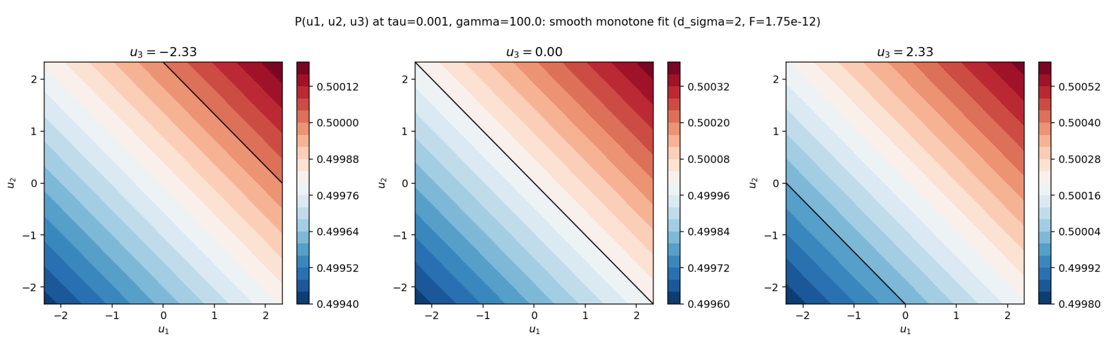

Figure E.16: Ansatz price slices at $\tau = 0.001$: visually indistinguishable from the analytic linear-Gaussian limit.

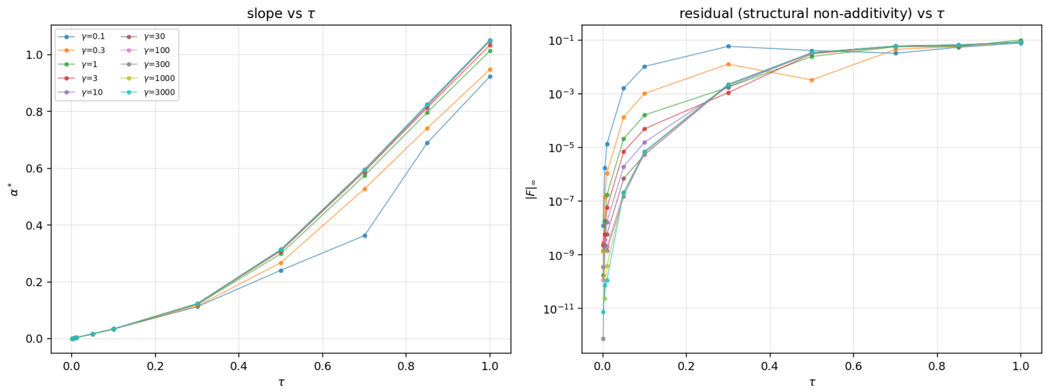

Figure E.17: Grid cross-sections: every γ row shows the same τ-transition — the factorization $\|F\|(\gamma,\tau) \approx \|F\|(\tau)$ of the ansatz residual.

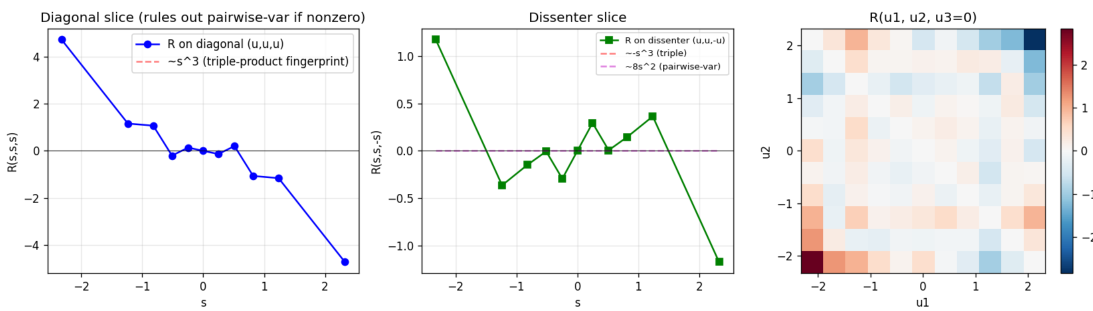

Figure E.18: The original v3 fingerprint (against the R4-era ansatz fit): diagonal and dissenter slices of the residual with candidate polynomial overlays — the first evidence that no simple symmetric polynomial captures the non-additive structure.

E.4 The pipeline iterations (v2–v4) and negative controls

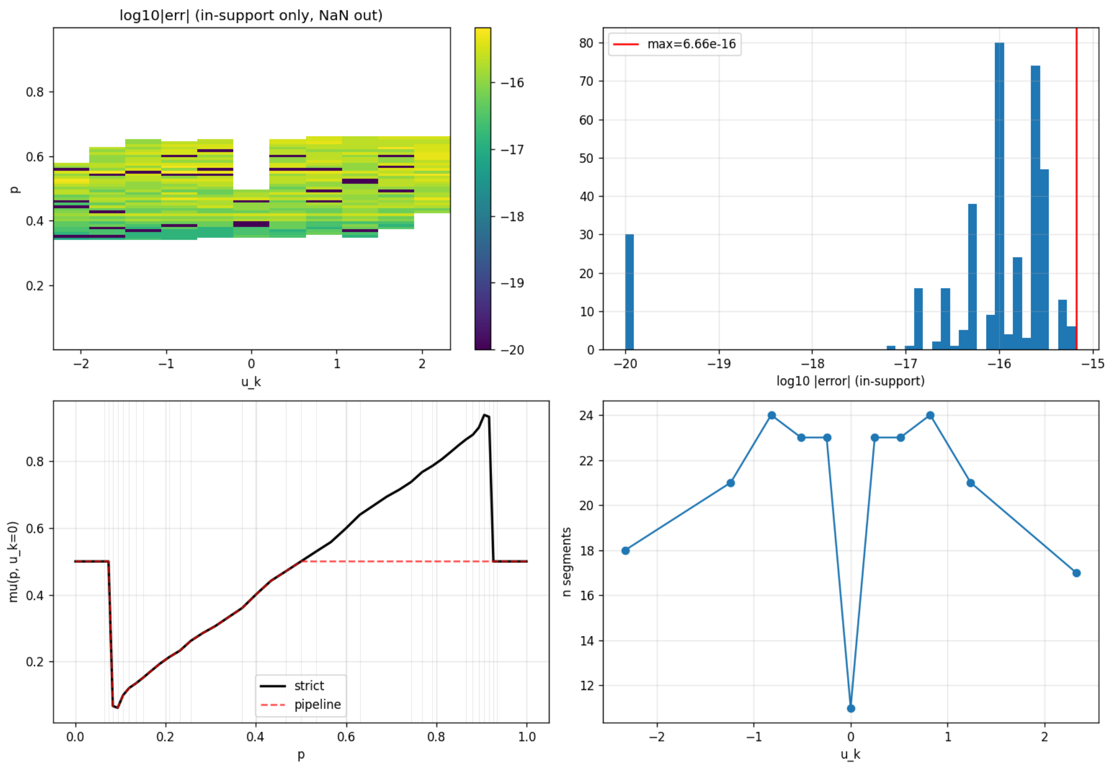

Figure E.19: Pipeline v2 at $(100, 1)$, degree 6: machine epsilon in-support, fallback plateau out-of-support (pre-tail-extrapolation stage).

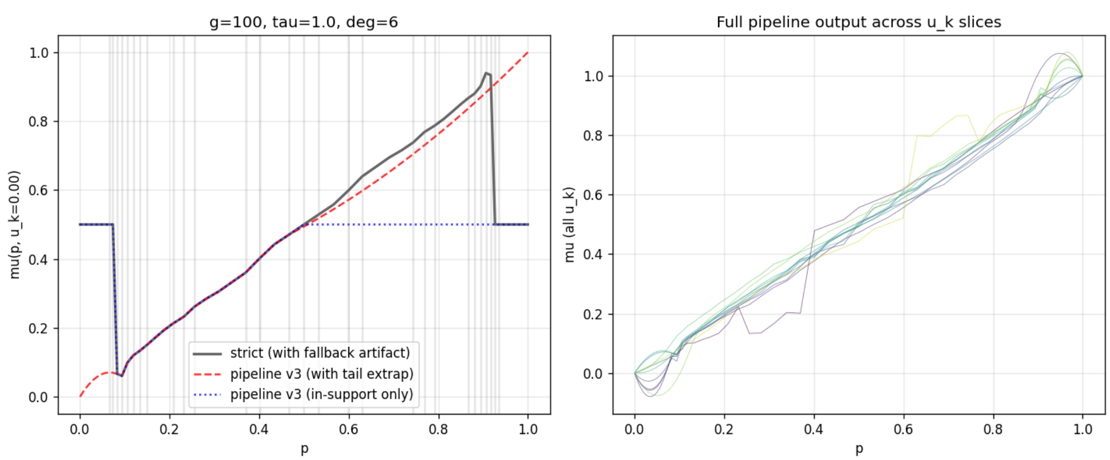

Figure E.20: Pipeline v3: Chebyshev tail extrapolation with Bayesian BCs — boundary-correct but with degree-2 tail overshoot (non-monotone), motivating v4's PCHIP tails.

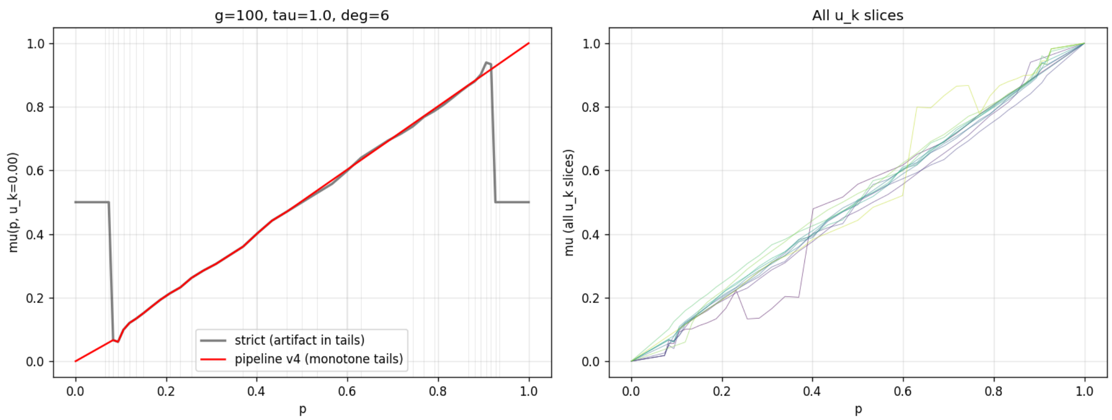

Figure E.21: Pipeline v4: shape-preserving tails. Monotone where the input is monotone — which exposed that the raw input was not (the fallback artifact), leading to Finding 14.

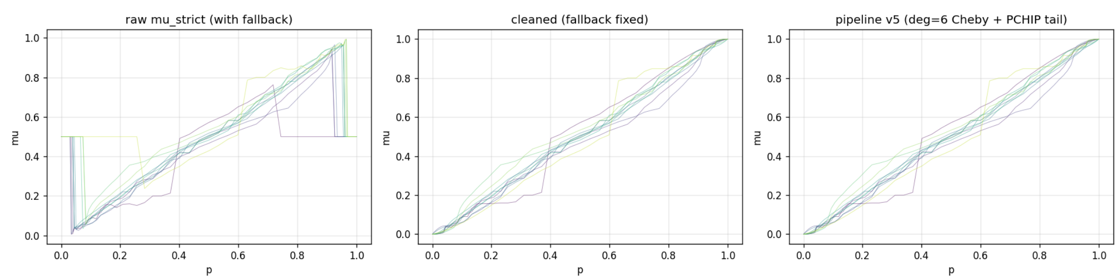

Figure E.22: Pipeline v5 at the $(1000, 1)$ corner: the full clean-detect-fit-extrapolate chain on the hardest reference cell.

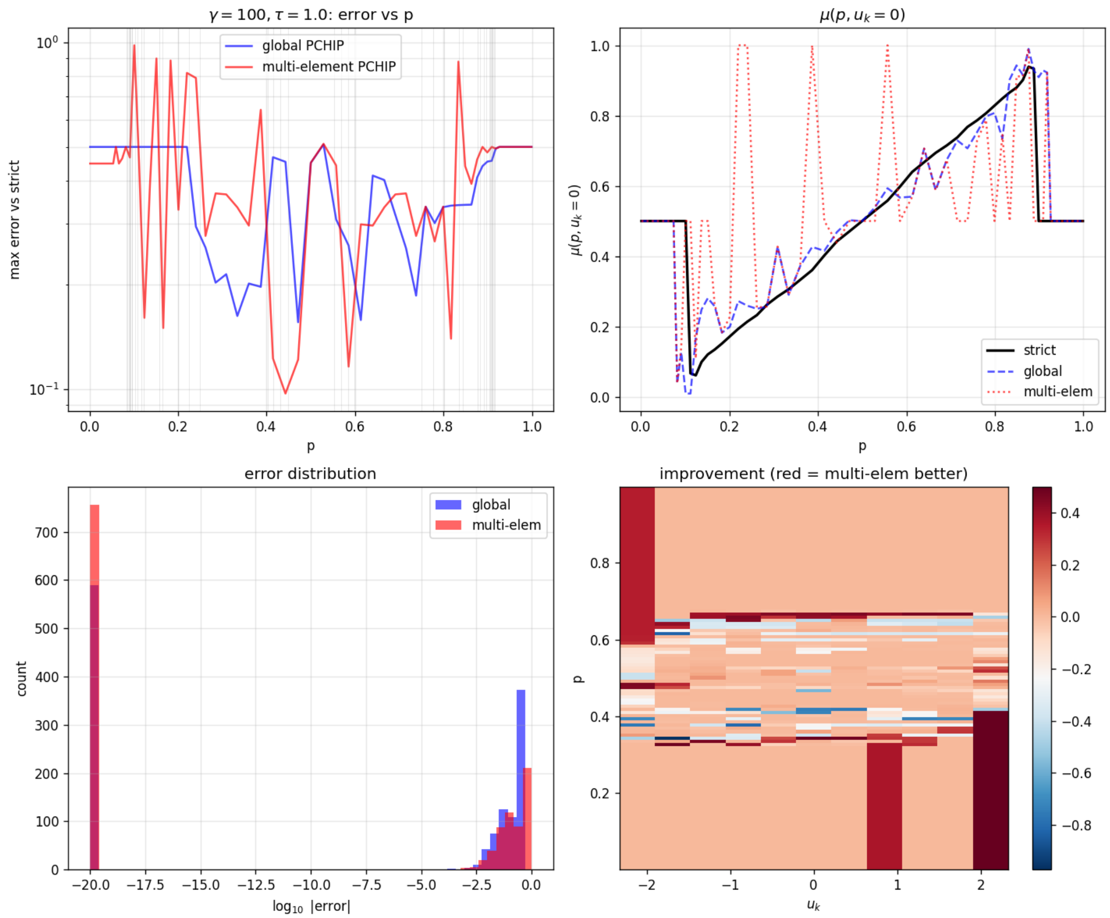

Figure E.23: Multi-element OptB+ (Finding 8): median pinned to zero at inserted knots, max unmoved — the diagnosis that the max error lives in the tails, not at the bulk cusps.

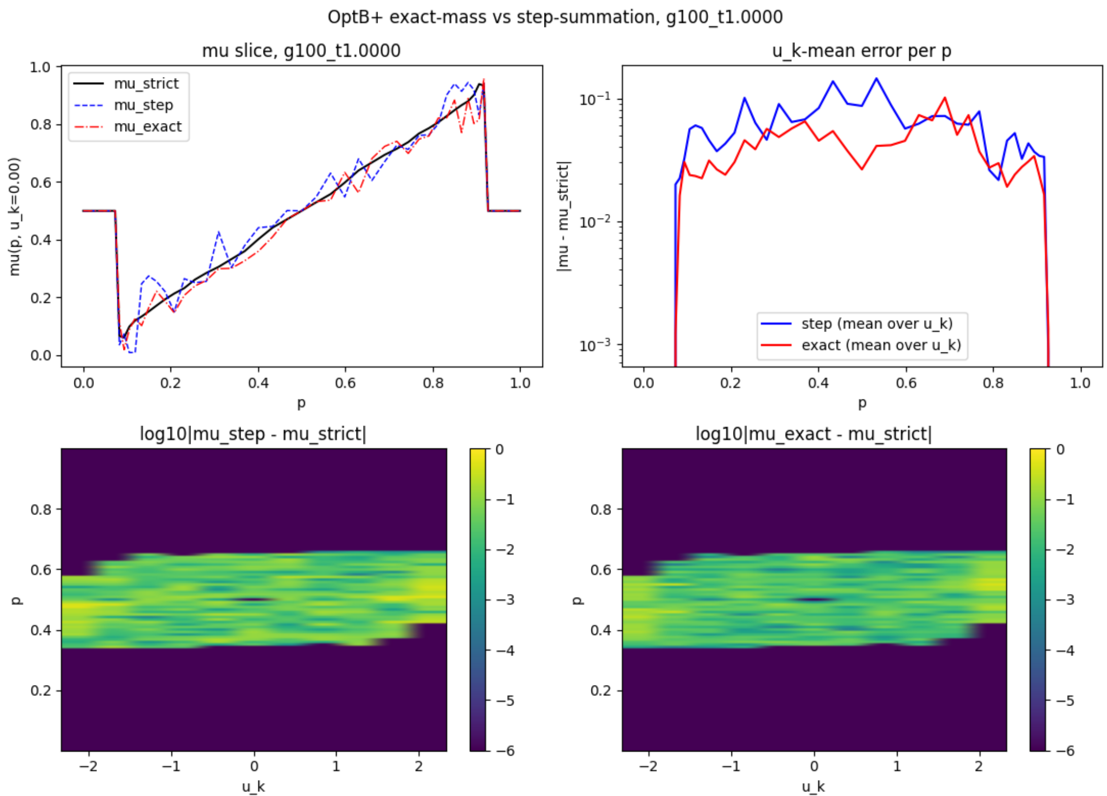

Figure E.24: Exact cell-mass integration vs step summation (Finding 12) at $(100, 1)$: the in-support median halves; the tails do not improve.

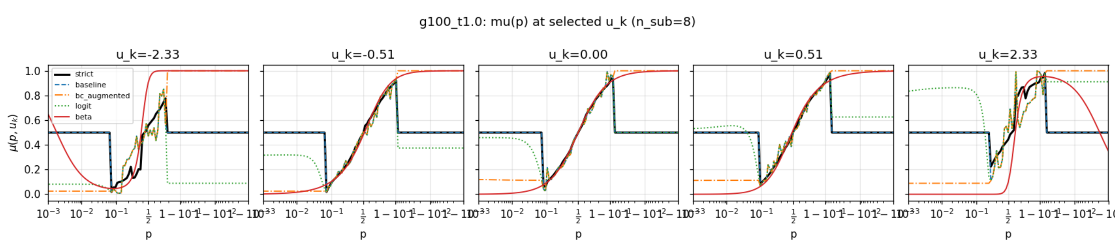

Figure E.25: Tail-BC variants (Finding 11): baseline, BC-augmented, logit-space, and Beta-CDF lookups overlaid — the three BC variants are indistinguishable from baseline in the bulk, and the bulk is where the max error lives.

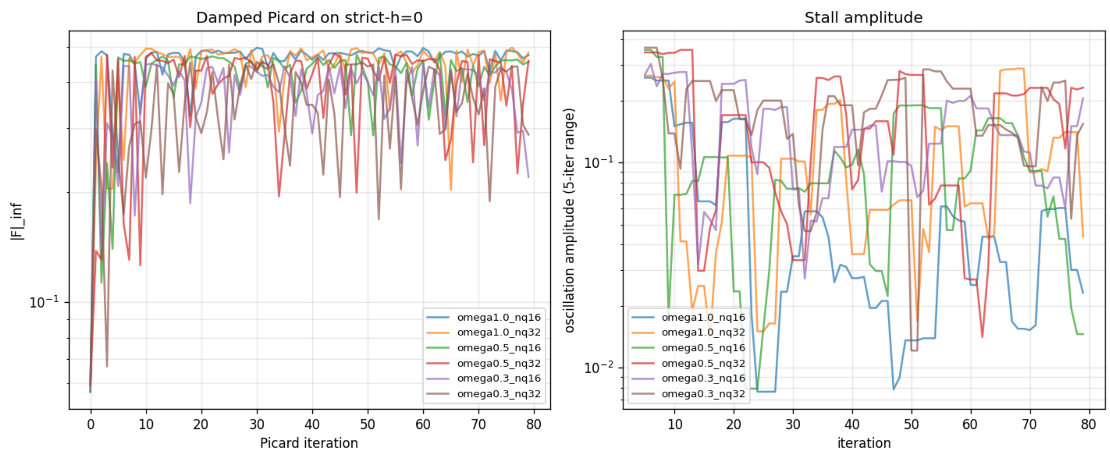

Figure E.26: The stall trace that opened Finding 9: damped Picard *diverges from the saved "fixed point"* at every damping tested — proof that the saved strict-h=0 object was a best iterate.

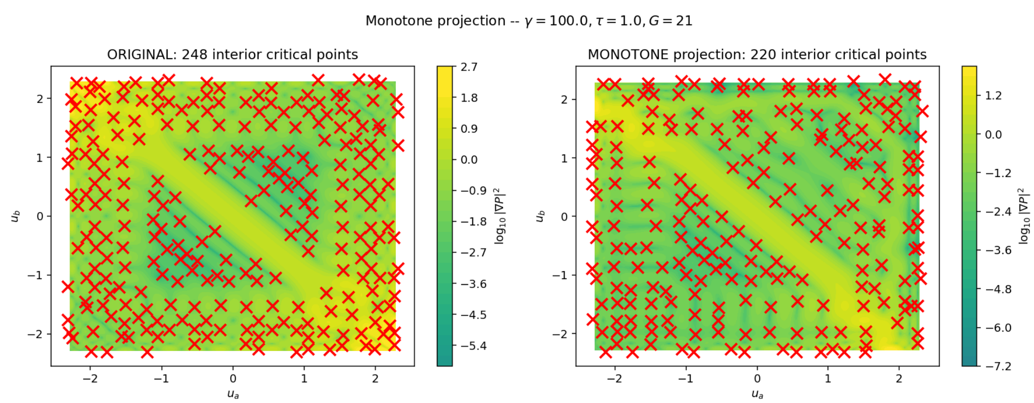

Figure E.27: Monotonicity projection experiment on the R4 fixed points: enforcing axis-monotonicity post hoc reshapes the surface materially — corroborating that the R4 violations were not cosmetic.

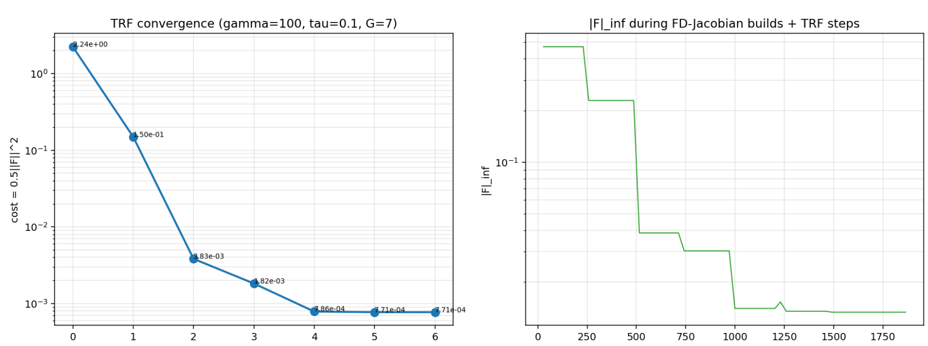

Figure E.28: Trust-region convergence of the ansatz fit: quadratic basin behavior, hitting the ansatz expressiveness floor in tens of iterations — solver speed was never the v2/v3 bottleneck.

E.5 The log-odds configuration, continued

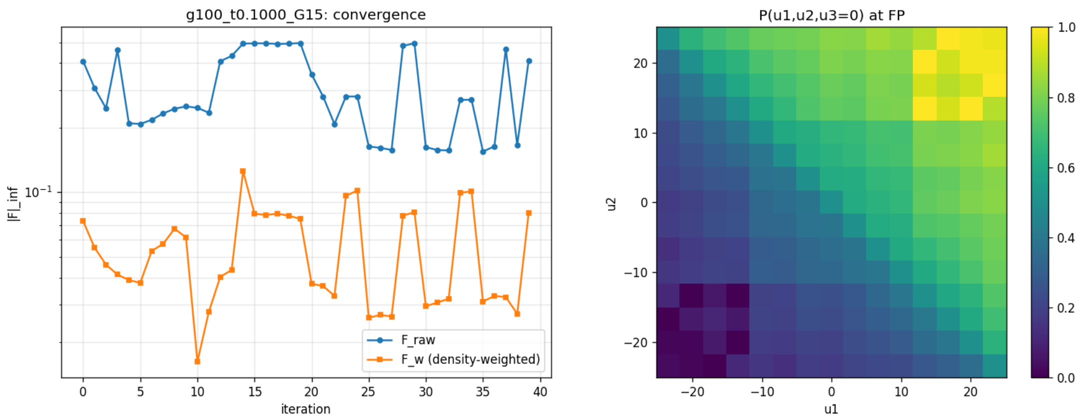

Figure E.29: Log-odds configuration at $G = 15$: the weighted residual reaches 10^{-4} before the (frozen) L_{pub} refresh perturbs it — the best surface obtained in the new coordinates at $\tau = 0.1$.

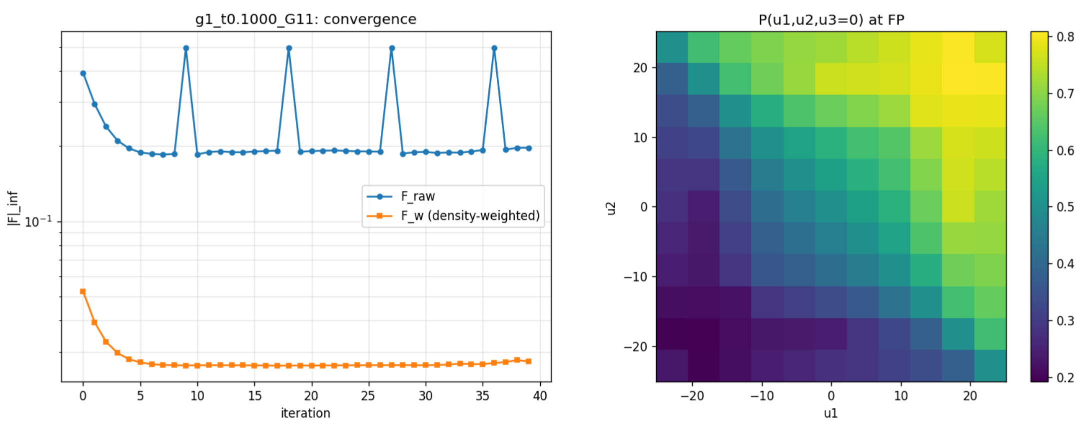

Figure E.30: $\gamma = 1$ in the log-odds configuration: the low-risk-aversion cell — largest Jensen wedge, widest price range, and the noisiest convergence of the sweep, consistent with the $1/\gamma$ scaling of the curvature term.

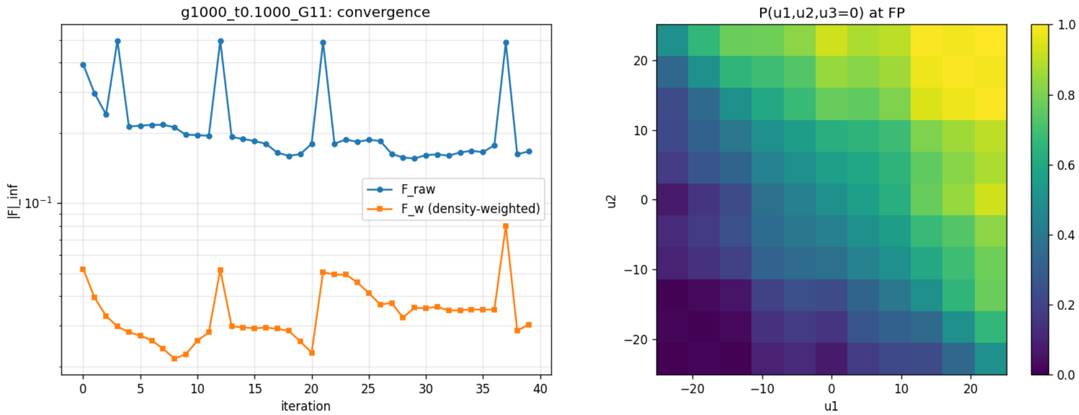

Figure E.31: $\gamma = 1000$: approaching the CARA limit — tighter price function, cleaner convergence; the wedge shrinks as $1/\gamma$.